



REB | RWANDA BASIC
EDUCATION BOARD

GEOGRAPHY AND ENVIRONMENT

Senior Three



Student's Book
Experimental version

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FOREWORD

Dear Student,

Rwanda Basic Education Board is honored to present to you the Geography and Environment book for Senior Three, which serves as a guide to competence-based teaching and learning to ensure consistency and coherence in the learning of geography and Environment subject. The Rwandan educational philosophy is to ensure that you achieve full potential at every level of education, which will prepare you to be well integrated in society and exploit employment opportunities.

The government of Rwanda emphasizes the importance of aligning teaching and learning materials with the syllabus to facilitate your learning process. Many factors influence what you learn, how well you learn and the competence you acquire. Those factors include quality instructional materials available, assessment strategies for the learners among others. Special attention was paid to activities that facilitate learning process, develop your ideas and make new discoveries during concrete activities carried out individually or with peers. In competence-based curriculum, learning is considered as a process of active building and developing knowledge and meanings by the learner where concepts are mainly introduced by an activity, a situation or a scenario that helps the learner to construct knowledge, develop skills and acquire positive attitudes and values. For effective use of this textbook, your role is to:

- Work on given activities which lead to the development of skills
- Share relevant information with other learners through presentations, discussions, group work and other active learning techniques such as role play, case studies, investigation and research in the library, from the internet or from your community;
- Participate and take responsibility for your own learning;
- Draw conclusions based on the findings from the learning activities.

To facilitate you in doing activities, the content of this book is self-explanatory so that you can easily use it by yourself, acquire and assess your competences. The book is made of units whereby each unit comprises: the key unit competence, followed by introductory activity before the development of geography concepts that are connected to real world situations.

I wish to sincerely extend my appreciation to the REB staff who organized the editing process of this textbook. Special gratitude also goes to lecturers, teachers, illustrators and designers who supported the exercise throughout. Any comment or contribution would be welcome to the improvement of this textbook for the next edition.



Dr. MBARUSHIMANA Nelson

Director General, REB

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I wish to express my appreciation to all the people who played a major role in editing process of this Geography and Environment book for Senior Three. It would not have been successful without their active participation.

Special thanks are given to those who gave their time to read and refine this textbook to meet the needs of competence-based curriculum. I owe gratitude to different Universities and schools in Rwanda that allowed their staff to work with REB to edit this book. I therefore wish to extend my sincere gratitude to lecturers, teachers, illustrators, designers and all other individuals whose efforts in one way or the other contributed to the success of this edition.

Finally, my word of gratitude goes to the Rwanda Basic Education Board staff particularly those from Curriculum, Teaching and Learning Resources Department who were involved in the whole process of editorial work.



Joan Murungi,

Head of Curriculum, Teaching and Learning Resources Department/REB

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UNIT 1

MAP READING AND INTERPRETATION



Key Unit Competence:

Interpret the relationship between physical and human features on maps.

Learning objectives

At the end of this unit you should be able to:

Knowledge and understanding

1

- Identify physical and human features on a map
- Reduce and enlarge maps

Skills

2

- Differentiate the physical and human features on a map with the help of a key.
- Interpret the relationship between human and physical features on a map.
- Apply appropriate scale calculations to enlarge or reduce a given map accurately.

Attitudes and values

3

- Appreciate the use of topographic maps to interpret physical and human features.
- Develop the ability to associate physical and human phenomena using maps.
- Appreciate the importance of reducing and enlarging maps.

1.0. Introduction

Map interpretation is the skill of reading and analyzing maps to understand the location, distribution, and relationships of physical and human features. It involves identifying features, understanding their interactions, drawing sketch maps and diagrams, and applying techniques for reducing or enlarging maps. These skills enable accurate representation and analysis of geographical information at different scales.

Introductory activity

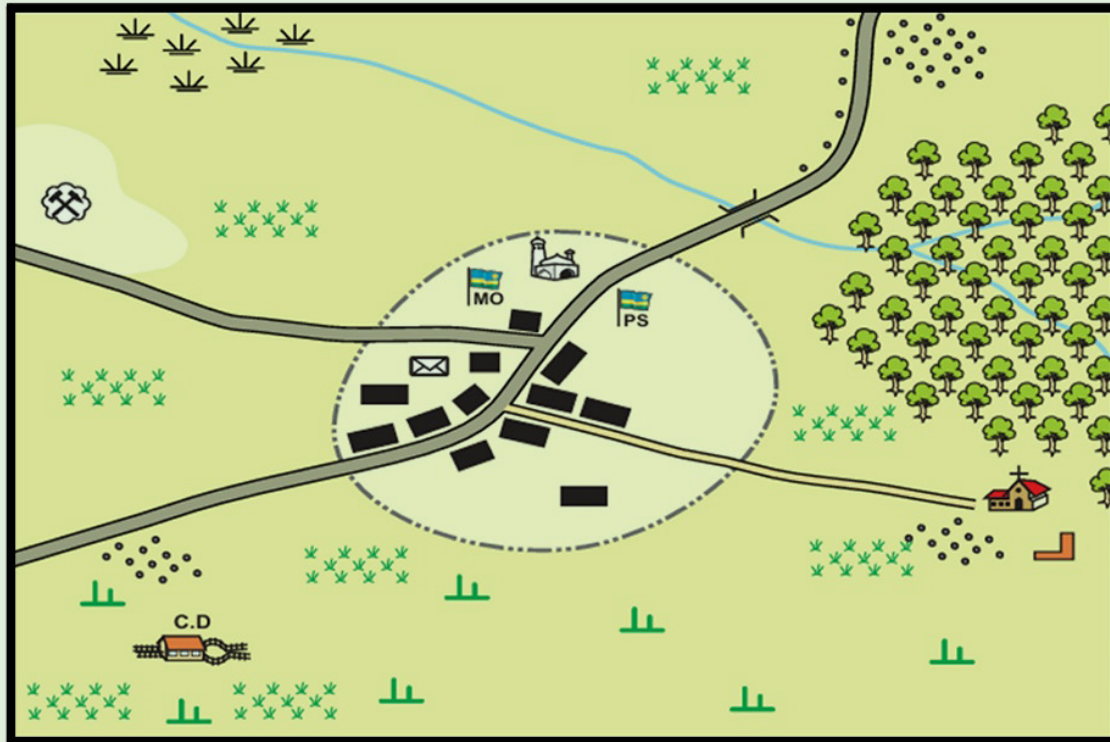
One morning during a Geography lesson, the teacher displayed a map of a nearby district. A learner named Aganze explained that she wanted to visit her relatives and needed the map to choose the best route. As she studied the map, she noticed rivers, hills, roads, settlements, and farmlands marked with various symbols and colors. She wondered which of these were physical features (natural) and which were human features (man-made). She also questioned why many settlements were located near rivers and along roads.

1. What features did Aganze notice on the map?
 - a) Rivers, hills, roads, settlements, and farmlands
 - b) Only roads and settlements
 - c) Mountains and deserts only
 - d) Lakes and oceans only
2. Which statement correctly shows natural and human-made features?
 - a) Natural (rivers and hills) and human-made (roads, settlements, and farmlands)
 - b) Natural (roads and settlements and human-made (rivers and hills)
 - d) **a** and **b** are correct
3. How can you tell the difference between physical and human features on the map?
4. What symbols, colours, or shapes are used to represent these features?
5. Why do you think some human features are found near certain physical features?

1.1. Identification of human and physical features on a map

Learning activity 1.1

1. During a Geography lesson, Ganza was given a sketch map by his teacher. She asked him to explain to his classmates how he would identify the physical and human features represented on it.

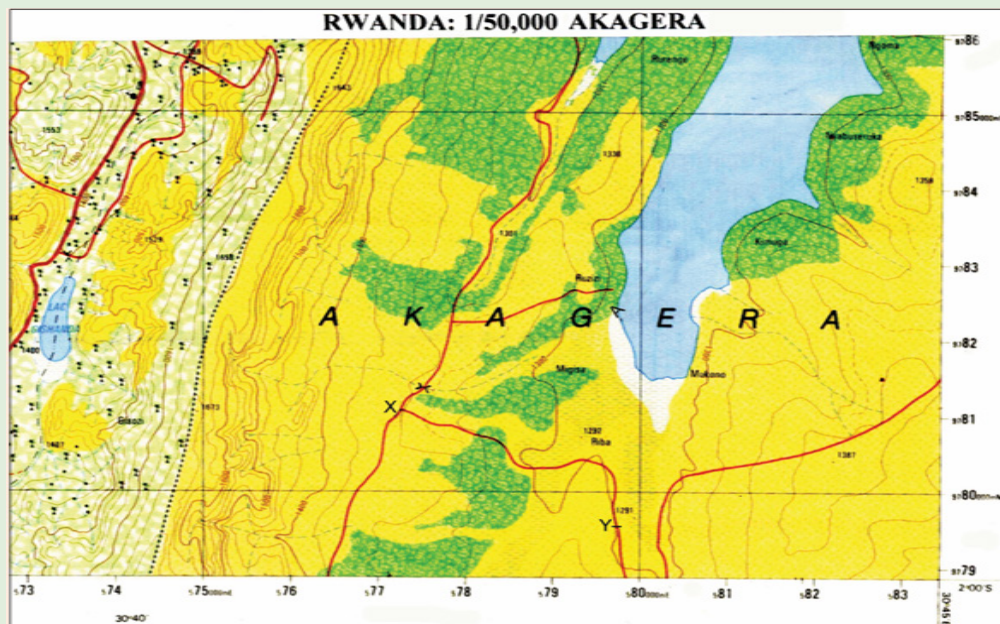


KEY

Tarmac road	Church	Mosque
Murram road	Forest	School
River	Grass	Quarry
Bridge	Scrub	Mayor's Office
Built-up area	Swamp	Police Station
Human Settlements	Cattle dip	
	Post Office	

- a) Suggest the physical features he could identify.
- b) List down the human-made features she could see on the map.
- c) Using evidence from the map, suggest the activities taking place on the map Ganza may have identified.

2. Identify the relief features shown on the topographical map below and prepare a class presentation based on your findings.



Topographic map of Akagera

The above **learning activity 1.1** shows that a map has both physical and human features, which relate to each other.

- The physical features may include ridges, ranges, hills, valleys, spurs, cliffs, plains, different types of vegetation, rivers, swamps, lakes, and steep slopes.
- Human features on a map include infrastructure such as transport systems and boreholes, as well as land use like crops, settlements, and towns.

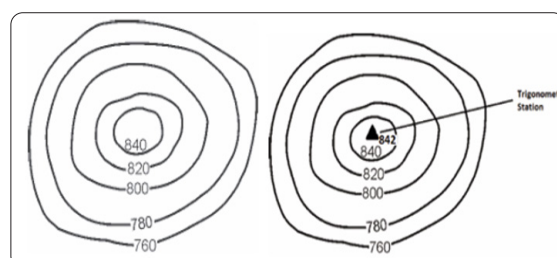
Topographic maps are often known as topo sheets. Simply, topographic maps are maps that show both natural features and human-made.

Topographic maps are often used as general reference maps because of the great variety of information included on them. They are used to show qualitative details as well as quantitative representations of natural and human-made features.

1. How to identify physical features from a topographical map

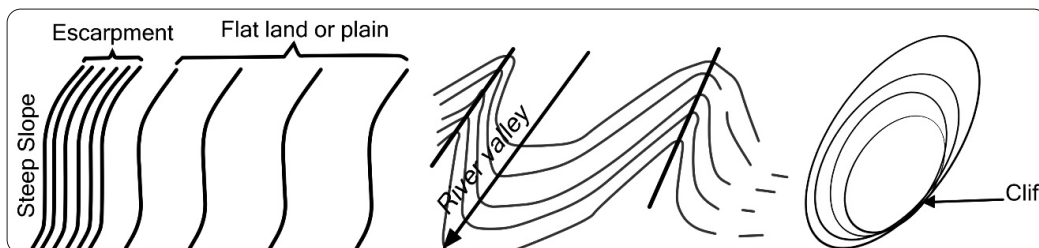
1. Relief

- Where contours form concentric circles, they indicate **a hill or a mountain**. Trigonometrical stations are used to show the actual height of a hill or a mountain.



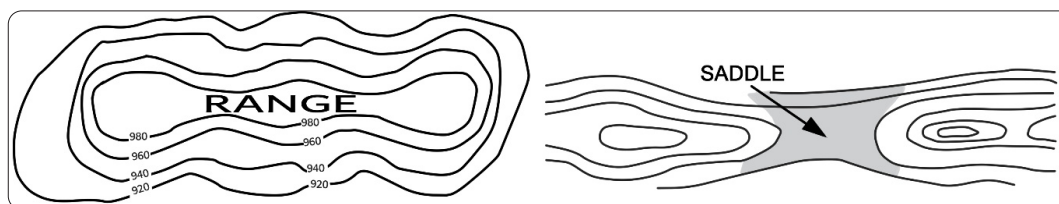
Contours and trigonometric stations indicate a hill or a mountain and its actual height.

- Where contours are close to one another, they indicate a **steep slope**. Where they are far apart, they indicate a **plateau, plain, or flat land**.
- Where contours are V or U-shaped, they indicate a **valley**. The shape of the contours usually point to the source of the river.
- Where contours appear to overlap or cross each other, they form a **cliff**.



A steep slope and a plain, a valley and a cliff

- A **range** is a series or chain of mountains or **ridges** that are connected and extend over a long distance. It is shown on a topographic map using a combination of contour lines and specific patterns.
- A **saddle** is a depression found between two hills. On a map, it is shown as an area of lower ground between two higher contour peaks.



A ridge and saddle

1. Drainage

Drainage refers to rivers, lakes, swamps, and dams. These water features are usually shown on a map using **blue colour**.

2. Vegetation

Vegetation is represented on a topographical map using different **colours** or using **symbols**.

II. Human features shown on a topographical map

Human features are made by people or come about because of people's activities. Some of the human features commonly shown on the map are settlement, means of transportation, agriculture, trade, and mining. To identify human and economic activities on a topographical map, look for evidence detailed in the table below:

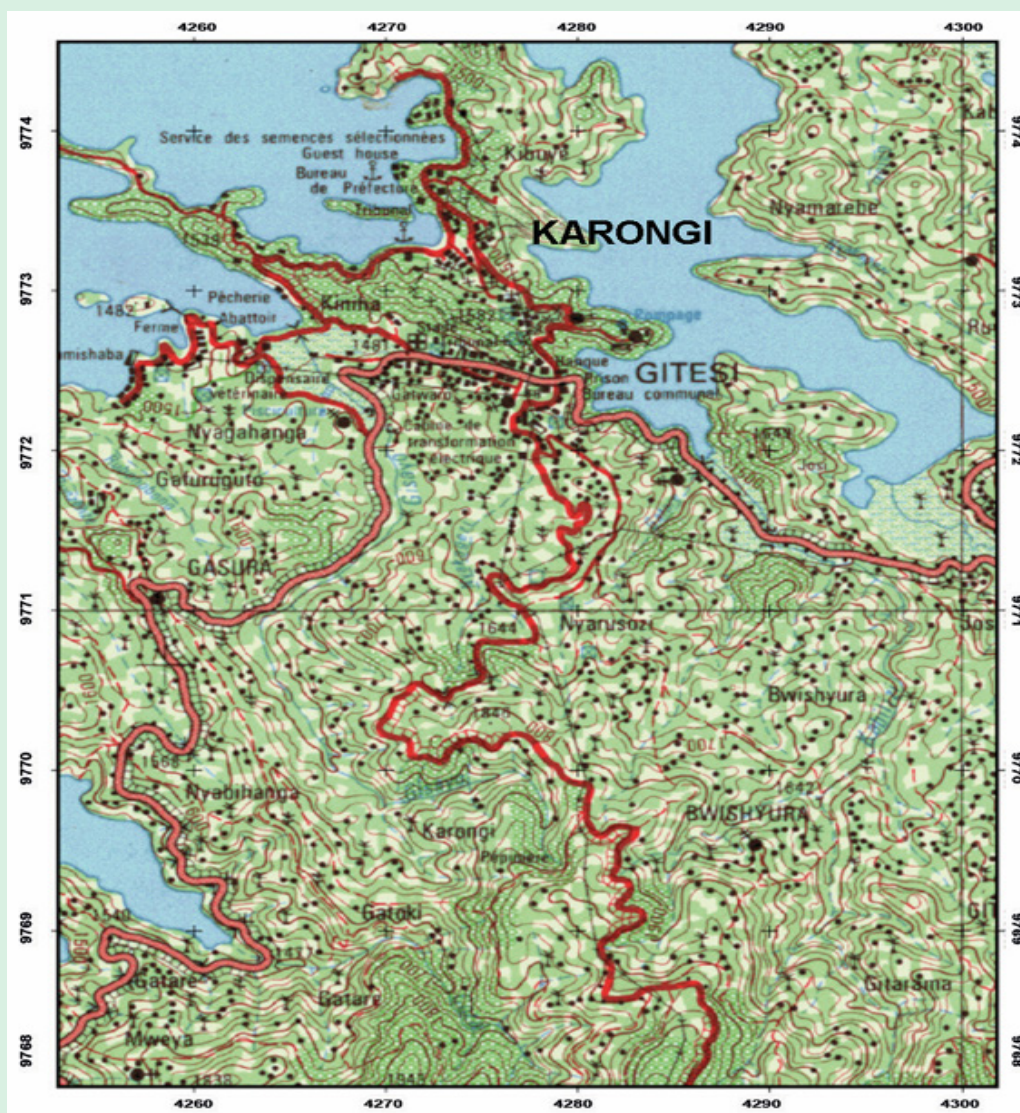
Table 1.1. Identification of human and economic activities on a topographic map.

Human activity	Evidence / Infrastructure/features
Trade & Commerce	Market, shops, stores
Manufacturing & Industry	Quarry, name of minerals, Power lines
Transportation & Connectivity	Railways, airstrips, Power lines (for transmission networks)

Education & Services	Schools, colleges, university
Tourism & Leisure	Hotels, game parks, mountains, Stadium, swimming pool
Primary Activities	Lumbering, Fishing, Livestock rearing
Religion & Culture	Religion (Activity - assumed to refer to religious sites/buildings)

Application activity 1.1

1. Study the map of Karongi (1:50,000), then answer the questions that follow:



Topographic map of Karongi

2. Describe:

- Relief features on the map.
- Human-made features on the map.

2. Find out how relief features relate to human activities on the map of Karongi.

1.2. Relationships between different features (human and physical aspects)

Learning activity 1.2

Read the story of Mutesi's journey from Congo-Nile to Muhanga carefully and then answer the questions that follow:

1. Mutesi from Congo-Nile travelled to Muhanga and observed how relief influenced settlement. Steep land near her home had few people, while flat areas near Lake Ruhondo were densely populated. In Gakenke, she saw hills with few houses on gentle slopes. She also observed that swampy land near Nyabarongo River was uninhabited. Throughout her journey, she observed that shopping centres were crowded, and settlements lined on the shores of Lake Ruhondo and Lake Burera. Discuss why you think people are not settled in swamps and on steep slopes.
2. Identify and list down the physical features mentioned in the story above that can be shown on a topographic map.

Relationships between physical and human features on a topographic map

Human Activity / Feature	Related physical feature(s)	How they are connected/related
Settlements (villages, towns)	Gentle slopes, water sources, fertile soils	People settle where land is flat, water is available, and farming is possible.
Roads, Railways, Airstrips	Valleys, gentle slopes, river crossings	Transport avoids steep slopes and follows valleys; bridges appear where transport meets rivers.
Agriculture (crop farming)	Fertile soil, flat/ gentle land, water sources	Good soil and flat land encourage farming; irrigation uses nearby rivers.
Livestock rearing	Grasslands, open plains	Animals graze in wide, gently sloping grassland areas.
Fishing	Lakes, rivers	Fishing villages develop along water bodies.
Lumbering	Forests, woodlands	Timber cutting happens where forests are located; sawmills may appear nearby.
Mining/Quarrying	Rocky areas, hills with minerals	Mines and quarries are located where mineral deposits exist.
Trade and services (markets, shops)	Flat land	Markets grow where roads meet and where many people live (Transport routes and settlements)
Tourism (hotels, parks)	Mountains, lakes, forests, scenic areas	Tourists visit beautiful landscapes, attracting hotels and camps.

Education (schools, colleges)	Accessible land	Schools are built where people live (Settlements) and where land is easy to build on.
Power lines & communication	Ridges, gentle slopes	Power lines avoid steep terrain and run through open, elevated areas.
Recreation (stadiums, playgrounds)	Flat land	Recreational facilities need large flat spaces, usually in towns/urban areas

Application activity 1.2

- Why are settlements few or absent in swampy areas and on steep slopes?
 - Because the land is difficult to build on and prone to flooding or erosion.
 - Because these areas are very close to markets.
 - Because soil is very fertile and attracts too many farmers.
 - Because people prefer living in remote areas.
- Which of the following lists correctly identifies physical and human features mentioned in Mutesi's journey that can be represented on a topographic map?
 - Mountains, rivers, forests; villages, shopping centres, roads.
 - Lakes, rivers, hills; schools, power lines, recreational facilities.
 - Swamps, gentle slopes, hills; settlements, markets, roads.
 - All of the above.

1.3. Reduction and enlargement of maps

Learning activity 1.3

During her Geography lesson, Ganza was given a topographic map of Karongi (scale 1:50,000). Her teacher asked her to redraw the map so that it was twice the size of the original and another version half the size, keeping the shapes of all features accurate. She carefully labeled each version with its correct scale factor.

- Why did Ganza need to keep the shapes of the features unchanged when redrawing the map?
 - To make the map look more colorful
 - To ensure distances and proportions are accurate
 - To reduce the number of features shown
 - To make the map easier to read
- What scale factor should Ganza write for the map that is twice the size of the original?

A. 1:100,000 B. 1:25,000 C. 1:50,000 D. 1:10,000

3. What scale factor should Ganza write for the map that is half the size of the original?

A. 1:100,000 B. 1:25,000 C. 1:50,000 D. 1:10,000

4. How does enlarging or reducing a map help in understanding the area represented?

A. It allows comparison of distances and features at different levels of detail

B. It makes the map more colorful and attractive

C. It removes unnecessary features from the map

D. It allows people to draw freely without using scales.

Procedures for the reduction and enlargement of the map

a) Method of squares

1. Trace the same part of the water body the way it is on a plain piece of paper. This is the actual size of the lake as per the map scale.
2. Draw grid squares on the map of the lake. The closer the grid, the more accurate the result will be.
3. Now draw another similar network of squares, enlarged or reduced as desired and copy the details of the lake square by square, by eye, on the drawing paper. Note every important point where the details intersect with the grid lines.
4. What do you see? Either an enlarged or a reduced map of the water body.

a) Method of similar triangles:

1. Use a plain piece of paper to draw either a position of a river or a road running vertically.
2. Mark point A at the top and point B at the bottom end.
3. Join the two points AB using a straight line to follow the major bends and curves.
4. Choose any point T at a convenient distance to the left of line AB.
5. Join each end A and B to T and any other significant point.
6. To reduce the size of road AB by half, measure distance TA and divide by 2, and mark the point C. Do the same for TB and mark point D. Notice line CD is parallel to line AB.
7. To enlarge the size of road AB, draw a parallel line to the right of guideline AB and label GH.
8. Trace the road using the points of relevant intersection of the road with lines originating from point T.
9. GH is an enlargement of AB.

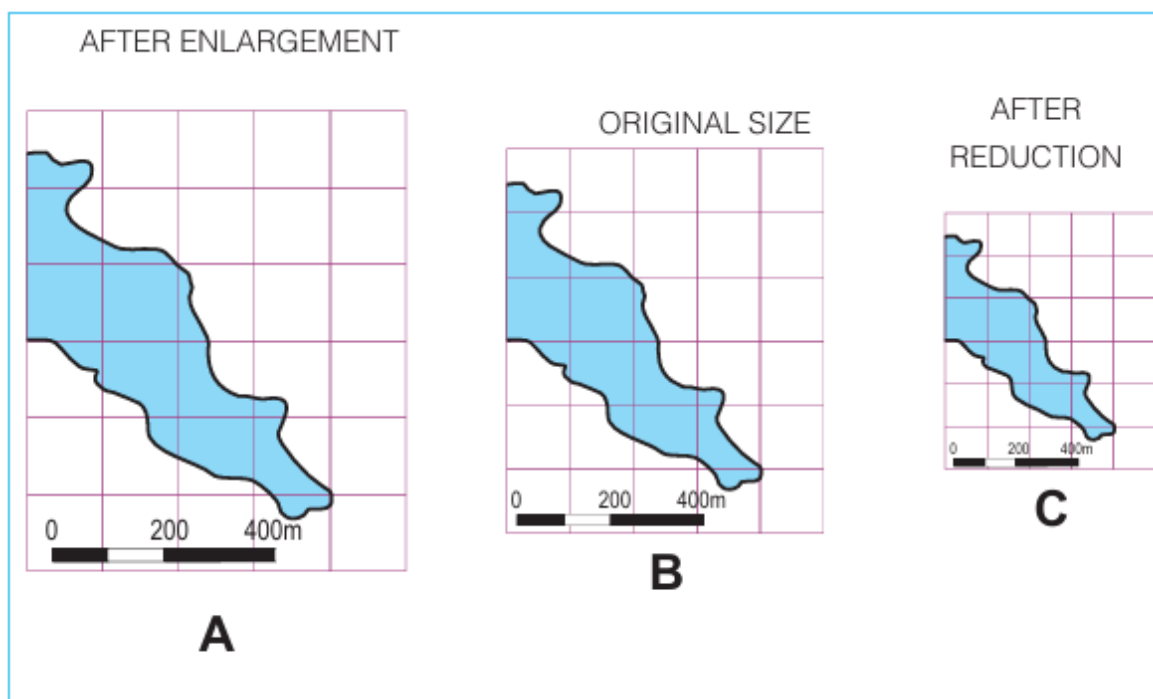
Reduction or enlargement of the map

Reduction or enlargement is where a feature or area on a map is drawn either smaller or bigger than the original size. Reduction or enlargement should result into shapes or areas that are identical to the original, without a change in form. An enlargement provides the same map but proportionally larger than the original.

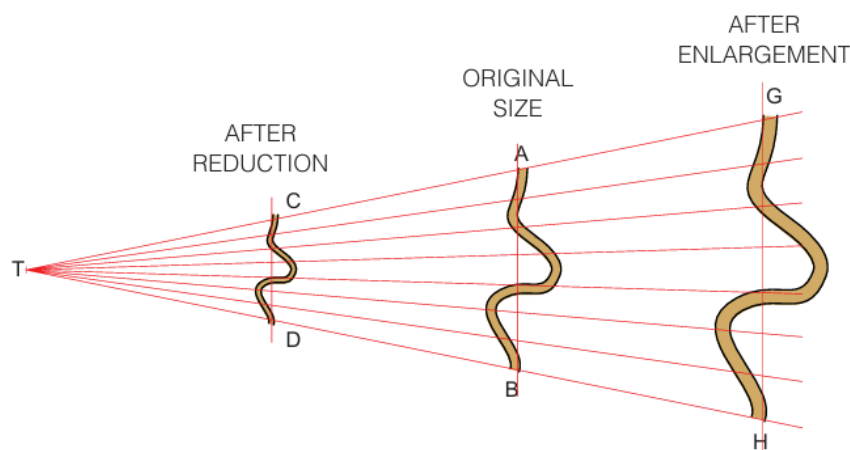
A reduction gives the same map that is proportionally smaller than the original. If the map has been reduced by $\frac{1}{2}$. The amount by which the original image/ map has been enlarged or reduced is called **a scale factor**, or **an enlargement or reduction factor**. It is the size you increase or decrease of the shape. The reduction and enlargement can be done using the method of squares or the method of similar triangles.

For example:

- (i) Using the method of squares, the following figures show how the original map of a lake, marked B, has been enlarged to 1.5 and reduced to 0.75 of its original size.



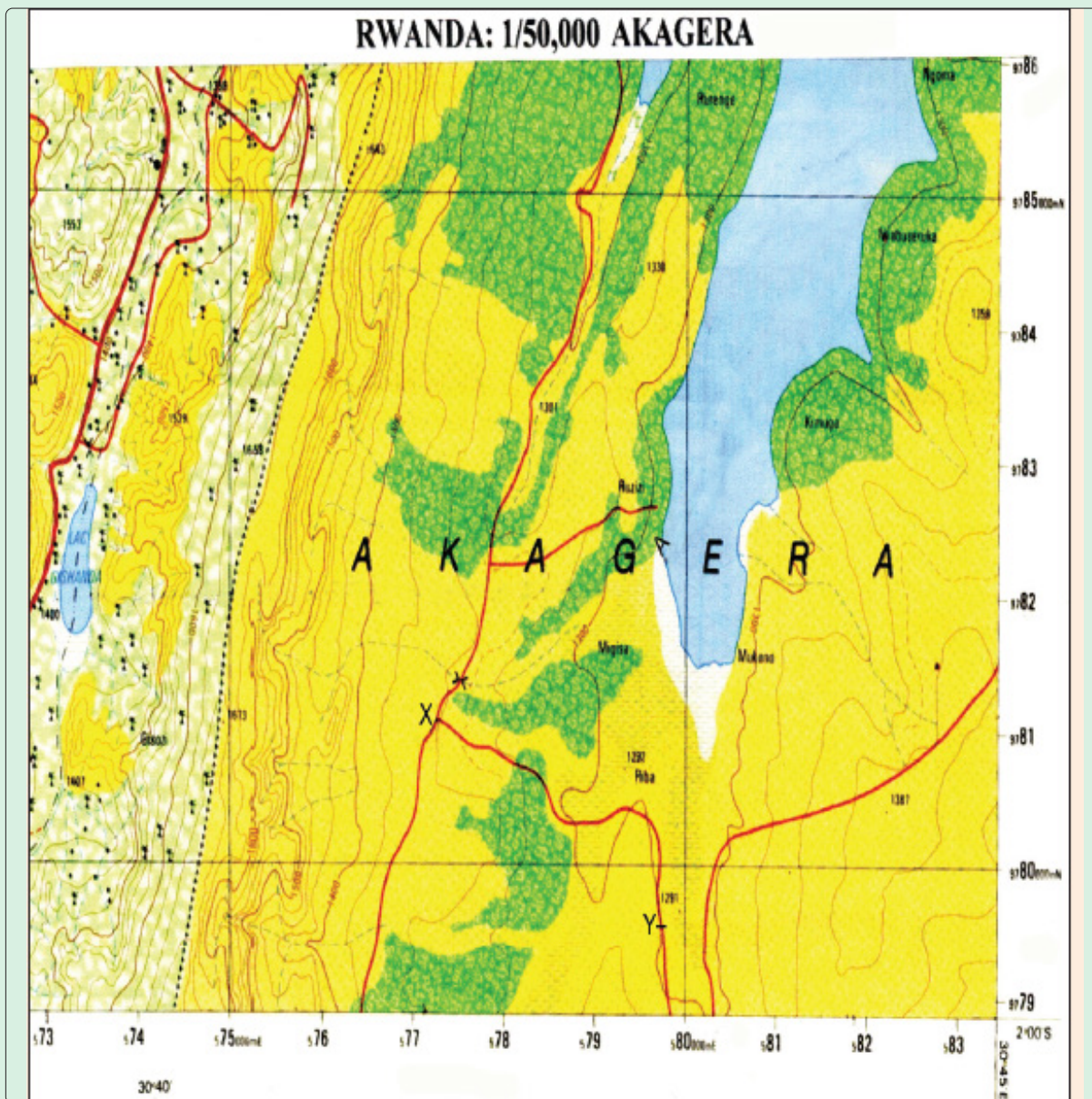
- (ii) When a river is enlarged or reduced using similar triangles, its appearance may change because narrow features do not scale properly.



Note: When reducing or enlarging maps, excessive reduction causes a loss of details, therefore, features must be simplified, as narrow elements like roads, railways, rivers cannot scale proportionally on the final map.

Application activity 1.3

Observe the map provided below and answer the questions that follow:

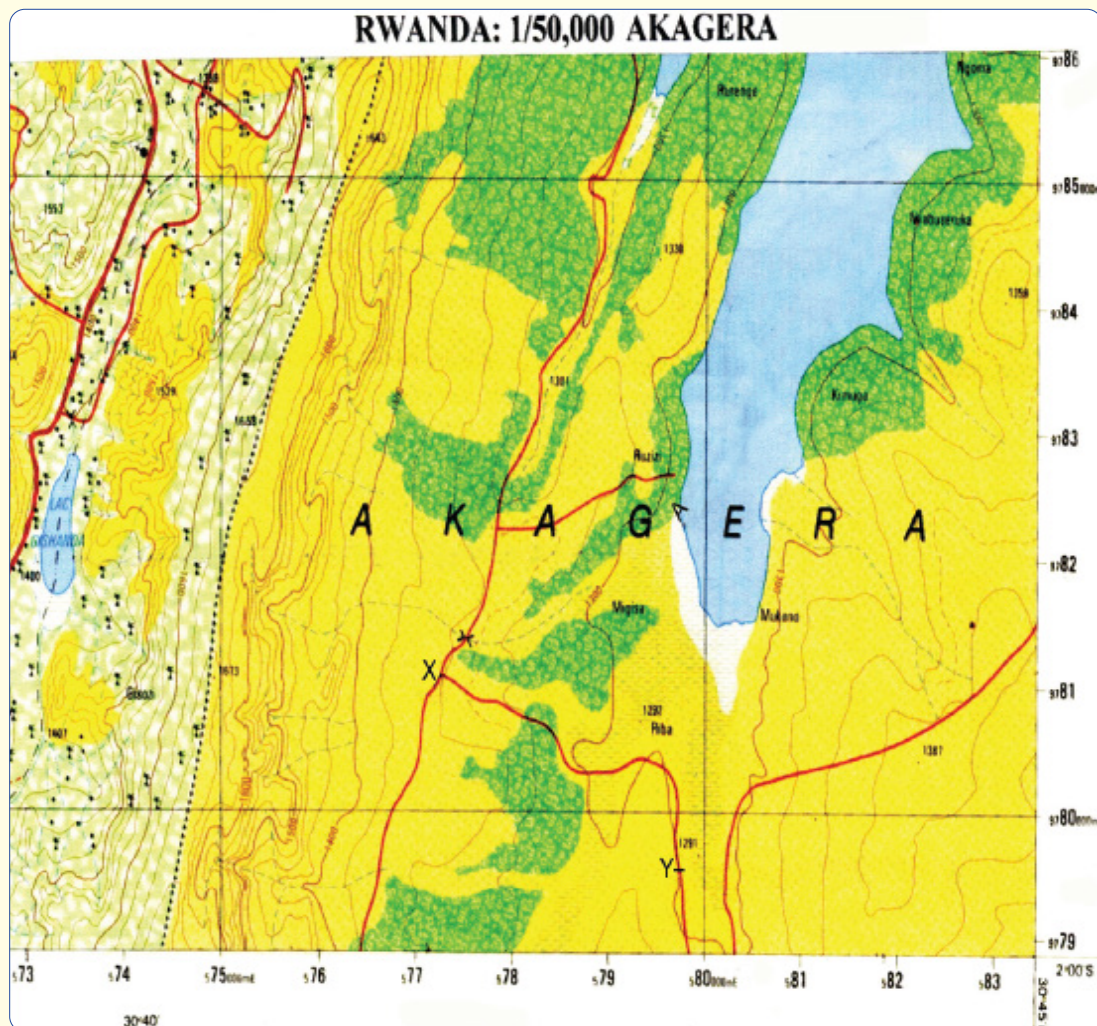


- Use the above given topographical map, enlarge and reduce the lake found in the northeastern part of the map using the method of squares. (use factor 2 for enlargement and $\frac{1}{2}$ for reduction).

End of unit assessment

❖ Knowledge and Understanding

1. With reference to the map of Akagera given:



- (i) Name the main means of transport likely to be used in the area covered by the map.
- (ii) List the types of vegetation.

❖ Skills

2. Describe the effects of vegetation on human settlement.
3. Explain why steep and hilly areas are sparsely populated.

❖ Attitudes and Values

4. Appreciate how vegetation and terrain influence human settlement patterns and development.

UNIT 2

RELIEF IN AFRICA AND THE WORLD



Key Unit Competence:

Analyse the impact of relief features on climate and human activities in Africa and the world

Learning objectives

At the end of this unit you should be able to:

Knowledge and understanding

1

- Identify the area, location, and population of Africa.
- Name the major forms of relief in Africa
- Identify the relationship between different forms of relief and human activities in Africa.

Skills

2

- Draw a sketch map of Africa and show the major relief features
- Describe the relationship between different forms of relief and human activities in Africa

Attitudes and values

3

- Appreciate the beauty of the physical landscape of the world.
- Show concern for the impact of human activities on relief features and climate.
- Appreciate the importance of the various relief forms.

2.0. Introduction

Relief refers to the physical landscape of the earth's surface. It describes different features like mountains, hills, plateaus, plains, and valleys, and how they change in height above sea level." This unit covers an introduction to Africa, the major relief forms of Africa and other continents, and the impacts of relief features on the environment.

Introductory activity

Carefully read the scenario below and answer the following questions:

Peter, a Geography student, is studying Africa using an atlas and online maps. He notices that Africa has a wide variety of landforms, including high mountains, broad plateaus, wide plains, deep valleys, and basins. As he compares Africa with other continents, he observes that Europe lies to the north of Africa and Asia to the north-east. While examining physical maps, Peter also realizes that relief features greatly influence climate, vegetation, settlement, and human activities across different regions of Africa.

1. Which of the following lists correctly identifies the major relief features mentioned in the scenario?
 - A. Mountains, plateaus, plains, basins, valleys, hills
 - B. Rivers, lakes, oceans, deserts
 - C. Volcanoes, earthquakes, floods, landslides
 - D. Forests, savannahs, wetlands, deserts
2. Using an atlas or the internet, which continents are nearest to Africa?
 - A. Asia and Europe
 - B. Europe and South America
 - C. Asia and Antarctica
 - D. North America and Australia
3. Explain how variations in relief and altitude can influence the environment and human activities in Africa.
4. Give two examples of how relief features in Africa affect settlement or economic activities

2.1. General presentation of Africa and its major relief forms of Africa (location and description)

Learning activity 2.1

❖ **Read critically the statement below and answer the following questions:**

A group of geography students explored Africa's diverse relief. They observed mountains, plateaus, basins, coastal plains, and the Rift Valley, noting their location and heights. Using maps, atlases, and online sources, they recorded Africa's size, stretching from 37.3° North to 34.5° South, and its population of 1.55 billion, analyzing how relief and location influence climate, vegetation, settlement, and economic activities.

1. Multiple Choice: Which of the following relief features is mostly found in eastern Africa?
 - A. Sahara desert
 - B. Coastal plains
 - C. Rift valley
 - D. Kalahari desert
2. Fill in the blank:
 - a) Africa extends from ____ ° North to ____ ° South of the Equator.
 - b) Major relief features observed by the students include mountains, _____, basins, and coastal plains.
3. Explain how Africa's relief and location influence human settlement, agriculture, and economic activities.

a) General presentation of Africa

Location: The points whose names you have found are located 37.3° North and 34.5° South of the equator. The farthest point to the east is 51° East, and on the west is 23.6° West.

Area (size): Africa occupies approximately 30 300 300 square kilometres. It is the second largest continent in the world. This area is about 23% of the total land area of the world.

Population: By 2025, Africa's population was estimated at 1.5 billion, making it the second most populous continent in the world after Asia. It accounts for 16.1% of the world's population.

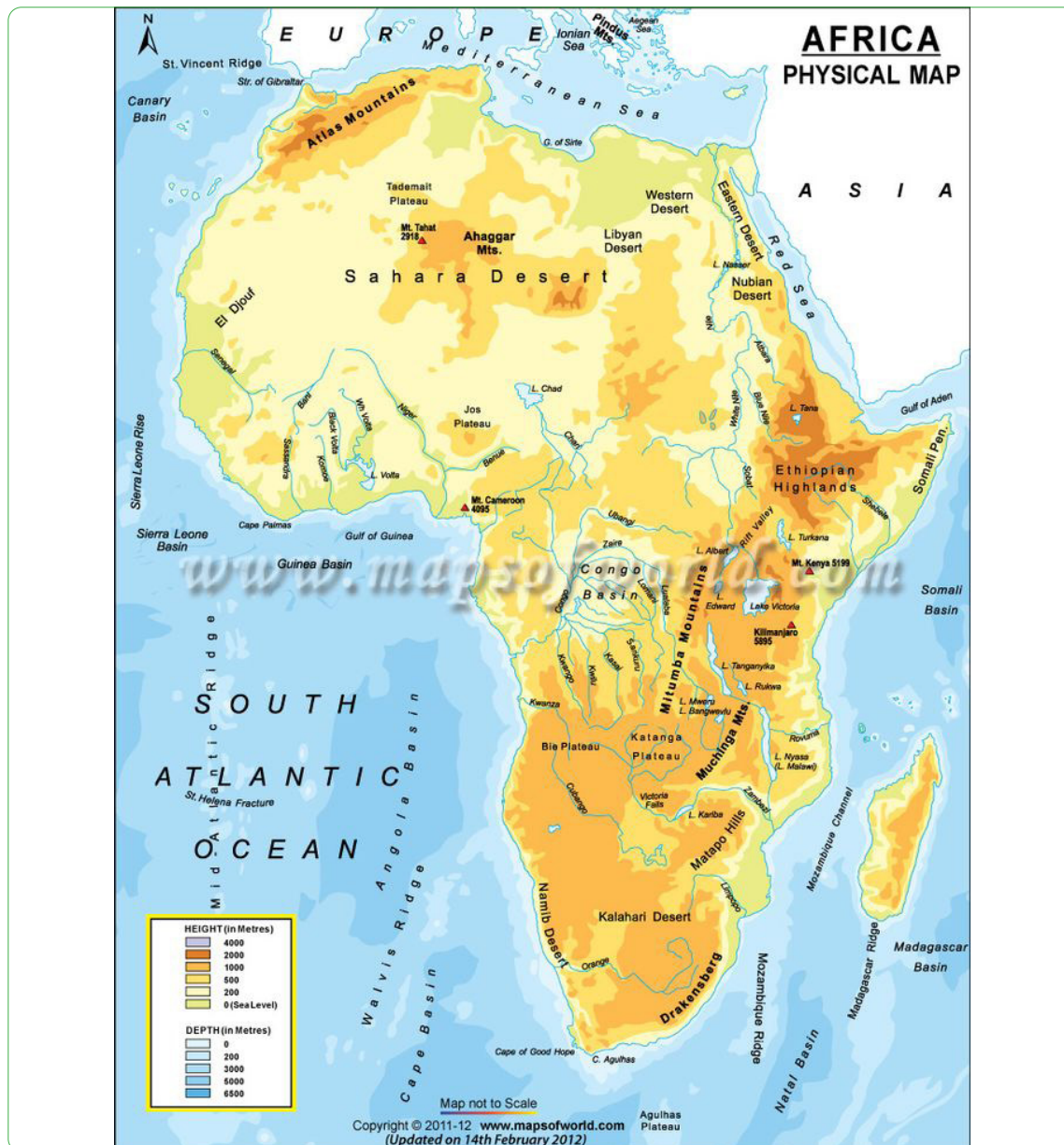
In comparison to other continents of the world, the table below shows the area and the estimated population in millions:

Table 1.2. Comparing Africa's Size and Population with other continents

Continent	Area (million km2)	Population in 2025
Africa	30.4	1,549,867,579
Europe	10.2	744,398,832
Asia	44.6	4,835,320,060
North America	24.7	617,311,580
South America	17.84	438,105,374
Australia & Oceania	8.5	46,609,644

Source : Statistics Times, 21 September 2025.

b) Major relief forms of Africa: Location and description



Physical map of Africa

The relief of Africa consists of a series of plateaus that are higher in the east but become lower westwards. In many parts, these plateaus have been modified to form new landforms. Those landforms may include:

Coastal plains, low plateau lands, basins, high plateau lands, the rift valley system and mountains

1. The coastal plain

This is a land bordering the ocean and is about 200 metres above sea level. In some parts of Africa, the coastal plain is narrow while in other parts it is relatively wide. Compared to other continents, the coastline of Africa has few inlets. The common features along the coast are deltas, estuaries, beaches, coral reefs, sand bars and lagoons.

2. Low plateau lands

The average height of the low plateau lands in Africa is 900 to 1200m. The low plateau lands in Africa include: Ahaggar (900m), Tibesti (1200m), Fouta Djallon (914m), Jos Plateau (1280m), and Ennedi (1400m).

3. The high Plateau lands:

These are in the eastern and southern areas of the continent. They include Ethiopian Plateau (4500m), East African Plateau (1500m), Zimbabwean High Veld (2500m) and Drakensberg (3400m).

4. Basins:

Basins are broad but shallow depressions on the earth's surface. Examples of basins in Africa include Lake Victoria, El Djouf, Kalahari basin, the Congo basin, Tchad basin, Sudan basin.

5. The mountains of Africa:

African mountains are scattered across the continent, and they include 4 main types:

- Fold mountains: The **Atlas fold mountains** in the Northwest Africa and **Cape Ranges** in South Africa
- Block mountains: **Mt. Rwenzori** between Uganda and DRC, **Pare** and **Usambara** mountains in northeastern Tanzania and **Damakil Alps** in Ethiopia and Eritrea
- Volcanic mountains: Mt. Kilimanjalo, Erta Ale (Ethiopia), Kilimanjaro (Tanzania), Meru (Tanzania), and the Virunga volcanoes such as Nyiragongo (DR Congo) and Kalisimbi (Rwanda), Mt Cameroon (Cameroon), Fogo (Cape Verde), Karthala (Comoros), and Piton de la Fournaise (Réunion).
- Residual mountains/inselbergs: **Namuli** Mountains in Mozambique and **Brandberg** in Namibia.

6. The Great Rift Valley:

In East Africa forms two branches, that is the Western branch (Albertine Rift) along the Tanzanian boarder and the Eastern branch which is the main one. **Table 1.3. Major relief features of Africa and their locations**

No.	Relief Feature	Location
1	Atlas Mountains	Northwest Africa (Morocco, Algeria, Tunisia)
2	Ethiopian Highlands	Eastern Africa (Ethiopia, Eritrea), also Roof of Africa
3	Drakensberg Mountains	Southeastern Africa (South Africa, Lesotho)
4	East African Highlands	Along East African Rift (Kenya, Uganda, Tanzania, Rwanda, Burundi)
5	Mount Kilimanjaro	Northern Tanzania, near Kenya border

6	Mount Kenya	Central Kenya
7	Rwenzori Mountains	Between Uganda and DRC
8	Sahara Desert	Northern Africa (Algeria, Libya, Egypt, Mali, Niger, Sudan, Chad, Mauritania)
9	Sahel Region	South of the Sahara (Senegal to Sudan)
10	Sudan Plateau	Central Africa (Sudan, South Sudan, Chad)
11	Congo Basin	Central Africa (DRC, Republic of Congo,)
12	Niger Basin	West Africa (Guinea, Mali, Niger, Nigeria)
13	Kalahari Desert	Southern Africa (Botswana, Namibia, South Africa)
14	Namib Desert	Southwestern Africa (Namibia coast)
15	East African Rift Valley	From Red Sea through Ethiopia, Kenya, Tanzania, Malawi, Mozambique
16	Great Rift Valley Lakes	East Africa (Tanganyika, Victoria, Malawi, Turkana)
17	Lake Victoria Basin	Tanzania, Uganda, Kenya
18	Lake Tanganyika Rift Valley	Burundi, Tanzania, DRC, Zambia
19	Lake Malawi Rift Valley	Malawi, Mozambique, Tanzania
20	Guinea Highlands	West Africa (Guinea, Sierra Leone, Liberia)
21	Cameroon Mountain	West Africa (Cameroon)
22	High Veld & Low Veld	South Africa, Lesotho, Eswatini
23	Angola Highlands	Angola and parts of Namibia
24	Zambezi Basin	Zambia, Angola, Zimbabwe, Mozambique
25	Victoria Falls	On the Zambezi River between Zambia and Zimbabwe
26	Madagascar Highlands	Central Madagascar
27	Great Escarpment	From Ethiopia down eastern and southern African coasts
28	Coastal Plains	Along all African coastlines

Application activity 2.1

1. By 2025, Africa's population is estimated at 1.55 billion. What percentage of the world's population does this represent?
2. Compare Africa's area and population with Asia and Europe and explain your observations.
3. Why might a continent with a smaller area have a larger population than a larger continent? Give examples.
4. Which of the following is a high plateau in Africa?
A. Jos Plateau B. Fouta Djallon C. East African Plateau D. Ahaggar Plateau
5. The Great Rift Valley in East Africa forms:
A. A single branch along the Red Sea
B. Two branches: Western (Albertine Rift) and Eastern (main branch)
C. Only in Southern Africa
D. Only the Eastern branch
6. Which basin is located in Central Africa?
A. Chad Basin B. Kalahari Basin C. Congo Basin D. L. Victoria Basin
7. Mount Kilimanjaro is an example of a:
A. Fold mountain B. Block mountain
C. Volcanic mountain D. Residual mountain
8. The Atlas Mountains are located in:
A. Southeastern Africa B. Northern Africa C. Central Africa D. Eastern Africa
9. Which relief feature generally influences settlement and farming by providing fertile soils and flat lands?
A. Basins B. High plateaus C. Mountains D. Coastal plains.

2.2. Major relief forms of other continents (Europe, America, Asia, and Oceania) and their impact on relief features on the environment.

Learning activity 2.2

A young geographer uses atlases and online maps to identify coastal plains, deltas, and major relief features across Europe, Asia, America, and Oceania, analyzing their locations and explaining how relief shapes human activities, development, and environments worldwide, with examples provided.

- List three positive and three negative ways relief features affect human activities and the environment based on the examples provided.
- How do volcanic mountains support both agriculture and tourism simultaneously?
- Copy and complete the table below by naming the continent where each feature is found.

Features	Continent
Mississippi delta	
Ganges Delta	
The great barrier reef	
Himalayas	
Andes	
Appalachians	
Alps	

a) Major relief forms of other continents (Europe, America, Asia, and Oceania)

1. Major coastal plains of the World

Coastal plains are low-lying landforms found along the edges of continents where the land meets the ocean. They vary in size and formation processes but are generally characterized by gentle slopes, fertile soils, and extensive wetlands.

The major coastal plains across different continents

Continent	Coastal Plains / Regions	Countries
North America	Gulf Coast; East Coast Lowlands	United States, Mexico, Canada (East Coast section)
South America	Coastal Lowlands	Venezuela, Guyana, Suriname, Brazil, Argentina
Europe	Atlantic Coastal Lowlands	Western Europe (e.g., France, Belgium, Netherlands, Germany)
Asia	Arctic Coastal Lowlands; Coastal Lowlands	Russia (Arctic coast), India, Bangladesh, Thailand, Indonesia, Northern China
Oceania	Southern New Guinea Lowlands; Northeast Coast and Southern Australia Lowlands	Papua New Guinea, A

2. Plateau lands

The following are examples of plateau lands in each of the other continents with their approximate heights above sea level.

- **High Plateaus (between 350m and 2400m):** These include Yukon Plateau (over 950m), Plateau of British Columbia (over 6000m), Snake River Plateau (average height of 1000m), Colorado Plateau (over 3500m high), Plateau of Mexico (rises to about 2700m) and Allegheny in the Appalachians (over 1200m high) in North America.
- **Mato Grosso (over 600m high):** This include Bolivian Plateau (average of 3750m) in South America.
- **The Spanish Meseta (average of 2165m):** These include Middle Rhine Highlands (highest point at 900m), the Fjeld Plateau (ranges between 1000m and 2500m) of Scandinavia in Europe.
- **Tibetan Plateaus** (over 4500m high): These include Iranian Plateau (over 5500 m high), Central Siberian Plateau (slightly over 1650m) in Asia.

3. Basins

A basin is a depression, or dip, in the Earth's surface. Some of the areas in other continents where basins are found include the Mississippi basin opens into the Gulf of Mexico in North America, the Amazon Basin in South America, the London Basin in Europe, the Murray-Darling basin and Lake Eyre basin in Australia.

4. Mountains

A mountain is a landform that rises at least 300 meters or more above its surrounding area. The following are some of the major mountains found in other continents:

In North America, the Rocky Mountains (over 4,400m) and Appalachians (average of about 2,000m) are fold mountains. The region also hosts volcanic mountains, for example Mount St. Helens (2,500m) in the Rocky Mountains. Mexico also has a variety of volcanoes, including El Chichon (1,200m) and Citlaltepetl (5,400m).

In North America, the Rocky Mountains (over 4,400 m) and the Appalachian Mountains (with an average height of about 2,000 m) are fold mountains. The region also contains volcanic mountains, for example Mount St. Helens (about 2,500 m) in the Rocky Mountain region. Mexico also has a variety of volcanoes, including El Chichón (about 1,200 m) and Citlaltépetl (about 5,400 m).

In South America, the most prominent are the Andes Fold Mountains (6,900m).

In Europe, there are several high mountains ranges, including the Pyrenees (3,400m), the Massif Central (1,800m), the Alps (4,800m), the Apennines (2,900m), the Dinaric Alps (2,600m), the Sudeten Mountains (1,600m), the Carpathian Mountains (2,600m), the Transylvanian Alps (2,000m) and the Balkan Mountains (2,300m).

Most of the Balkan Mountains are located in the southern Europe, while the Kjolen Mountains (2,400m) are found in Scandinavia. To the east are the Ural Mountains (1890m).

"In Europe, there are several high mountain ranges, including the Pyrenees (3,400 m), the Massif Central (1,800 m), the Alps (about 4,800 m), the Apennines (2,900 m), the Dinaric Alps (2,600 m), the Sudeten Mountains (1,600 m), the Carpathian Mountains (2,600 m), the Transylvanian Alps (2,000 m), and the Balkan Mountains (2,300 m). Most of the Balkan Mountains are located in southern Europe, while the Kjolen Mountains (2,400 m) are found in Scandinavia. To the east of Europe are the Ural Mountains (1,890 m)."

The Himalayan fold mountains (8800m) are centrally located **in Asia**.

In Oceania, the main mountain ranges are the Great dividing range (2,200m) in Australia to the east and the Southern Alps (3,700m) in New Zealand.

b) Impact of relief features on the environment

Positive influence of relief features on the environment and human activities

Relief features significantly affect the physical environment and various human activities in different ways:

- They shape patterns of settlement, farming practices, transportation networks, tourism development, and mining activities
- Alluvial plains are suitable for farming because rivers deposit fertile soil during floods. Plateaus support mechanised agriculture due to their wide and fairly flat surfaces.
- Volcanic mountains provide fertile soil on their lower slopes that are suitable for farming. Different relief features create beautiful scenery that attracts tourists. **Tourism** brings **foreign exchange** to countries and provides activities such as mountain climbing and skiing in snowy areas.

Negative influence of relief features on the environment and human activities

Relief features can also cause problems for the environment and human activities:

- **Mountains:** Steep slopes make settlement, building, and road or railway construction difficult. They increase the risk of landslides and soil erosion.
- **Volcanic mountains:** Eruptions can destroy settlements, farms, and roads. Ash and cinders disrupt air travel and affect human health.
- **Plateaus:** High altitudes cause low temperatures that limit crop growth. Some have poor or shallow soils unsuitable for farming.
- **Deserts:** Extreme heat and lack of water make farming and settlements difficult. Sand dunes make transport and communication hard.
- **Swamps and wetlands:** Waterlogged soils prevent construction and mechanized farming. They can also be breeding grounds for mosquitoes and other disease-causing insects.
- **Coastal lowlands:** These areas are prone to flooding, storms, and coastal erosion. High soil salinity in some areas reduces crop growth.
- **River valleys:** Seasonal flooding can destroy crops and homes, and may cause waterborne diseases like cholera.

Application activity 2.2

Imagine you are a planner in Africa, assigned to develop a new settlement and farming project in Africa. You must consider the relief features and their impacts on human activities and the environment:

- a) Which relief feature presents the most challenges for transportation and construction, and why?
- b) Explain the positive and negative effects of relief features in Africa?
- c) Suggest strategies to maximize the benefits of mountainous relief and to minimize the negative impacts of other relief features. For example, constructing levees or using flood-resistant crops for river valleys that are prone to flooding. Using your atlas, geographical documents or the internet:
- d) Locate the areas that are basins in other continents.
- e) Compare the size of the extent of basins in other continents to those of Africa
- f) Locate the areas with high plateaus of the other continents.
- g) Compare the size of the high plateau lands in other continents to those of Africa.

End of unit assessment

❖ Knowledge and understanding

8. "Africa occupies a central position among the continents". Explain this statement using relevant examples.
9. Explain six positive impacts of relief, two on the physical and four on the human environment.

❖ Skills

1. Draw a sketch map of Africa and on it, mark and label:
 - a) The Atlas b) Lake Victoria c) The Great Rift Valley d) Cape ranges
 - e) Drakensberg f) Ethiopian highlands g) Mount Rwenzori.
2. Using a physical map of Africa, draw your own sketch map and accurately plot the following features: Atlas Mountains, Ethiopian Highlands, Drakensberg Mountains, Mount Kilimanjaro, Sahara Desert, Sahel Region, Congo Basin, Kalahari Desert, Namib Desert, East African Rift Valley, Lake Victoria Basin, and Victoria Falls.

❖ Attitudes and values

3. Suggest three negative impacts of relief on the environment.

UNIT 3

SOILS IN AFRICA AND THE WORLD



Key Unit Competence:

Analyze the major types of soils in Africa and the world.

Learning objectives

At the end of this unit you should be able to:

Knowledge and understanding

1

- Outline and locate different soil types in Africa and the major soil types in Africa
- Identify the relationship that exists between soil types in Africa and those of other continents.
- Outline the factors favoring soil formation.

Skills

2

- Locate the major soil types in Africa
- Compare the types of soil found in Africa with those of other continents
- Analyze the main factors influencing soil formation in Africa.

Attitudes and values

3

- Appreciate the importance of major types of soil in Africa and other continents
- Show concern for the different soil types in Africa and develop a desire for soil conservation

3.0. Introduction

Soil is made of rock particles, minerals, water, air, and organic matter. It forms a thin layer on the Earth's surface that supports all plant and animal life. In this unit, you will explore the major soil types and the factors that influence soil formation in Africa. You will also briefly compare these to soils found on other continents around the world.

Introductory activity

Read critically the statement below and answer the questions that follow:

Imagine you are an environmental scientist studying soil across the world. You notice that soil is not just "dirt"; it is made of rock particles, minerals, water, organic matter, and air. It forms a thin but vital layer on the Earth's surface, supporting plants, animals, and human life. Different regions have different soils, influenced by climate, vegetation, relief, and human activities. To understand how soil affects life and agriculture globally, you are tasked with investigating soil types and their formation in Africa and other continents.

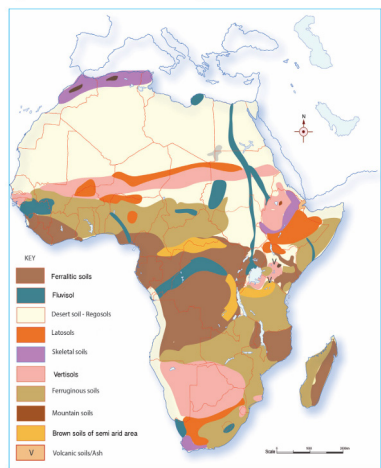
1. Identify and list the major soil types that you know.
2. Explain how each soil type supports or limits agricultural activities.
3. Identify the main factors that influence soil formation in Africa

3.1. Major types of soil in Africa

Learning activity 3.1

Study the map of Africa showing the distribution of different soil types. Then copy the table below and complete it in your notebook.

The map below shows the various types of soils found in Africa.



Source: *Africa Soil Information Service (AfSIS), 2016*

Regions of Africa	Dominant types of soils
Northern Africa	Desert soils
Central Africa	-----
Southern Africa	-----
Western Africa	-----
Eastern Africa	-----

The major types of soils found in Africa are ferralitic and ferruginous soils, fluvisols, regosols and brown soils of semi-arid areas, latosols, skeletal soils, vertisols, as well as mountain and volcanic soils. Ferralitic and ferruginous soils are described below.

- (i) Ferralitic and Ferruginous soils:** Ferralitic humid tropical forest soils; ferruginous savannah soils under alternating wet-dry climates.

(ii) Fluvisols: Young fertile alluvial soils on floodplains and river valleys, periodically flooded.

(iii) Regosols and brown soils: Eroded arid to semi-arid reddish-brown soils, iron-rich, formed under water deficiency.

(iv) Latosols: Tropical rainforest soils rich in iron and aluminium oxides.

(v) Skeletal soils: Shallow soils with abundant rock fragments, gravel, and ironstones.

(vi) Vertisols: Clay-rich soils cracking seasonally, shrinking-swelling, deep A horizon, no B horizon.

(vii) Mountain and volcanic soils: Mountain and volcanic slope soils, humus-rich but nutrient-poor, varying by altitude.

Application activity 3.1

1. Assume that you are an agricultural and environmental officer assigned to advise communities in different parts of Africa on the best land use practices based on the types of soils found in their regions. Your task is to apply your knowledge of African soils to real-life situations.
2. Draw the sketch map of Africa and locate on it the following soils types: Ferralitic / ferruginous soils, fluvisols, regosols and brown soils of semi-arid areas, latosols, skeletal soils, vertisols and mountain and volcanic soils.

3.2. Factors and processes of soil formation (weathering, humification, leaching)

Learning activity 3.2

In Senior 2, you learnt how rocks break down as a result of weathering to form soil. Use geographical documents or the internet and prepare a presentation for class discussion on how the following processes lead to soil formation:

(a) Humification (b) Leaching (c) Eluviation (d) Illuviation (e) Mineralisation

a) Factors for soil formation

There are several factors that influence the formation of different soils found in Africa and other parts of the world. These factors include:

- **Nature of the parent rock:** Determines soil texture, mineral composition, colour, fertility, and overall soil characteristics.
- **Climate:** Rainfall and temperature influence weathering rate, leaching intensity, and organic matter decomposition.

- **Living organisms:** Plants, animals, and microorganisms add organic matter and improve soil structure.
- **Topography and relief:** Slope affects drainage, erosion, soil depth, and accumulation of materials.
- **Time:** Longer periods allow soils to develop mature profiles and well-defined horizons.

Note that: The factors of soil formation are interdependent. None of the factors acts independently and a region may have the influence of all the factors. However, one factor may be more dominant in a particular region as compared to another region.

b) Processes of soil formation

Apart from weathering, there are several processes involved in soil formation. They include the following:

- **Humification:** Humification is the process where decomposers convert organic matter into humus, enriching soil through rainfall action
- **Leaching:** Leaching is the downward removal of dissolved minerals from topsoil to subsoil by heavy rainfall.
- **Eluviation:** Eluviation is the downward washing of minerals and organic matter from upper soil layers by water.
- **Illuviation:** Illuviation is the accumulation of leached minerals, salts, and fine particles in a lower soil horizon.
- **Mineralisation:** Mineralisation is the decomposition or oxidation of the chemical compounds in organic matter into plant-accessible forms.

Application activity 3.3

1. Why are soil formation factors considered interdependent?
 - a) Each factor acts alone without influencing others
 - b) One factor completely replaces the role of others
 - c) Factors interact and influence each other in soil development
 - d) Soil formation depends only on climate
2. What is the importance of humification on soil fertility?
 - a) It removes nutrients from the soil
 - b) It increases soil acidity only
 - c) It adds humus, improving soil structure and nutrient availability
 - d) It hardens the soil and reduces productivity

3. What is the main difference between eluviation and illuviation?
 - a) Eluviation adds minerals; illuviation removes them
 - b) Eluviation is weathering; illuviation is erosion
 - c) Eluviation removes materials from upper layers; illuviation deposits them in lower layers
 - d) Both processes occur only in deserts
4. Which region in Africa shows climate as the dominant factor in soil formation?
 - a) Congo Basin
 - b) Sahara Desert
 - c) Ethiopian Highlands
 - d) Nile Delta

3.3. Major types of soil in other continents (Europe, North America, South America, Asia, Oceania)

Learning activity 3.3

Conduct research via internet and other geographical documents about soils in Africa and other continents, and use the findings to answer the below questions:

5. Which continents have tundra soils?

a) Africa and South America	b) South America and Europe
c) Asia and Antarctica	d) Australia and South America
6. Which part of South America has mainly Latosols?

a) Amazon Basin	b) Andes Mountains
c) Patagonia	d) Atacama Desert
7. Where are **podzols** typically found, and why are they fertile?
 - a) In tropical rainforests; rich in iron and aluminium oxides
 - b) South of tundra regions; fertile due to accumulated organic matter and leaching
 - c) Desert regions; fertile because of high sand content
 - d) Mountain slopes; fertile due to volcanic ash

Major types of soils in other continents:

It is important to note that some of the soils that are found in Africa are also found in other parts of the world. For example, in continents that have deserts such as Asia

(Gobi Desert), Australia (Australian Desert) and South America (Atacama Desert) have desert soils.

The soils that are found in other continents that are far from the tropics are not found in Africa include:

- **Tundra soils:** These soils occur in polar regions, they are formed under permanently cold climates with frozen ground.
- **Podzols:** These soils, including peat and gray brown podzolics, occur south of tundra with cold winters and short summers.
- **Chestnut soils (clay loams):** These types of soils are found in areas that experience tropical hot mid-latitude steppe climate.
- **Chernozems:** These soils are found in areas that experience warm temperature continental climates such as the prairies of North America and the steppes of Asia.

Application activity 3.3

Referring the information covered above describing that Africa has diverse soils influenced by climate, vegetation, and topography; major types include ferralitic, ferruginous, fluvisols, regosols, latosols, skeletal, vertisols, and mountain/volcanic soils:

1. Which type of soil is common in desert regions like the Sahara, Gobi, Atacama, and Australian deserts?
a) Latosols b) Desert soils c) Alluvial soils d) Volcanic soils.
2. What is the main difference between ferralitic and ferruginous soils?
a) Ferralitic soils form in deserts, while ferruginous soils form in cold regions
b) Ferralitic soils develop under humid tropical forests, while ferruginous soils form in savannah regions with alternating wet and dry seasons
c) Ferralitic soils are rich in sand, while ferruginous soils are rich in clay
d) Ferralitic soils occur in polar regions, while ferruginous soils occur in mountains
3. Describe the location of the following types of soils: Chernozems, Chestnut and Tundra soils.

End of unit assessment

❖ Knowledge

1. Which factor primarily influences the formation of Latosols?
 - A. Cold climate with permafrost.
 - B. High rainfall and chemical weathering in tropical regions.
 - C. Low rainfall and sandy parent material.
2. Chernozem soils are mainly found in:
 - A. Polar regions with tundra climate.
 - B. Temperate grasslands in Asia and South America.
 - C. Tropical rainforests in Africa.
3. Which of the following is a common characteristic of tropical soils?
 - A. High fertility due to humus accumulation.
 - B. Deep weathering with high iron and aluminium content.
 - C. Frozen topsoil and poor drainage.

❖ Skills

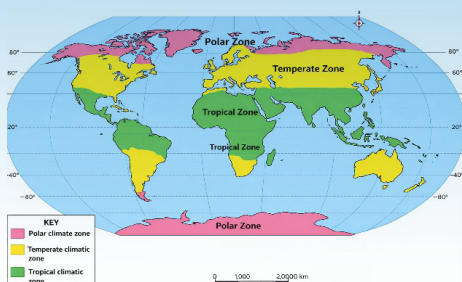
4. Analyse the similarities of soils found within the tropics, considering climate, vegetation, and fertility.

❖ Attitudes and Values

5. Explain why soil conservation is important for sustainable agriculture and environmental protection.

UNIT 4

CLIMATE IN AFRICA AND THE WORLD



Key Unit Competence:

Analyze the types of climates of Africa and other continents and its impact on human activities.

Learning objectives

At the end of this unit you should be able to:

Knowledge and understanding

1

- Outline and locate different soil types in Africa and the major soil types in Africa
- Identify the relationship that exists between soil types in Africa and those of other continents.
- Outline the factors favoring soil formation.

Skills

2

- Locate on a map the climatic regions of Africa
- Analyze the climatic regions of the world.
- Describe the major world climatic zones
- Explain the relationship between climate and human activities

Attitudes and values

3

- Appreciate the relationship between climate and human activities
- Show a continual desire to protect the environment

4.0. Introduction

Climate is the long-term average of weather over about thirty years, varying by region; the world is divided into climatic zones, each with distinct weather, vegetation, and human activity patterns. In this unit you will cover the types of climates in Africa and other continents and the relationship between climate and human activities.

Introductory activity

Read the statement below and answer the following questions:

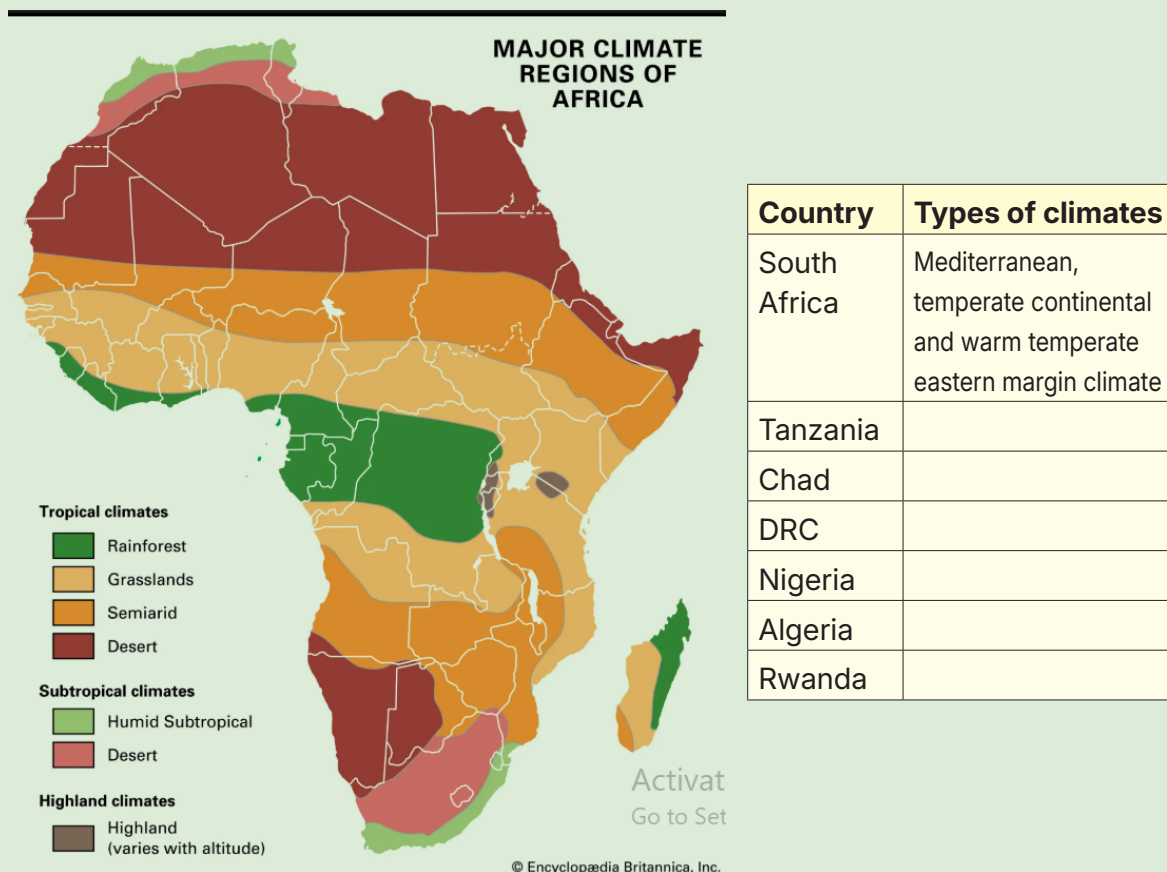
Africa lies the equator, dividing it almost equally into north and south, causing climatic and physical conditions in the north to mirror those in the south. Examples include the Sahara and Kalahari deserts, the Karoo and Maghreb, and the Cape and Mediterranean regions. The continent's shape reduces maritime influence in the north, creating a larger Sahara, a smaller Kalahari, and a cooler southern climate.

- Africa is divided almost equally by the equator. How does this unique position influence the similarity of climatic conditions between the northern and southern parts of the continent?
- Based on the shape and position of Africa, why do some regions such as the Cape area experience climate conditions similar to the Mediterranean region despite being far apart geographically?

4.1. Climate of Africa: types and characteristics

Learning activity 4.1

Use the map shown below and other maps from the atlas showing the major climatic regions of Africa, copy and complete the table below:



Types and characteristics of climate in Africa

The following are the major types of climates in the various regions of Africa:

1. Hot climates that include the equatorial climate, tropical continental climate and tropical desert and semi-desert climates
2. Temperate climates include the warm temperate eastern margin climate, warm temperate western margin or mediterranean climate and warm temperate continental climate
3. Mountain climates occur in high altitude areas of Africa where temperatures decrease with altitude.

1. Hot climates

a) Equatorial climate

Location: Lowlands within 10° north and 10° south of the equator (Congo basin and the lowlands of West Africa).

Characteristics of Equatorial climate

- High temperature (24° - 27°C).
- Small annual range of between 3° - 5°C.
- Diurnal mean temperature of 26°C.
- Diurnal range of temperature of 6° - 7°C.
- Mean annual rainfall of 1500mm to 1800mm or more in some areas.
- Rainfall is evenly distributed throughout the year.
- Rainfall regime is a double-maxima.
- Relative humidity is over 80% throughout the year.
- There is very little difference between the wettest and the driest months.

b) Tropical continental/Savanna climate/ Sudan climate

Location: Found between the tropic of Cancer and the tropic of Capricorn (5 ° and 15 ° North and South of the equator).

Characteristics of tropical continental/Savanna climate

- High temperatures throughout the year; 21°C during the cool season and 32°C during hot season.
- Moderate annual range of temperature, 11°C.
- Highest temperatures occur before the rainy season.
- Moderate rainfall varying between 760mm and 1200mm.
- Rainfall is more abundant in regions bordering the equatorial climatic belts and much less in regions of the outer margin of the belt.
- Rainfall is seasonal and mainly convection type; falling during the hot season.

- Where there are highlands, it is orographic.
- There are three distinct seasons: a rainy season, a cool dry season and a hot season extending into rainy season.
- Trade winds are the prevailing winds

Tropical desert and semi-desert climates

Location: African tropical deserts lie between 15°–30° north and south, mainly on western continental margins, within the tropics; the Sahara, largest, stretches across North Africa from west to east coast areas.

Note: Equator is very hot, why doesn't experience desert?

- The sun's rays are more concentrated at the equator, so it is the hottest part of the earth.
- This intense heat warms the land, the air and the sea and causes water to evaporate into the air.
- This very warm, moist air then rises.
- Once the warm air rises it cools and the water turns back into rain, creating clouds.
- This means that although it is hot, there is lots of at the equator (hence rainforests are located here, **not deserts**)

Characteristics of tropical desert and semi-desert climates

- High mean annual temperature of 27°C.
- Annual range of temperature, sometimes reaching 28°C.
- Deserts have very high day temperatures and very low night temperatures.
- The sky in the tropical desert remains cloud-free due to the subsiding air of high pressure.
- Very low annual rainfall, less than 250mm.
- Rainfall is erratic and unreliable.
- Humidity is very low.
- Frequent strong winds.

2. Temperate climates

a) Warm temperate eastern margin climate

This type of climate is also referred to as humid sub-tropical climate (China Type).

Location: Coastal belt of Natal.

❖ Characteristics of warm temperate eastern margin climate

- This climate is experienced in the southeastern part of Africa around Natal.

- It is characterized by hot summers with temperatures of about 26° C and a mean annual rainfall of about 1000 mm per year.
- The rain is brought about mainly by onshore trade winds in summer.
- Winters are cool and dry.

b) Warm temperate western margin climate

This type of climate is also called the mediterranean climate.

Location: Occurs between 30°–45° latitudes on western continental margins, notably Mediterranean coasts, North Africa and southwestern Cape regions areas.

❖ Characteristics of warm temperate western margin climate

- The average temperatures range from 21°C in summer to 10°C in winter.
- Temperature increases from warm to hot as one moves further away from the coast towards desert regions
- Moderate annual range of temperature, about 11°C.
- Mean annual rainfall between 500 – 900mm.
- Hot winds from the desert increase temperatures of this region and increase the risk of veld fires
- Rainfall is cyclonic, falling in showers during the cold season.
- Trade winds are offshore during the dry season.
- Westerlies dominate during the cold season and are onshore.

c) Warm temperate continental climate

Location: The high Veld of South Africa.

❖ Characteristics of warm temperate continental climate

- The dry season is hot, with average temperature of 26°C
- It experiences a mild, cold season with temperatures of about 10°C.
- The annual range of temperature is moderate, about 16°C
- Mean annual rainfall is low to moderate, between 380 – 700mm
- It receives convectional type of rainfall and falls mainly during the dry season.

3. Mountain climates

Location: Found in East African highlands, Ethiopian highlands, Drakensberg Mountains, Mount Cameroon and the Atlas Mountains.

❖ Characteristics of mountain climates

- Temperature decreases with increasing altitude.
- In temperate regions, the equator-facing slopes are warmer than those facing the poles side.

- Rainfall increases with increasing altitude up to 3000m above sea level, when it starts to decrease.
- In general, windward slopes receive far much more rain than leeward slopes.
- Atmospheric pressure decreases with increasing altitude.
- Local winds blow upslope during the day and downslope at night.

Note: Within the tropics, snow can exist on top of very high mountains.

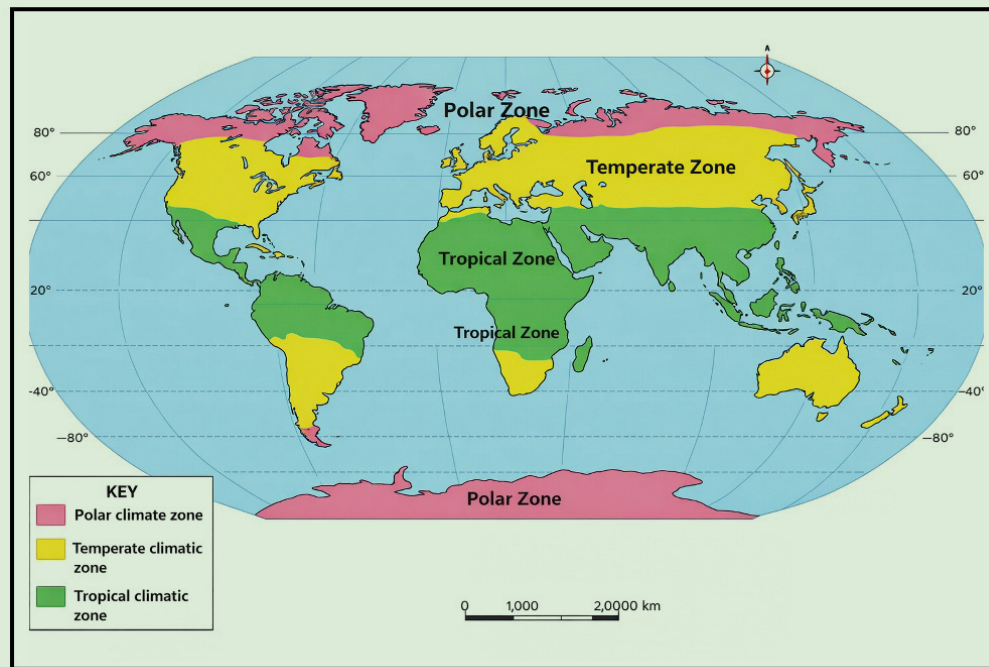
Application activity 4.1

- Which of the following is classified as a hot climate in Africa?
 - Mediterranean climate
 - Mountain climate
 - Equatorial climate
 - Warm temperate continental climate
- The equatorial climate in Africa is mainly found in:
 - Atlas Mountains
 - High Veld of South Africa
 - Congo Basin and lowlands of West Africa
 - Coastal Natal
- Tropical continental (savanna) climate is mainly located:
 - Within 10° north and south of the equator
 - Between 5° and 15° north and south of the equator
 - Beyond 30° latitude
 - Only in desert regions
- Which season occurs just before the rainy season in the tropical continental climate?
 - Cool dry season
 - Cold season
 - Hot season
 - Windy season
- African tropical deserts are mainly located:
 - Along eastern coasts
 - Around the equator
 - Between 15° and 30° north and south
 - Above 3000m altitude
- The warm temperate eastern margin climate in Africa is best developed in:
 - Southwestern Cape
 - Sahara Desert
 - Coastal Natal
 - Ethiopian Highlands
- Which statement best describes mountain climates?
 - Temperature increases with altitude
 - Rainfall decreases on windward slopes
 - Temperature decreases with altitude
 - Pressure increases with altitude.

4.2. Climate of other continents: Climatic zones and their characteristics

Learning activity 4.2

1. Use the map provided below to locate the areas that lie within the different climatic zones. For each continent (apart from Africa).



2. In which zones do the following areas fall?
(a) Australia (b) Antarctica (c) North America

Climate of other continents (Zones, types, locations and characteristics)

The climate experienced throughout the world can be divided into four zones: Tropical climate, temperate climate, arctic (very cold) climate, mountain climate.

4.2.1 Tropical climate

This is the region found between the Tropic of Cancer and the Tropic of Capricorn and include:

Equatorial climate, tropical continental climate, tropical marine (maritime) climate, hot desert and semi-desert climate and tropical monsoon climate.

a) Equatorial climate

Location: Central America and part of North America, South America (Amazon forest) and Asia (Islands of East Indies)

Characteristics

- No distinct dry season
- No summer or winter because they are typically hot and wet throughout the year
- There is heavy and frequent rainfall
- Larger change in temperature between the day and night.

b) Tropical continental climate/Savannah climate or Sudan type

Location: It occurs within tropical inland areas far from the sea, including parts of the Americas, Australia, and India, generally between 5° and 20° north and south of the Equator. In Mexico (Central and North America), Mato Grosso Plateau of Brazil, Colombia and Venezuela lowlands, Guyana highland in South America, Northern Australia and parts of Deccan plateau of India (Oceania).

Characteristics

- There are distinct wet and dry seasons
- Rainfall increases towards the equatorial regions and decreases away from the equator.
- Temperatures are high throughout the year varying between 26°C to 32°C with an annual range of about 11°C
- The convectional rainfall falls during the hot season and varies between 750 to 1000mm.
- The prevailing winds are trade winds
- There is low relative humidity
- The daily temperature range may be 15°C or even more, with much colder nights than days.

c) Tropical marine (maritime) climate

Location: It is experienced in tropical regions close to the oceans, including Central America and the Caribbean (Mexico, Guatemala, Venezuela, Panama, Jamaica, Cuba, Puerto Rico, and Haiti), the east coast of South America (Colombia, Suriname, and Brazil), the Philippines in Asia, and the east coast of Queensland in Australia.

Characteristics

- High mean annual temperature of about 25°C with an annual range of about 8°C
- High mean annual rainfall of between 1100-2000mm
- High humidity throughout the year
- Prevailing winds are trade winds

- Tropical cyclones occur towards the end of the hot season.
- Maximum temperatures occur when the sun is in overhead position.

d) Hot desert and semi-desert climate

Location: Hot desert and semi-desert climates are found in many parts of the world: Asia (Gobi, Karakum, Taklamakan, and Thar deserts, Middle East in Jordan, Iran, Iraq, Saudi Arabia, Afghanistan, and Israel), North and Central America (Mojave, Colorado, Mexican, Great Basin deserts and Thomson Plateau), South America (Atacama, La Guajira, Monte, Patagonian, and Sechura deserts), and Australia (Great Australian, Simpson, Gibson, Great Sandy, Little Sandy, and Tanami deserts).

Characteristics

- High mean annual temperature of about 27°C with an annual range of 26°C to 28°C.
- Large diurnal range of temperature
- Very low annual rainfall (less than 250mm) which is erratic and unreliable
- Humidity is low and strong winds are frequent.
- Atmospheric pressure is low during the hot season and high during the cool season.
- There are no clouds in the atmosphere to scatter the insolation and to water vapor to absorb it.
- Presence of fog and mist on coastlands
- There is little or absence of vegetation.

e) Tropical monsoon climate.

Location: The main areas are:

- Southeast Asia, south China, Indian sub-continent (Asia)
- Northern parts of Australia (Oceania)

Characteristics

- It occurs in areas where seasonal changes take place on a continental scale. This involves changes in land and sea winds which affect both oceans and continents.
- There is seasonal reversal of winds. The winds reversal is due to pressure changes over the land masses.
- During one season, the winds are onshore, bringing heavy rainfall to the coastal areas. During another season, the winds are offshore, thereby bringing little or no rainfall.
- The temperatures range from about 32°C in the hot season to about 25°C in the cool season

- The mean annual temperature range is about 7°C.
- Summers are hot and wet while winters are warm and cool.
- Both rainfall and temperatures vary a lot depending on altitude and direction of the onshore winds.
- There is high humidity in summer and low humidity in winter.

4.2.2 Temperate climate

This zone is subdivided into warm temperate climate, cool temperate climate and cold temperate climate. There are found between 30°&45° North and South of the Equator.

I. WARM TEMPERATE CLIMATE

a) Warm temperate western margin (Mediterranean type)

Location:

- Central coast of California in North America
- Central Chile in South America
- Southern Europe bordering Mediterranean Sea

b) Warm temperate continental climate

Location:

- In **North America**: Oklahoma and Texas
- In **South America**: Central Argentina and Paraguay
- In **Australia**: Murray-Darling basin

Characteristics

- Hot summers with an average temperature of 26°C
- Mild cool winters with temperatures of about 10°C
- Annual temperature range is moderate, of about 16°C
- Low to moderate mean annual rainfall, between 380 to 700mm
- Convectional rainfall falling mainly in summer.

c) Warm temperate eastern margin climate

Location:

- South-Eastern states of USA in **North America**
- Southern Brazil and Uruguay, Eastern Paraguay and coastal Argentina in **South America**
- South-East China and Southern Japan in **Asia**
- South-East part of Australia and the North Island of New Zealand in Oceania.

d) Warm temperate desert climate (Mid-latitude desert climate)

Location: Utah and Nevada in USA, Patagonia in South America and Gobi Desert in Asia

Characteristics:

- Very hot summers with temperatures between 25°C and 37°C
- Cold winters with temperatures dropping below -7°C
- Very large diurnal and annual range of temperature
- Low and unreliable rainfall of about 250mm
- Rainy season is experienced during late winter and early spring.

II. COOL TEMPERATE CLIMATES

It is experienced between 35° and 60°N&S of the Equator.

a) Cool temperate western margin climate

Location:

- In **North America:** coastal British Columbia (Canada), southwestwards up to the state of Oregon (USA).
- In **South America:** the coastal parts of Chile.
- In **Europe:** British Isles, central and Western Europe, and southern Scandinavia.
- In **Oceania:** Tasmania and South Island of New Zealand.

b) Cool temperate continental climate

Location:

- In **North America:** Alberta (Canada), Manitoba, the Midwest and North Central USA.
- In **Europe:** Central and Eastern Europe and western Russia.

Characteristics:

- Warm summers with temperature of about 18°C
- Very cold winters with temperatures below -19°C
- Very large annual range of temperature; up to 37°C
- Low annual rainfall (400-550mm)
- Convectional rainfall received throughout the year.

c) Cool temperate eastern margin climate

Location:

- Maritime provinces of Canada and North-East of USA in **North America**
- Northern China, Korea and Central Northern Japan in **Asia**

Characteristics:

- Warm to hot summers of 12 to 18°C (in North America) and 24°C in Asia
- Cool to cold winters with temperatures of -4 to -15°C
- Large annual temperature range of 28 to 39°C
- Precipitation is a mixture of snow and rain
- High precipitation of more than 1000mm in North America and Northern Japan; and lower (700mm) in Asia.
- Occurrence of precipitation throughout the year.

III. COLD TEMPERATE CLIMATES

They border the cool temperate climates from latitudes 60° N to 68°N, but in the interior they start at about 55°N.

a) Cold temperate western margin climate

Location: Alaska in North America and Central and Northern Norway and Sweden in Europe

Characteristics:

- Cool but short summers with mean temperature of 12°C
- Long and cold winters with temperature between -2°C and 4°C
- The mean annual rainfall is about 750mm and falls mostly in summer
- In winter, the precipitation is in form of snow
- The dominant winds are Westerlies.

b) Cold temperate continental climate

Location:

- In North America: a belt across the continent between 55°N and 68°N, from Alaska border in West to Labrador in the East
- In Eurasia: From Finland, through Siberia to the Pacific Coast in the East.

Characteristics:

- Warm short summers with temperatures not exceeding 21°C
- Very cold long winters with temperatures going below -45°C
- Annual rainfall is about 380mm and most of it falls in summer with snow falling in winter.
- Rainfall is convectional and falls in summer.

c) Cold temperate eastern margin climate

Location: North East Pacific Coast of Russia

Characteristics:

- Short but hot summers with mean temperatures of about 21°C
- Long but severe winters, with temperatures of about -20°C
- Annual range of temperature is very high
- The mean annual rainfall ranging from 500mm to 1000mm
- Most rain falls in summer
- The prevailing winds are moist south easterly during summer
- There is strong, dry and cold north westerlies in winter.

4.2.3. Arctic (very cold) climates

They are also called **polar desert climates**. They are found in regions beyond the Arctic Circle ($66\frac{1}{2}^{\circ}$ N&S of the Equator).

a) Tundra climate

Location:

- In **North America**: Along the Coast bordering the Arctic Ocean from Alaska, Northern Canada and South Coast of Greenland.
- In **Eurasia**: Along the Arctic Ocean Coast from Scandinavia to the North-eastern Coast of Russia.

Characteristics:

- Short and cool summers, with average temperatures of 20°C
- Long and very cold winters, with temperatures varying between -29°C and -40°C.
- Annual range of temperature is very high
- Mean annual rainfall is about 380mm
- Most rain falls in summer while snow falls in winter.
- The type of rainfall is convectional

b) Polar climate

Location: Interior of Greenland and Iceland, the Arctic ice lands of Canada, Russia, Northern Norway and Finland in **N America** and **Eurasia** and the whole continent of Antarctica.

Characteristics:

- Very cold temperatures throughout the year, normally below 0°C
- It has the world's lowest temperature of -88.3°C recorded at Vostok in Antarctica.
- Blizzards are frequent
- Average annual precipitation is below 200mm and is in form of snow.
- Winter consists of continuous nights and the summers of continuous day.

4.2.4. Mountain climate

- It is experienced in mountainous areas with the same characteristics as those identified in Africa.
- The main areas are: The Rocky Mountains and Appalachian Mountains in **North America**, **Andes mountains** in **South America**, Alps, the Balkans, the Scandinavian and Ural Mountains in **Europe** and Himalayas in **Asia**.

Application activity 4.2

1. Which of the following is a climate type shared between Africa and other continents?
A. Equatorial climate B. Polar climate
C. Warm temperate continental D. All of the above
2. The tropical monsoon climate is mainly experienced in:
A. Sahara Desert B. West Africa and coastal India
C. South Africa D. Antarctica
3. Fill-in-the-blanks
a) One area with Mediterranean climate is _____.
b) Two regions where warm temperate continental climate occurs are _ and ____.
4. True or False: Latitude and altitude influence global climate distribution by affecting temperature and rainfall patterns.

4.3. Relationship between climate and human activities

Learning activity 4.3

Your school is organizing a geography exhibition on "Climate and Human Activities." Different groups of students are asked to collect information from the internet and geographical documents to show how climate affects people's lives and economic activities in different regions of the world. One group studies the Sahara Desert, a region with extreme climatic conditions. Another group examines how climate influences tourist movements and activities in the World.

To prepare your presentation for the exhibition, respond to the following questions:

1. How does the climate in Rwanda influence agricultural activities
A. It limits farming to only a few types of crops.
B. It provides favorable conditions for diverse crops and harvests.
C. It prevents any form of farming in the country.
D. It causes frequent droughts across all regions.
2. Why is the climate of the Sahara Desert unfavourable for human settlement and agriculture?
A. It has abundant rainfall and fertile soils.

- B. It experiences extreme heat, low rainfall, and poor soils.
- C. It has mild temperatures and rich vegetation.
- D. It is located near rivers and lakes

3. In what ways does climate influence tourism in different parts of the world?

Relationship between climate and human activities

Climate positively and negatively affects human activities, influencing settlement, forestry, agriculture, fishing, and tourism. In turn, human activities such as land use and pollution can alter the climate, contributing to climate change and affecting environmental conditions.

- **Influence of climate on settlement:** Very hot and very cold areas discourage settlement, whereas warm regions provide favorable conditions for people to live.
- **Influence of climate on forestry:** Climate determines forest tree species; tropical regions support hardwoods, cooler climates favor conifers.
- **Influence of climate on agriculture:** Climate determines crops and livestock; some thrive in warm, hot, or cool regions
- **Influence of climate on fishing:** Cool waters support plankton growth, increasing fish abundance, while high tropical temperatures reduce plankton and fish.
- **Influence of climate on tourism:** Warm, wet climates support diverse wildlife and grasslands, attracting tourists to herbivores and carnivores worldwide widely.

For example:

- Low rainfall results in extensive grasslands on which a large variety of herbivores live.
- Carnivores such as lions and cheetahs, which prey on the herbivores, are therefore found in plenty in the grasslands.
- The arid areas have limited plant cover and therefore few wild animals.
- The high and cold mountain regions are inhabited by a limited number of animals such as mountain gorillas.

Application activity 4.3

Read the below statement and answer the following questions:

Climate directly influences human activities such as settlement, forestry, agriculture, fishing, and tourism, while human activities like land use and pollution can in turn affect climate and environmental conditions.

1. How does climate influence where people choose to settle? Provide examples from very hot, very cold, and warm regions.
2. Explain how climate affects agriculture, forestry, and fishing, giving at least one example for each.

End of unit assessment

❖ Knowledge and understanding:

1. Which of the following is a major climatic zone in Africa?
A. Equatorial B. Desert C. Mediterranean D. All of the above
2. Rwanda's climate is similar to some global climatic zones because:
A. It has extreme deserts and polar conditions
B. It experiences moderate temperatures and rainfall patterns
C. It has year-round snow and ice
D. It is completely arid

❖ Skills

1. Explain the similarities between the tropical continental climate and the warm temperate continental climate.
2. Explain the differences in characteristics between the warm temperate western margin and the cool temperate western margin climates.

❖ Attitudes and Values

3. Discuss the impact of climate on the various human activities in Africa and the rest of the world.

UNIT 5

VEGETATION OF AFRICA AND THE WORLD



Key Unit Competence:

Examine the relationship between vegetation types and human activities in Africa.

Learning objectives

At the end of this unit you should be able to:

Knowledge and understanding

1

- Identify major vegetation types in Africa and their characteristics.
- State the different types of vegetation in other continents.
- Identify the relationship between vegetation and human activities in Africa.

Skills

2

- Locate on the map of Africa, the major vegetation types
- Describe the distribution of major vegetation types in Africa and the world.
- Examine the relationship between vegetation and human activities in Africa

Attitudes and values

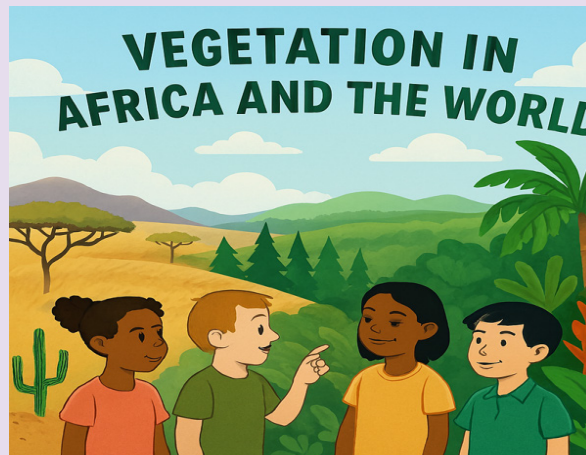
3

- Show concern for the impact of vegetation on human activities.
- Participate voluntarily in the conservation of the forests.

5.0. Introduction

Vegetation refers to all plant life in an area and is classified as natural, semi-natural, or planted. In Africa, major vegetation types include equatorial rainforest, savanna, desert vegetation, Mediterranean vegetation, montane vegetation, and mangrove swamps, whose distribution is influenced by climate, soils, relief, drainage, human activities, policies, rainfall, temperature, slope, water availability, wind, and sunlight.

Introductory activity



Observe the picture carefully and then answer the questions that follow. It shows learners standing in an environment with different vegetation types, including grasslands, forests, and desert plants, representing varied regions of Africa and the world.

Questions

1. The vegetation with scattered trees and grasses shown in the picture represents the _____ vegetation type.
2. Dense green trees with broad leaves in the background indicate a ____ climate region.
3. In your own words, explain how climate and human activities can influence the types of vegetation found in different parts of Africa and the world.

5.1. Vegetation in Africa: types, characteristics and reasons for their distribution

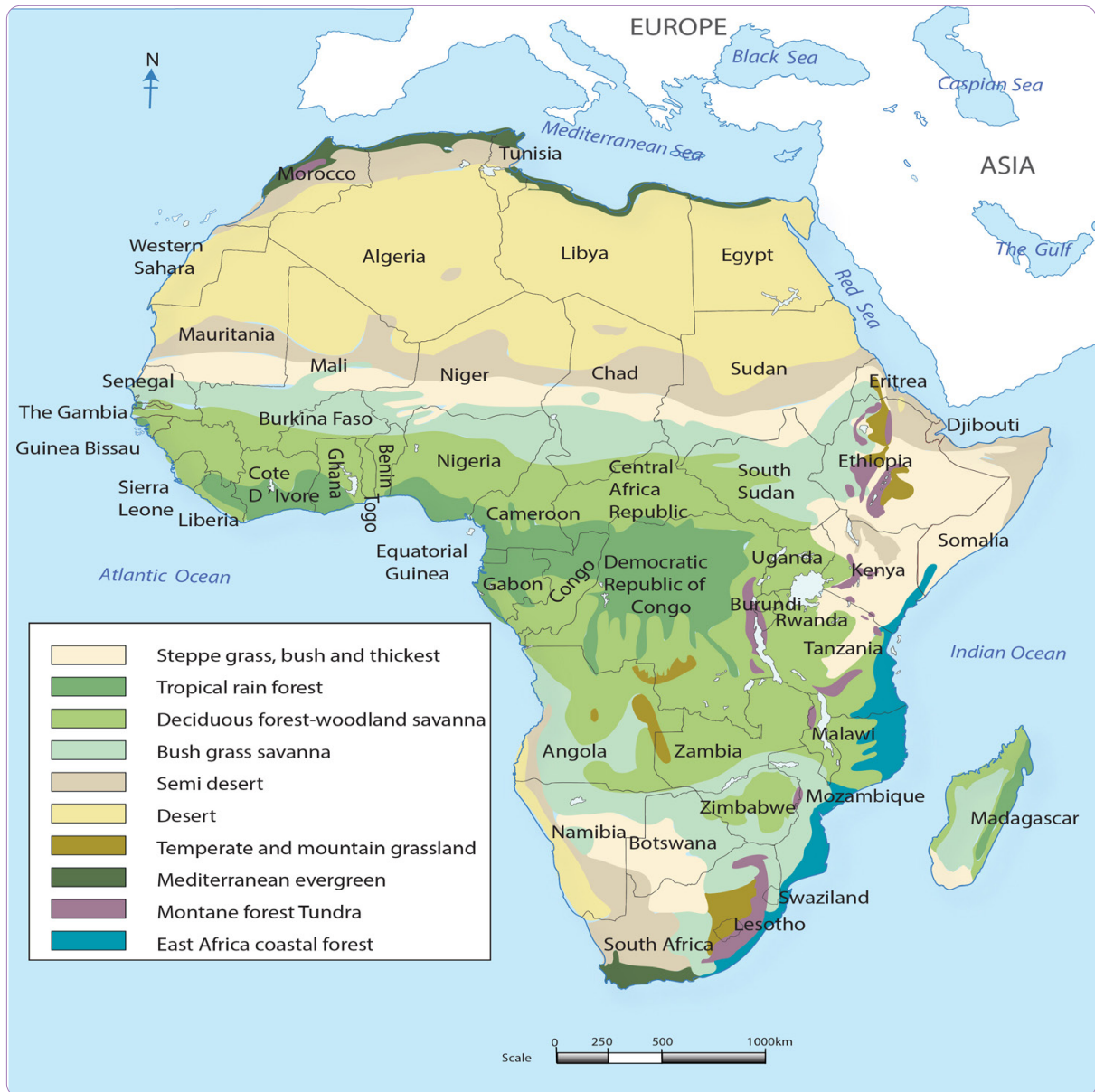
Learning activity 5.1

During a school trip across Africa, Umuhire observed tall dense forests in the Congo Basin, scattered trees and grasses in the savannah, thorny plants in the desert, and changing vegetation on mountains. She noticed that rainfall, temperature, and altitude strongly influenced how plants grow and survive.

1. Which two types of vegetation did Umuhire observe during her trip?
2. Why are the forests in the Congo Basin very dense?
3. Savannah areas have scattered trees and grasses because rainfall is _____ and occurs in _____ seasons.
4. One vegetation type that is very fragile to climate change is the _____ because it depends heavily on _____ rainfall.

Types of vegetation in Africa

The major types of vegetation in Africa are: Forest vegetation, grassland vegetation, desert vegetation and mountain vegetation



1. Forest vegetation

The four main types of forests in Africa are: Mangrove forests, mediterranean forests, tropical rainforests and temperate evergreen forests.

a) Mangrove forests

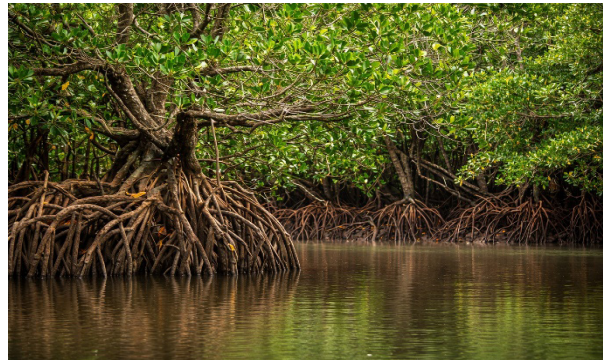
These are also called **mangrove swamps**.

They grow in the salty muddy waters of tropical seas.

Location:

- The east coast from East Africa to Mozambique
- Northwest coast of Madagascar

- The Gulf of Guinea especially in the Niger delta.
- The coast of West Africa from Gambia to Sierra Leone.



Mangrove

Characteristics:

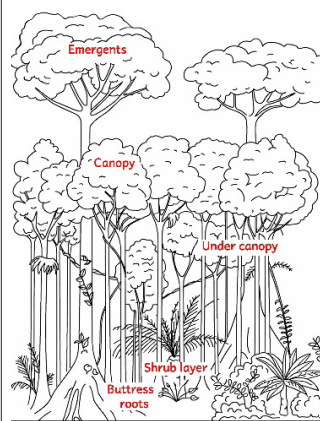
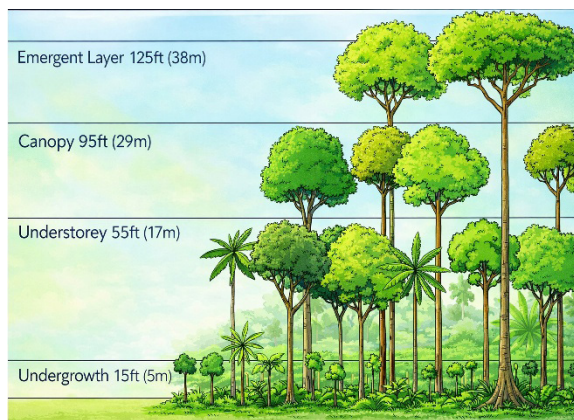
- Trees have special roots that are partly aerial
- Roots grow horizontally above water and then turn vertically into muddy water.
- At high tide the lower trunk and aerial roots are submerged.
- Their roots are for anchorage and for breathing
- At low tide the trees appear to be supported on slits of roots.

b) Tropical rainforests or equatorial forests

Location:

- Congo basin
- West African coast from Senegal up to Ivory coast
- Eastern Madagascar.





Characteristics:

- There is heavy convectional rainfall of about 2000mm per year.
- There are so many species of plants which exceed 15000 plant species.
- Trees are closely set and evergreen throughout the year
- There is presence of creepers subdivided into three types: **climbers, epiphytes and lianas.**
- There is abundance of food supply for animals there.
- There is poor climate for human health
- Trees have buttress roots
- There is dense vegetation.
- Trees have 3 or 4 distinct canopies (layers) namely ***undergrowth (shrub layer), understorey (undercanopy), canopy and Emergent Layer.***

2. Mediterranean forests

Location: There are found in areas with Mediterranean type of climate. The main areas are:

- The coastal lands of Morocco, Algeria and Tunisia
- Southwest part of South Africa around Cape Town.

Characteristics



- Some trees are cone-shaped
- There is thorny creeping plants
- Trees have thick barks
- Some plants have spiny leaves
- Some trees have shiny/waxy leaves
- Some trees are deciduous
- The grasses dry off during summers and germinate during winters
- Some trees have long roots
- Some plants have bulbous roots.
- They are often referred to as open woodlands.

3. Temperate evergreen forests

Location: There are found in regions with warm temperate Eastern margin type of climate and alongside the coast of Natal in South Africa

Characteristics:

- The forests are evergreen
- Trees have broad leaves
- Some areas of forest have undergrowth of other plant species
- Some trees are deciduous
- Many evergreen tree species are hardwoods.

4. Grassland vegetation

There are two types of grasslands in Africa. These are **tropical grasslands** and **temperate grasslands**.

a) Tropical grasslands

These are commonly referred to as **savannah**. There are subdivided into **open grasslands** with few scattered trees or no tree at all, and **woodlands** where trees dominate.

Location: West Africa between Sahel region and the tropical rainforest and extends across Eastern and central Africa.

Characteristics:

- Some areas are dominated by open grasslands
- Trees are of medium height and grow wide apart
- Space between the trees is either occupied by grass or shrubs
- Many trees have umbrella-shaped crowns
- Most trees are deciduous and shed their leaves in the dry season.

- Trees have deep roots and thick barks
- Grasses can grow up to 3m tall in wet areas.
- The elephant grass variety grows up to 4.6m
- Grasses dry off during the dry season but sprout at the onset of rain.
- Regions receiving rainfall below 750mm have much shorter grass and low growing trees and shrubs.
- Trees are scattered due to low rainfall
- Trees are flat topped due to strong prevailing winds blowing in the area
- The main types of trees growing in this vegetation area are acacia and baobab.

b) Temperate grasslands

Location: In the Veldt region in central parts of South Africa.

Characteristics:

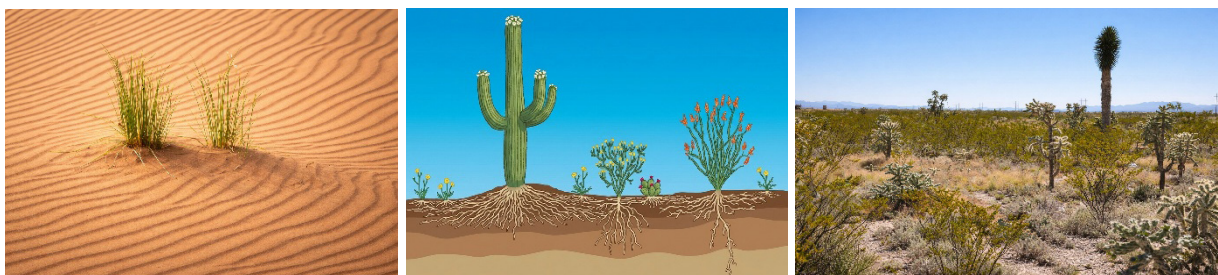
- The grass spreads out over extensive areas
- There is little or no mixture of trees and shrubs
- The wet areas support tall grasses
- A "carpet" of short **red oat grass** forms on the high plateau.

5. Desert vegetation

Location: In hot and dry regions of Africa, temperate regions that receive very low rainfall.

The main areas are: Sahara, Namib and Kalahari deserts.

Characteristics:



- There are few varieties of plants
- Some plants are perennial and have succulent stems with spines but no leaves.
- The stems are used for breathing surfaces and for water storage
- Shrubs and short stunted trees are evergreen with hard leaves.
- There are drought resistant deciduous shrubs

- There are salt-tolerant plants in areas with poor drainage
- Date palms are common in oases.

6. Mountain vegetation

Mountainous regions have different types of vegetation. The vegetation is not uniform all over the mountain due to the effects of temperature and moisture variation. The main factors influencing vegetation zonation are altitude, aspect, slope, temperature, and moisture availability.



- With change in elevation (height); there is rapid vertical change in forms of vegetation.
- Plant cover is influenced by climatic conditions, from the base to the peaks.
- Each belt of mountain vegetation includes several different plant communities because of a variety of local conditions like relief and degree of weathering.

Application activity 5.1

- Fill in the blanks with an appropriate concept:
 - Mangrove forests are adapted to coastal environments through __ roots that allow them to breathe in muddy soils.
 - Tropical rainforests receive __ rainfall throughout the year, resulting in dense vegetation cover.
 - As altitude increases on mountains, temperature ____, leading to changes in vegetation types.
- Explain how human activities can affect vegetation types in Africa.

5.2. Major types of vegetation in other continents

Learning activity 5.2

During an educational journey, Ishimwe traveled from equatorial rainforests to monsoon forests, grasslands, mangroves, Mediterranean regions, and temperate zones. He observed how rainfall, temperature, seasons, and latitude influence forest density, grass height, leaf types, and plant adaptations, from evergreen trees to deciduous forests and cold desert vegetation.

1. Fill in the blanks with an appropriate concept:
 - a) One vegetation type Ishimwe observed during his journey was the ____ forest.
 - b) Forests that remain green throughout the year are called ____ forests.
 - c) Vegetation with short grasses and scattered trees is known as ____.
 - d) Differences in plant height and leaf type are mainly influenced by __ and ____.
2. In your opinion, which vegetation type mentioned in the story is most sensitive to seasonal climate changes? Give one reason

5.2.1. Tropical Vegetation

This includes: The tropical rainforests, tropical monsoon forests, tropical grasslands vegetation and mangrove forest

a) Tropical rainforests

Location: In **North America:** Central America from Guatemala through Panama, Colombia up to Ecuador.

In **South America:** Amazon Basin and East coast of Brazil.

In **Asia:** Western India, Myanmar, Vietnam, Malaysia, Indonesia and Philippines.

In **Australia** and **Oceania:** Guinea Island and coastal region of Queensland.

Characteristics:

- There is heavy convectional rainfall of about 2000mm per year.
- There are so many species of plants which exceed 15000 plant species.
- Trees are closely set and evergreen throughout the year
- There is presence of creepers subdivided into three types: **climbers, epiphytes and lianas.**
- There is abundance of food supply for animals there.
- There is poor climate for human health

- Trees have buttress roots
- There is dense vegetation.
- Trees have 3 or 4 distinct canopies (layers) namely *undergrowth (shrub layer)*, *understorey (undercanopy)*, *canopy* and *Emergent Layer*

b) Tropical Monsoon forests

These are adapted in areas with tropical Monsoon type of climate having a dry season with high temperatures.

Location: In North America in Southern and southwest coast of Mexico, **Asia** in Northeastern India, Bangladesh, Myanmar and parts of Indonesia **and Australia** in Southern New Guinea, Northern Australia.

Characteristics:

- Most trees are deciduous
- There are tall trees up to 30m height but are far apart
- The trees grow more branches than those of tropical rainforests
- There is less variety of tree species than in the tropical rainforests.
- The dominant plant is bamboo due to enough sunlight reaching the ground.
- There is dense undergrowth on the forest floor.

c) Tropical grassland vegetation

Location: Brazilian highlands, parts of Guyana, Lowlands of Venezuela and Colombia in **South America**, India in **Asia** and Northeast of the Australian desert in **Australia**.

Characteristics:

- Some areas are dominated by open grasslands
- Trees are of medium height and grow wide apart
- Space between the trees is either occupied by grass or shrubs
- Many trees have umbrella-shaped crowns
- Most trees are deciduous and shed their leaves in the dry season.
- Trees have deep roots and thick barks
- Grass can grow up to 3m tall in wet areas.
- The elephant grass variety grows up to 4.6m
- Grasses dry off during the dry season but sprout at the onset of rain.
- Regions receiving rainfall below 750mm have much shorter grass and low growing trees and shrubs.
- Trees are scattered due to low rainfall
- Trees are flat topped due to strong prevailing winds blowing in the area
- The main types of trees growing in this vegetation area are acacia and baobab.

d) Mangrove vegetation

Location of mangrove forests

The forests are common in shallow estuaries, bays and lagoons found in areas such as:

- The east coast from East Africa to Mozambique
- Northwest coast of Madagascar
- The Gulf of Guinea especially in the Niger delta
- The coast of West Africa from Gambia to Sierra Leone

Characteristics of mangrove forests:

- Trees have special roots that are partly aerial.
- Roots grow horizontally above the water and then turn vertically into muddy water.
- Some tree species grow horizontally in the mud and then bend upwards above the water as breather roots.
- At high tide the lower trunk and aerial roots are submerged.
- At low tide the trees appear to be supported on stilts of roots.
- Their roots are for anchorage and for breathing.

5.2.2. Mediterranean vegetation

Location: Central coastal region of California in **North America**, Central Chile in **South America**, southern Europe, around Mediterranean Sea in **Europe** and southwest coastal area around Perth and to the south around Adelaide in **Australia**.

Characteristics

- This type of forest includes the Mediterranean woodlands.
- It has greatly been influenced by human activities such that most of it is now secondary.

5.2.3. Temperate Vegetation

This is subdivided into six categories of vegetation:

- a) Temperate evergreen forest
- b) Temperate deciduous forest
- c) Temperate coniferous forest
- d) Mixed temperate forests
- e) Temperate grasslands
- f) Temperate and cold desert vegetation

a) Temperate evergreen forests

Location:

In **North America**: it is found in Southern and Southeastern states of USA

In **South America**: it is found in Southern Brazil, Eastern Paraguay and Northern Uruguay.

In **Asia**: Southern China and south Japan.

In **Australia**: Southeastern Australia and North Island of New Zealand.

There are four types of temperate evergreen forests which are:

- Tropical monsoon forest
- Temperate deciduous forest
- Coniferous forests
- Mixed forests.

Characteristics (features)

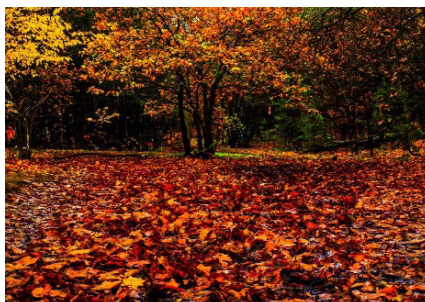
- The forests are evergreen
- Trees have broad leaves
- Some areas of the forest have thick undergrowth of other plant species.
- Some trees are deciduous
- Many evergreen tree species are hardwoods.

b) Temperate deciduous forests

Location:

- In **North America**: it is found in the Eastern States of USA
- In **South America**: it is found in Chile
- In **Europe**: it is located in western and central Europe.
- In **Asia**: it is located in Northern Japan, Korea and Eastern China.
- In **Australia**: it is found in Tasmania and South Island of New Zealand.

Characteristics:



- There is broad-leaved deciduous trees
- Trees are hardwoods

- Trees shed their leaves in autumn and remain bare in winter.
- The forests are quite open hence many undergrowth plants
- Some areas have mixed tree species while other areas may have pure stands.
- The coniferous trees also are common in some areas.

c) Temperate coniferous forests

Location: It is found in cool and cold climates between 45° and 75°N&S of the Equator

- In **North America:** West coast of Canada to Hudson Bay and up to the East Atlantic coast.
- In **South America:** it is limited to a narrow portion of the continent beyond 45°S.
- In **Eurasia:** From Scandinavia, through Russia up to the Pacific coast.

Characteristics:



- Most tree species are fast-maturing softwoods
- Most trees are evergreen but a few shed their leaves in winter.
- Most trees have thin needle-like leaves.
- The leaves have tough and waxy skin.
- The leaves and branches form a conical shape.
- Tree trunks are flexible.
- Trees have widely spread shallow root system
- Trees occur in large pure stands.

d) Mixed temperate forests

It is a mixture of both deciduous and coniferous forest tree species.

Location:

- In **North America:** Alberta, Saskatchewan and Nova Scotia provinces of Canada. From North of New York, Northern Pennsylvania, up to Minnesota (USA).
- In **Europe:** The Scottish Highlands. From North Alps through central and North Germany to South and Central Sweden up to Siberia.

- In **Asia**: North and East of Manchuria, North Korea, Northern Kyushu, and South Hokkaido islands of Japan.

Characteristics:

- There is a mixture of broad-leaved deciduous and coniferous trees.
- Tree species found in temperate deciduous and coniferous forests are present.

e) Temperate grasslands

This has different characteristics depending on their location.

Location: In North America: it is called **Prairies**.

They are found in the central lowlands, Manitoba and Alberta (in Canada) and extends to neighboring states of USA.

Characteristics:

- Grasses form a continuous cover of green tuft grass called true prairies.
- Short grass with leguminous plants, thus mixed prairies
- Mixture of grass species growing to different heights
- Tough and shorter grass grows where rainfall is below 500mm.
- Drying and dying of grass in summer but growing again in spring.

In South America: there are found in Argentina and are called Pampas.

Characteristics:

- Species of feather-like grass dominate
- Drought-resistant shrubs are common in the drier areas
- In most parts, the grass forms tussocks in wet areas, with patches of bare soil in between.

In Europe and Asia: they are called steppes.

Characteristics:

- The vegetation forms a rich layer of grass resembling a carpet.
- Some trees grow along water courses
- It is divided into 3 categories according to characteristics:
- **The wooded steppes:** where rainfall is more than 500mm
- **The true steppes:** where rainfall is between 250mm and 500mm
- **The desert steppes:** where rainfall is less than 250mm.
- The grass is usually short and grows in tussocks.

In Australia: they are found in the Murray-Darling basin and are called Downs.

Characteristics:

- The grass is tall and mixed with trees
- Areas with little rainfall have patches of short grass
- Vegetation is a mixture of tropical and temperate species.
- In New Zealand grass is in tussocks and uniform in structure.

f) Temperate and cold desert vegetation

Location: Antarctica, Greenland, North and West China, Turkestan, Iran and near Arctic area.

Characteristics:

- There is a mixture of grasses and woody plants
- Many plants have spiny leaves and are drought resistant.
- Plants are xerophytic and where soils are salty, plants are halophytic.
- The vegetation of tundra desert grows for only 3 months.
- Vegetation is composed of arctic meadow grass, mosses and small flowering plants.
- Shrubs have twisted and woody stems and shallow roots.

Application activity 5.2

1. Which of the following is NOT a main type of tropical vegetation
 - A. Tropical rainforest
 - B. Savannah grassland
 - C. Tropical monsoon forest
 - D. Temperate coniferous forest
2. Tropical rainforests have dense, evergreen vegetation mainly because:
 - A. They experience low temperatures throughout the year
 - B. High and well-distributed rainfall and high temperatures
 - C. They have long dry seasons
 - D. Human activities protect all forest areas
3. Fill in the blanks:
 - a) Plants in tropical grasslands adapt to seasonal rainfall by developing ____ roots and shedding ____ during the dry season.
 - b) Tropical rainforests occur in ____ climates with heavy rainfall, while temperate deciduous forests are found in ____ climates with seasonal rainfall.
 - c) The distribution of Mediterranean vegetation is influenced by ____ factors such as rainfall and temperature, as well as ____ activities like farming and deforestation.

5.3. The relationship between the different types of vegetation and human activities.

Learning activity 5.3

While visiting Nyungwe forest and nearby savannah grasslands, Gasana noticed people collecting fruits, honey, firewood, and medicinal plants. In eastern Rwanda, he saw herders grazing livestock on natural grasslands and families using grasses, trees, and other vegetation for food, fuel, construction and other daily needs.

Match the questions from column A to their related answer in column B

Column A	Column B
a) Two products people collected from forests in Nyungwe	(i) Provides grazing areas for animals and supports livestock farming
b) How forests support people's livelihoods in Rwanda	(ii) Fruits, honey, firewood, medicinal plants
c) Usefulness of grasslands to livestock keepers	(iii) Forests provide food, medicine, fuel, and income
d) Reason desert vegetation areas are sparsely populated	(iv) Limited resources and harsh living conditions make settlement difficult
e) Most important vegetation type for human survival (from the story) with one reason.	(v) Forests, because they supply essential resources for daily survival.

a) Relationship between forests and human activities

- Tropical hardwoods of tropical rainforests are valuable for timber
- Forests provide nuts and fruits medical plants and fibres
- They are habitat for bees which provide honey
- They provide poles for building and construction
- The bark of the oak tree provides cork for bottling wine
- Olive trees provide fruits which give olive oil for various purposes.
- They are source of fuel in form of firewood, charcoal and saw dust for cooking.

b) Relationship between grasslands and human activities

- The rotting grass gives humus to soil, hence increasing grain cultivation.
- Attraction of tourists due to herbivores and carnivores found there.
- Harvesting of honey because it is habitat for bees

- They are widely used for grazing livestock
- They provide plants used for medical purposes
- Source of food in form of fruits and vegetables
- Source of fuel in form of firewood and charcoal.

c) Relationship between desert vegetation and human activities

- They are sparsely inhabited by people
- They provide pastures for nomads of sheep and camels
- Some plants provide edible fruits
- Source of fuel wood to people
- Source of food from Date palms growing around oases.

Application activity 5.3

1. Which of the following is a key benefit of grasslands in Rwanda?
 - A. Mining opportunities
 - B. Agriculture and tourism
 - C. Industrial development
 - D. Desertification prevention
2. Explain how desert vegetation supports human survival despite harsh environmental conditions.
3. Assess the importance of forests in supporting human livelihoods through products such as timber, food, and fuel.

End of unit assessment

❖ Knowledge and understanding:

1. Which types of vegetation did the family observe in Rwanda?
 - A. Tropical rainforest and savannah grassland
 - B. Desert and tundra
 - C. Mediterranean and coniferous forests
 - D. Mangroves and alpine
2. Over-harvesting firewood in forests can lead to:
 - A. Increased biodiversity
 - B. Deforestation and reduced resources for humans
 - C. Improved soil fertility
 - D. Expansion of grasslands

❖ **Skills:**

3. Analyze how over-harvesting firewood could affect human activities in the area.
4. Using a map of Africa, identify and mark the areas where tropical forests and grasslands are located.

❖ **Attitudes & Values**

5. Suggest two ways the family can use forest and grassland resources sustainably while conserving the environment.

UNIT 6

DRAINAGE IN AFRICA AND THE WORLD



Key Unit Competence:

Evaluate the importance of water bodies and wetlands in Africa and the world.

Learning objectives

At the end of this unit you should be able to:

Knowledge and understanding

1

- Identify the major water bodies in world
- List features related to rivers
- Recall the importance of water bodies
- State the challenges of obtaining sustainable and clean water in Africa.

Skills

2

- Locate on a map the major water bodies found in Africa and the world.
- Examine the various physical features resulting from river erosion and deposition.
- Evaluate the importance of water bodies.
- Examine the challenges faced in obtaining sustainable and clean water in Africa.

Attitudes and values

3

- Appreciate the importance of water resource management and conservation (e.g. rainwater harvesting) in Africa.
- Show a continual desire to seek other perspectives related to the misuse of water in Africa.

6.0. Introduction

Drainage is the movement and distribution of water across the Earth's surface, primarily through rivers and streams that eventually flow into lakes, seas, or oceans. Rivers grow as tributaries join them. This unit explores Africa's major rivers and lakes, lake formation, river erosion and deposition features, their importance, and challenges to sustainable clean water, including scarcity, pollution, and sedimentation.

Introductory activity

A group of Rwandan students visited the Nyabarongo River. They traced its flow, observed small streams joining it, and noted a nearby wetland. Their teacher explained how the river supports farming, fishing, and local communities. They also saw signs of pollution and soil erosion, prompting a discussion on protecting water resources.

1. Apart from the Nyabarongo river name other three major rivers in Africa.
2. Explain how tributary streams affect the size of a river.
3. How can local communities benefit from rivers and wetlands in Rwanda?
4. Identify two challenges observed by the students that affect river health.
5. Suggest two ways the students or their community could help conserve the Nyabarongo River.

6.1. Major rivers in Africa and the rest of the world

Learning activity 6.1

A Rwandan geography class explored the Nyabarongo River, tracing its tributaries and noting nearby farming. Their teacher explained that, like the Zambezi, Congo, Niger, Volta, Senegal, Limpopo, Orange, and Nile, rivers originate in highlands and flow to oceans or lakes. Students also observed litter, prompting a discussion on river conservation.

1. Name three African rivers mentioned in the story besides the Nyabarongo River.
2. Explain the relationship between the origin and destination of rivers as discussed by the teacher.
3. Suggest two ways the students could help reduce litter and protect rivers during their field visits.

Major rivers in Africa

Some of rivers in East Africa are Ruvuma, Rufiji, Pangani, Tana, Akagera, Nyabarongo, Juba, and so on. The major river systems such as Niger, Congo, Orange, Zambezi and Nile have very large **drainage basins** and drain their waters into seas or oceans.

There are other rivers which drain their waters inland. These are called **basins of internal drainage**.

Examples: Chad basin, Okavango basin, Western Tanzania basin and Lake Turkana basin.

Some of the Africa's rivers in terms of length are: Nile (6,695km), Orange (2,092km), Congo (4,700km), Limpopo (1,800km), Niger (4,200km), Senegal (1,641km), Zambezi (2,693km) and Volta (1,600km)



Application activity 6.1

1. Fill in the blank: The longest river in Africa is the __, measuring 6,695 km in length
2. Which of the following rivers drains its waters inland instead of into the sea or ocean?
A. Nile B. Zambezi C. Okavango basin rivers D. Congo
3. Explain the difference between major river systems like the Nile and rivers in internal drainage basins, giving examples of each.

6.2. Major lakes of Africa and the world according to their mode of formation

Learning activity 6.2

A geography class studied major lakes around the world, including those in Africa, Asia, South America, Europe, and North America. The teacher explained that lakes like Victoria, Tanganyika, Malawi, Albert, Kyoga, and Turkana occupy depressions in eastern Africa, while lakes such as Baikal (Asia), Titicaca (South America), and the Great Lakes (North America) also occupy tectonic, volcanic, or glacial depressions. Students learned that glacial lakes form on mountains due to ice erosion, volcanic lakes form in craters, and lowlands rarely have glacial lakes. They discussed the countries hosting each lake.

1. Name three major lakes in eastern Africa and three major lakes from other continents.
2. Explain why lakes are concentrated in eastern Africa and discuss factors influencing lake formation in other continents.
3. Locate the following lakes on a sketch map: Victoria, Tanganyika and Malawi (Africa) Baikal (Asia), Titicaca (South America) and Lake Superior (North America)
4. Compare the sizes and differences that exist between: (a) Lake Victoria and Lake Chad (b) Lake Baikal and Lake Titicaca, explaining why differences exist.
5. Suggest ways local communities can sustainably use resources from lakes like Kivu, Victoria, Baikal, or Titicaca.

❖ Major lakes in Africa

A lake is a mass of water that occupies a large depression on the ground. A lot of lakes are concentrated on the eastern side of the continent, while they are absent in Sahara Desert.

The major lakes in Africa include Victoria (69,485km²), Chad (25,760km²), Kyoga (4,403km²), Tanganyika (32,893km²), Turkana (6,405km²), Malawi (30,044km²) and Albert (5,299km²). Other lakes include: Rukwa, Mweru, Kivu, Edward, and so on. There are both natural lakes and human-made lakes in Africa.

Note:

- Lake Victoria is the second largest fresh-water lake in the world and the First largest in Africa.
- Lake Tanganyika is the first deepest lake in Africa and the second in the world
- Lake Malawi contains the largest number of fish species than any other lake in the world
- Lake Volta in Ghana is the largest human-made lake in the world with **8,502 km²**
- Lake Kariba (**5,580km²**) in Zimbabwe is the 4th largest in the world among the human-made lakes.

- Lake Nasser (**5,248km²**) on River Nile is shared among Sudan and Egypt and is the 6th largest human-made lake in the world.
- Other important human-made lakes in Africa include: Kainji, Cabora Bassa, Masinga and Kossou.

❖ Lakes formed by tectonic movements

Earth movements cause folding, faulting, and warping. These processes lead to the formation of tectonic basins in which water collects to form lakes. They include:

- **Lakes due to faulting:** They are formed in deep valley called **rift valley**. When water accumulates within depression created by faulting, those lakes are created. These lakes are quite deep and long. **Examples:** Kivu, Tanganyika, Malawi, Albert, Edward and Turkana.
- **Lakes due to crustal warping:** these are formed when water accumulates within depressions created by down-warping. These lakes are fairly shallow, wide and have several inlets. **Examples:** Victoria, Kyoga, Muhazi and Chad Lakes.

❖ Lakes formed by vulcanicity

- **Lava-dammed lakes:** are formed when a lava flow cools, accumulates and solidifies by forming a barrier to the flow of a river, hence making the water of a river accumulates behind. **Examples:** Burera and Ruhondo lakes, Mutanda and Tana lakes.
- **Crater lakes:** these are formed when water accumulates in craters or calderas on the top of volcanoes or on the ground. **Examples:** - Crater lakes on Bisoke and Muhabura (Rwanda) and Magadi Caldera Lake in Ngorongoro (Tanzania)

❖ Lakes formed due to erosion

- **Lakes due to erosion by Glaciers:** these are formed when depressions created by moving ice (cirques) are occupied with water. These lakes are called cirque lakes or tarns. **Examples:** Lake Speke and Lake Catherine on Mount Rwenzori and Lakes Gallery, Hidden, Hanging and Teleki on Mount Kenya. There is also Ribbon or Finger Lakes on mountains such as Lake vert and Lake Noir on Mount Rwenzori.
- **Lakes due to wind erosion:** are formed when hallows formed by wind erosion in deserts are filled with water. **Examples:** Qattâra Depression in Egypt and Bodélé depression in Chad.
- **Solution lakes:** are formed in limestone regions when water accumulates within depressions formed by weathering and erosion. **Examples:** Lake Guinas in Namibia and Lake Azigza in Morocco.

❖ Lakes formed by deposition

- **Lakes formed by river deposition:** these are formed by both erosion and deposition in floodplains or simply by **alluvial deposition** in deltas. **Examples:** -Ox-bow lakes such as Utange (Tanzania) and Shakababo (Kenya) and Lakes due to alluvial deposition such as Mugesera and Sake.

- **Lakes formed by wave deposition:** these are formed in Lagoons that are cut off from the ocean or lake. **Example:** Lake Nabugabo and Sare from Lake Victoria.
- **Lakes formed by glacial deposition:** these are found in high mountains after deposition and melting of moving ice in hollows. **Example:** Lake Gris and Mahoma in Rwenzori.
- **Lakes formed by mass movement:** they are formed when a landslide blocks a river valley. **Example:** Lake Bujuku and Nyabihoko (Uganda), Lake Agulu (in Nigeria), and Lake Funduzi (South Africa).
- **Lakes formed due to human activities:** These are also called **human-made lakes**. They are formed when people build barriers such as dams across river valleys. **Example:** - Lake Kariba on River Zambezi (Zambia & Zimbabwe) and Lake Volta on River Volta (in Ghana)
- **Lakes formed by meteorites:** These are formed when water accumulates within depressions created by the falling of meteorites on the ground. **Example:** Lake Bosumtwi in Ghana.

6.3. Major rivers and lakes of other continents (Europe, Asia, Oceania, South and North America)

1. Asia

❖ Major rivers in Asia include:

- Yangtze River (China): Longest river in Asia, ~6,300 km; important for transport, agriculture, and hydroelectric power (Three Gorges Dam).
- Ganges River (India/Bangladesh): Sacred River, ~2,525 km; supports agriculture, fishing, and millions of people.
- Mekong River (Southeast Asia): Flows through six countries; vital for irrigation, fishing, and transport.
- Ob, Lena, and Amur Rivers (Russia/China): Major Siberian rivers important for drainage and hydroelectric power.

❖ Major Lakes in Asia include:

- Caspian Sea (border of Europe & Asia): Largest inland body of water in the world; rich in oil and gas.
- Lake Baikal (Russia): Deepest and oldest freshwater lake hosting the significant several biodiversity.
- Aral Sea (Kazakhstan/Uzbekistan): Once large; shrinking due to irrigation.

2. Europe

❖ Major rivers in Europe include:

- Volga River (Russia): Longest river in Europe; important for transport and irrigation.
- Danube River (Central & Eastern Europe): Flows through 10 countries; used for navigation, agriculture, and hydroelectric power.
- Rhine River (Western Europe): Important for trade and industry; flows through Switzerland, Germany, and the Netherlands.

❖ **Major lakes in Europe include:**

- Lake Geneva (Switzerland/France): Large freshwater lake used for tourism, drinking water, and hydropower.
- Lake Ladoga (Russia): Largest lake in Europe; important for transport and fisheries.
- Lake Balaton (Hungary): Largest lake in Central Europe; tourist attraction.

3. North America

❖ **Major rivers in North America include:**

- Mississippi River (USA): Major river system for transport, agriculture, and industry.
- Missouri River (USA): Longest river in the US; tributary of the Mississippi.
- St. Lawrence River (Canada/USA): Connects Great Lakes to the Atlantic Ocean; key for trade.

❖ **Major lakes North America include:**

- Great Lakes (USA/Canada): Superior, Michigan, Huron, Erie, Ontario; largest freshwater system in the world.
- Lake Winnipeg (Canada): Important for fishing and transport.
- Lake Mead (USA): Reservoir on the Colorado River; water supply and hydroelectric power.

4. South America

❖ **Major rivers in South America include:**

- Amazon River: Longest and largest by discharge; vital for transport, biodiversity, and local communities.
- Orinoco River (Venezuela/Colombia): Important for transport and agriculture.
- Paraná River (Brazil/ Argentina/ Paraguay): Supports hydroelectric power and irrigation.

❖ **Major lakes in South America include:**

- Lake Titicaca (Peru/Bolivia): Highest navigable lake in the world; cultural and economic importance.
- Maracaibo Lake (Venezuela): Rich in oil; largest lake in South America.

5. Australia (Oceania)

❖ **Major rivers in Oceania:**

- Murray River: Longest river in Australia; supports agriculture.
- Darling River: Tributary of Murray; important for irrigation.
- Sepik River (Papua New Guinea): Major river for transport and local livelihoods.

❖ **Major Lakes Oceania:**

- Lake Eyre: Largest salt lake; often dry, fills occasionally.
- Lake Taupo (New Zealand): Volcanic lake; important for tourism and hydroelectric power.

Application activity 6.2

Using an atlas, other reference materials, and the Internet, identify the names of the dams and rivers over which these lakes or reservoirs have formed, both in Africa and other continents. Copy the table in your notebook and fill in the missing information.

Continent	Name of Reservoir / Lake	River on which it was formed
Africa	Lake Volta	_____
Africa	_____	Zambezi River
_____	Kainji Lake	Niger River
Africa	Lake Nasser	Nile River
_____	Three Gorges Reservoir	Yangtze River
North America	Lake Mead	_____
North America	Lake Ontario (man-made sections)	St. Lawrence/Niagara River
_____	Itaipu Reservoir	Paraná River

a) Importance of wetlands, rivers, lakes, oceans and seas and challenges to obtain sustainable clean water in Africa and other continents.

Learning activity 6.3

Rivers and lakes are vital to people and nature. The Nile supports farming, the Amazon sustains forests and communities, and Lake Baikal protects unique species. However, pollution, poor sanitation, and overuse threaten water quality and availability worldwide.

- Fill in the blanks
 - The _____ River supports farming in Africa.
 - Lake _____ is known for protecting unique species
- Which challenge most threatens rivers and lakes today?
 - Sustainable use
 - Pollution and poor sanitation
 - Conservation activities
 - Natural beauty
- Which of the following is an important benefit of rivers and lakes?
 - Causing soil erosion
 - Providing drinking water and food
 - Increasing water scarcity
 - Destroying ecosystems

a) Importance of rivers, lakes, seas, ocean and wetlands in Africa

- They provide water for domestic and industrial purposes
- They provide water for irrigation, hence promoting agriculture
- They lead to generation of hydroelectric power

- Promotion of water transport
- Promotion of fishing for domestic and commercial purposes
- Attraction of tourists due to features formed and species of animals within them.
- Promotion of mining and quarrying
- Promotion of agriculture due to fertile soils in alluvial or floodplains
- They act as natural boundaries between countries or regions.
- Lakes and swamps act as the control of flooding of rivers.

Negative effects of rivers, lakes and wetlands in Africa

- They are habitats for diseases vectors and dangerous animals which are harmful to people and domestic animals.
- They are barrier to land transport
- Destruction of property, loss of human lives and displacement due to flooding.

b) Challenges to obtain sustainable clean water in Africa and other continents

- Africa and other continents face water scarcity due to irregular rainfall, prolonged droughts, and high population growth, which place pressure on limited freshwater resources.
- Pollution from industries, agriculture, and poor waste management contaminates rivers, lakes, and wetlands, making clean water unsafe and difficult to access.
- Silting and sedimentation caused by soil erosion reduce the capacity of water bodies, degrade water quality, and disrupt water supply systems
- There is increase in population in rural areas which interferes the natural water sources by contaminating them through washing clothes, vehicles and themselves.
- Lack of proper sanitation such as pit latrines and toilets, hence people urinating and defecating in open places, therefore contaminating water bodies due to erosion.
- Disposal of sewage and industrial wastes (chemicals) in water courses.
- Reduction of normal infiltration due to expansion of settlements hence decreased volumes of underground water.
- Use of chemical fertilizers and pesticides in agriculture which contaminate water.
- Swamp reclamation for agriculture and settlements
- Climate changes lead to contamination of water due to flooding or high rates of evaporation during droughts hence water shortages.
- Poor land use methods leading to soil erosion
- Misuse of water resources
- Lack of adequate and sufficient infrastructure to harvest and store rainwater.

Application activity 6.3

1. Fill in the blanks
 - a) A new water body can provide _____ for people, _____ for wildlife, and _____ for energy production.
 - b) Protecting water bodies is essential to ensure _____ and _____ resources for future generations.
 - c) Climate change and population growth increase pressure on freshwater resources by causing _____ and increasing _____ demand.
2. Which combination best improves water availability and quality in rural areas?
 - A. Deforestation, irrigation, and waste dumping
 - B. Rainwater harvesting, soil conservation, and pollution control
 - C. Overuse of rivers and poor sanitation
 - D. Industrial discharge and wetland drainage
3. Which factor is a major cause of water scarcity in Africa and other continents?
 - A. Proper water management
 - B. Conservation practices
 - C. Misuse of water resources and poor infrastructure
 - D. Abundant rainfall

End of unit assessment

❖ Knowledge and understanding:

1. Fill in the blanks
 - a) A drainage basin is an area of land where all rivers and streams drain into a _____ or lake.
 - b) Lake Tanganyika was formed mainly as a result of _____ (rift) movements in the Earth's _____.
2. Which of the following is a major drainage basin in Africa?
A. Amazon Basin B. Nile Basin C. Mississippi Basin D. Danube Basin
3. Which lake was formed mainly by tectonic (rift valley) activity?
A. Lake Victoria B. Lake Tanganyika C. Lake Volta D. Lake Chad

❖ Skills

4. Using an atlas or online map, identify and draw the drainage basins of the Nile, Niger, and Zambezi rivers. Explain how tributaries contribute to the size and flow of each river.

❖ Attitudes and values

5. Explain the importance of major rivers and lakes in the economic development of the countries where they are found, giving relevant examples from Africa and other continents.

UNIT 7

POPULATION IN AFRICA



Key Unit Competence:

Analyze the population problems and their solutions in Africa.

Learning objectives

At the end of this unit you should be able to:

Knowledge and understanding

1

- Identify the population distribution in Africa and the rest of the world
- Identify factors influencing the density and distribution of population
- Identify the population structure of Africa and the rest of the world.
- Outline reasons for rapid population growth and its effects in Africa.
- State possible ways to control rapid population growth in Africa
- Identify the types, causes and effects of migration in Africa
- State the major problems of population in Africa

Skills

2

- Describe the population distribution of Africa and the rest of the world.
- Illustrate using the population pyramid the population structure in Africa and the rest of the world
- Analyse the causes and consequences of rapid population growth in Africa
- Describe the possible solutions to the problems related to over population in Africa
- Examine the causes and effects of the different types of migration in Africa
- Analyse the major population problems in Africa and propose solutions.

Attitudes and values

3

- Support the promotion of awareness about factors affecting population distribution.
- Appreciate the importance of population control measures for sustainable development

7.0. Introduction

This unit covers human populations, their distribution, composition, and impact on development. Africa is the world's second most populous continent, with over 1.5 billion people characterized by rapid growth, a youthful population, and diverse ethnic and cultural groups, influencing economic and social development.

Introductory activity

In a geography class, students learned that Africa has over 1.5 billion people, making it the world's second most populous continent. The teacher explained that Population Geography studies where people live, how they are structured, and their impact on development. Africa's population grows rapidly, is largely youthful, and culturally diverse. Students discussed how population affects economy, education, health, and urban planning and how understanding population patterns helps governments and communities plan for the future.

1. Fill in the blanks
 - a) The total population of Africa is approximately _____ billion, making it the _____ most populous continent.
 - b) Population Geography studies where people live, how they are _____, and how they impact _____.
2. Why is Africa's population described as "youthful" and "rapidly growing"?
 - A. Most people are elderly and birth rates are low
 - B. Most people are children and young adults, with high birth rates
 - C. Population is evenly distributed across all age groups
 - D. Most people live in rural areas
3. Explain how Africa's population affects economic and social development, giving examples related to education, health, and urban planning.

7.1. Population distribution, density, and structure in Africa and the rest of the world.

Learning activity 7.1

Amina, a geography student, travels from a crowded city along a fertile river valley to sparsely populated highlands and swampy lowlands. She notices dense settlements on flat, well-drained lands, fewer people on steep slopes, and many children in villages. In cities, jobs and services attract migrants, shaping population patterns and structures.

1. Population density is the number of people living in one _____ kilometer.
2. The graph showing population structure by age and sex is called a _____
3. Explain how relief, drainage, and climate influence population distribution observed during Amina's journey.
4. Assess how Africa's youthful population structure affects future development and social services.

❖ Meaning of population distribution, density, and structure

- Population **distribution** shows how people are spread, while population **density** is number of people living on one square kilometer or $\frac{\text{People}}{\text{km}^2}$ (indicating crowdedness, with patterns being clustered, dispersed, or concentrated). Population **structure** shows a population's age and sex composition using pyramids.
- Population density** = $\frac{\text{Total population}}{\text{Total Area (Size)}}$
- The population distribution and density in Africa is uneven. There are areas which are densely populated, moderately populated, while others are sparsely populated.

❖ Factors influencing population settlement

There are both physical and human factors determining the population distribution:

Physical factors

- Drainage:** Well-drained soils support agriculture and settlements, while poorly drained swamps are avoided due to flooding and diseases.
- Relief:** Steep slopes and high altitudes are avoided; gentle slopes and flat areas attract population.
- Climate:** High rainfall and moderate temperatures favor settlements; extreme climates discourage human habitation.
- Vegetation:** Dense forests discourage settlements; grasslands attract people as they are easy to clear.
- Soils:** Fertile soils encourage settlements for agriculture, while infertile soils remain sparsely populated.

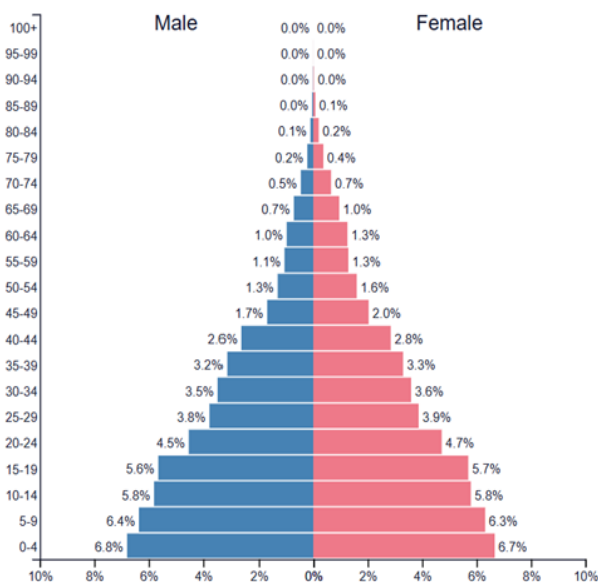
Human factors

- Historical events:** Historical events like slavery, conflicts, and colonization reduce origin population, increase destination population.
- Political and government policies:** Political issues, wars, and policies cause migration, altering population distribution in regions.
- Economic activities:** Economic activities such as mining, tourism, and trade attract people; plantations may discourage settlement.
- Urbanisation:** towns and cities attract many people due to presence of social amenities, modern lifestyle and jobs.

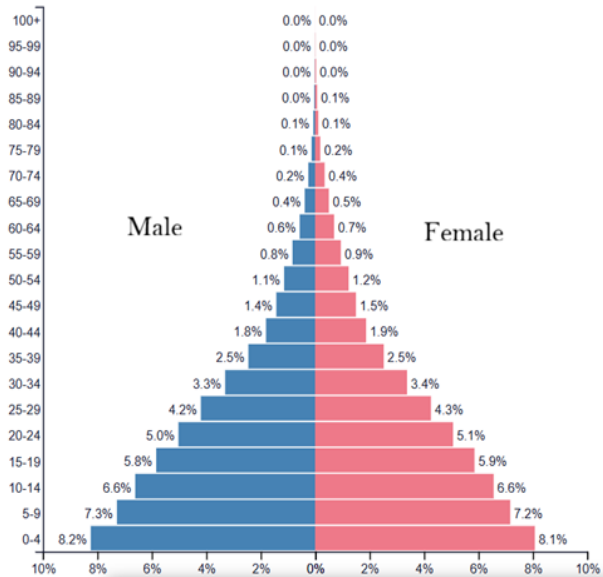
❖ Population structure in Africa and the rest of the world

Population structure is the composition of a population of a given area in terms of age and sex (gender).

The population structure of a given area is shown by a graph called **population pyramid (Age-sex pyramid)**.



2023 Rwanda population 13,954,471



2023 Uganda Population: 48,656,601

The population structure of African countries is characterized by:

- A large number of children and youth who are **dependents**
- The active (working) population is between 15-64 years.
- Those 65 years and above are old population who are also **dependent**.
- From the observation of the population pyramids of most African countries we can generalize that:
- About 45% of the total population is made up of children below 15 years.
- The old people of over 65 years is about 3% of the total population
- The population pyramid has very broad base to indicate a large proportion of children and young people.
- Large proportion of population under 15 years indicates a large number of potential parents.
- The population pyramids have narrow tops to indicate a low proportion of old people.

The low number of old population in African countries may be explained by a number of factors such as diseases, civil wars, accidents, poor living conditions, and so on.

Application activity 7.1

1. Fill in the blanks

Population data helps planners decide where to build more _____ and _____ to meet community needs.

2. Which strategy best helps manage rapid population growth in an African country?
 - A. Ignoring population data
 - B. Expanding education, healthcare, and employment opportunities
 - C. Encouraging early marriage
 - D. Reducing health services
3. Why is understanding population patterns important for sustainable development?
 - A. It increases population growth
 - B. It helps governments plan services and use resources efficiently
 - C. It discourages urbanization
 - D. It prevents migration completely.

7.2. Population growth, its effects, and control measures in Africa.

Learning activity 7.2

In 2025, Africa's population reached 1.6 billion, with a birth rate of 35 per 1,000 and a death rate of 8 per 1,000. Rapid growth caused unemployment for 20%, poor housing, and limited healthcare. Family planning and education were needed to stabilize growth.

5. According to the story, what was Africa's population in 2025?
 - A. 1.2 billion
 - B. 1.4 billion
 - C. 1.6 billion
 - D. 2.0 billion
6. Which set correctly shows the birth rate and death rate mentioned in the story?
 - A. Birth rate 8 per 1,000; Death rate 35 per 1,000
 - B. Birth rate 20 per 1,000; Death rate 10 per 1,000
 - C. Birth rate 35 per 1,000; Death rate 8 per 1,000
 - D. Birth rate 30 per 1,000; Death rate 15 per 1,000
7. Explain what "rapid population growth" means in the context of the story.
8. Describe two social problems caused by rapid population growth in Africa.

Population growth is the increase or decrease in the number of people in a given region within a given period of time.

Population explosion is the rapid positive population growth. The population growth is influenced by the **birth rate**, **death rate**, **fertility rate** and **growth rate**.

a) Birth rate: is the number of children born in a year per 1000 of persons of the total population.

$$\text{Formulae: BR} = \frac{\text{Total number of live births per year} \times 1000}{\text{Total population}}$$

b) Death rate: this is the number of deaths in a year per 1000 people of the total population.

$$\text{DR} = \frac{\text{Number of deaths} \times 1000}{\text{Total population}}$$

c) Fertility rate: this is the average number of children a woman would bear during her lifetime.

$$\text{FR} = \frac{\text{Total number of children born} \times 1000}{\text{Total number of women in reproductive age}}$$

From this; we can note that:

- A rate of 2 children per women is the replacement rate for a population which results in relative stability of population numbers.
- Rates above 2 children indicate the population that is growing in size
- Higher rates indicate families in difficulties such as to feed and educate children
- Rates below 2 children indicate the population decreasing in size and growing older.

d) Growth rate: is the percentage ratio of birth rates to death rates per 1000

$$\text{GR} = \text{Birth rate} - \text{Death rate}$$

❖ **Effects of population growth in Africa include:**

- **Decline in GDP:** National income decreases, reducing overall economic performance and citizens' living standards.
- **Slow economic development:** Economy grows slowly, limiting infrastructure, services, and population well-being improvement.
- **Unemployment:** Many people lack jobs, causing poverty and social challenges in society.
- **Poor housing and health facilities:** Inadequate homes and medical services harm living conditions and public health.
- **Reduction in cash crop production:** Decline in cash crops lowers farmer incomes and export earnings.
- **Scarcity of amenities:** Limited access to essential services reduces quality of life.
- **Slow industrial growth:** Industries develop gradually, restricting job creation and economic diversification.
- **Underdevelopment:** Country remains economically and socially backward, lacking infrastructure and basic services

❖ Possible ways to control rapid population growth in Africa

- Encouraging family planning programmes helps communities access and use birth control methods effectively.
- Discouraging early marriages and encouraging education of girls delays marriage and improves social outcomes.
- Carrying out campaigns that support small families promotes the benefits of smaller households to society.
- Encouraging women to get involved in gainful employment and offering career opportunities enhances economic independence.
- Encouraging inducement to parents to have fewer children provides rewards for families with smaller numbers.
- Discouraging polygamous marriage promotes monogamy to reduce population growth and social complications.
- Encouraging celibacy advocates voluntary abstinence to help limit family size and population growth.
- Encouraging outward migration supports relocation to less populated areas to balance population distribution.

Application activity 7.2

1. Which government action can best reduce unemployment caused by rapid population growth?
 - A. Reducing school enrollment
 - B. Creating jobs through industrial and skills development
 - C. Encouraging early marriage
 - D. Cutting healthcare services
2. Using the birth rate of 35 per 1,000 and death rate of 8 per 1,000, what is the natural increase rate?
 - A. 27 per 1,000
 - B. 43 per 1,000
 - C. 17 per 1,000
 - D. 8 per 1,000
3. Propose a plan that combines family planning and education to help stabilize Africa's population.
4. Evaluate how education and family planning can help control population growth, using examples from the story.

7.3. Migrations in Africa: types of migration, causes and effects of migration

Learning activity 7.3

In a geography lesson, students learned that migration is the movement of people from one place to another, either temporarily or permanently. Some people move within their country from rural to urban areas for jobs, while others cross borders as immigrants or emigrants due to conflicts, disasters, land pressure, or employment opportunities. Migration changes population size, labour supply, and development patterns in both origin and destination areas.

1. Which statement correctly distinguishes immigration from emigration?
 - A. Immigration is moving within a country, while emigration is moving to cities
 - B. Immigration is entering a country, while emigration is leaving a country
 - C. Immigration is temporary movement, while emigration is permanent movement
 - D. Immigration is rural movement, while emigration is urban movement
2. Which of the following is a push factor causing migration?
 - A. Availability of jobs
 - B. Better education facilities
 - C. Conflict or natural disasters
 - D. Improved healthcare services
3. Explain why many people move from rural areas to urban areas.
4. Describe two effects of migration on both the place of origin and the place of destination.

a) Types of migrations

Migration refers to the movement of people from one place to another that leads to a change of residence, either temporarily or permanently. **Immigration** involves people entering a country or region, while **emigration** refers to people leaving it. Migration occurs in different forms, including internal and international movements.

- **Internal migration:** it involves the movement of people from one place to another within the same country. It includes rural-urban migration (rural exodus), rural-rural migration, urban-rural migration, urban-urban migration.
- **External or international migration** is the movement of people from one country to another. It involves the crossing of international borders.

❖ Causes of migrations

There are both push and pull factors of migration. These include:

- Political persecution resulting in **internally displaced persons (IDP's)** and **international refugees**.
- Religious conflicts

- Search of welfare due to civil wars
- Natural calamities such as floods, volcanic eruptions, earthquakes, etc.
- Pressure on the land
- Search of employment opportunities.

❖ Effects of migration

- Shortage of labour (employees) to work in agriculture in rural areas
- General shortage of food grown in rural areas
- Land use consolidation in areas of origin, hence improved agricultural production
- Imbalance in the female-male ratio because the majority of migrants are males.
- General under-development of rural areas
- Decrease of population density and fertility rate in areas of origin
- In case of forced migration; some parts may be depopulated
- Increase in population in areas of destination.

Application activity 7.3

1. Analyse how migration causes rural labour and food shortages, and propose government measures to reduce forced migration
2. Evaluate the statement: *"Migration contributes more to development in destination areas than in areas of origin. "Support your answer with reasons"*

7.4. Major population problems and solutions in Africa

Learning activity 7.4

In Umutuzo village, many young people moved to the city looking for jobs. Farms were left unattended, food became scarce, and poverty increased. Some villagers fell sick due to HIV/AIDS, while others lacked education. Leaders later promoted schooling, healthcare, job creation, and family planning, slowly restoring hope.

1. List four population problems mentioned in the story.
2. Explain why food became scarce in Umutuzo village.
3. Suggest one-way education could help reduce poverty in the village.
4. Analyze how rural–urban migration affected both farming and employment in Umutuzo

Africa faces several population challenges, including HIV/AIDS, poverty, overpopulation, famine, illiteracy, rural–urban migration, and unemployment, which require effective and sustainable solutions to improve living standards and development:

Those problems and their possible solutions include:

Problems: Population growth and distribution present several challenges that affect people's livelihoods and development including:

- Famine caused by prolonged drought, hence shortage of food
- Rural to urban migration in search of employment and better lifestyles
- Unemployment due to large number of people in search of it.
- High incidences of diseases pandemics like Corona virus and HIV/Aids which lead to deaths of many people and high government expenditures.
- High population pressure on the land

Solutions: Understanding the above problems and applying appropriate solutions is essential for improving living standards, managing resources sustainably, and promoting balanced social and economic development in the following ways:

- Proper water management and storage
- Provision of social amenities and creation of employment in rural areas.
- Developing skills in people for self-employment
- To negotiate with needy countries to employ skilled unemployed people.
- Massive education on hygiene and preventive measures for diseases
- Provision of medical services
- Family planning to reduce birth rates.

Application activity 7.4

1. What is the relationship between unemployment and poverty in the story?
 - A. Unemployment increases poverty because people have no income to meet basic needs
 - B. Unemployment reduces poverty because people stay home
 - C. Unemployment has no effect on poverty
 - D. Unemployment encourages urbanization
2. Which combination of actions could help reduce overpopulation and unemployment in the village?
 - A. Promote family planning and provide education and job opportunities
 - B. Reduce schools and close healthcare centers
 - C. Encourage early marriage and limit employment
 - D. Ignore migration and population issues
3. Evaluate the role of healthcare, family planning, and education in reducing the effects of HIV/AIDS, overpopulation, and unemployment in African communities.

End of unit assessment

❖ Knowledge and understanding:

1. Which factor contributes most to rapid population growth in Africa?
 - A. High birth rates and declining death rates
 - B. Urbanization only
 - C. Migration only
 - D. Education improvement
2. Which statement best distinguishes immigration from emigration?
 - A. Immigration is leaving a country; emigration is entering a country
 - B. Immigration is entering a country; emigration is leaving a country
 - C. Both involve rural-urban movement only
 - D. Both occur only in Africa

❖ Skills

3. Describe Ugandan and Rwandan population age structure and gender composition.
4. Discuss how individuals and governments can promote awareness for sustainable development.

❖ Attitudes & Values

5. Examine two major population problems in Africa and suggest solutions for each.

UNIT 8

URBAN SETTLEMENT IN AFRICA



Key Unit Competence:

Assess the impact of urban settlement in the development of Africa.

Learning objectives

At the end of this unit you should be able to:

Knowledge and understanding

1

- Explain the characteristics of urban centres in Africa
- Outline the functions of urban centres
- State the factors influencing the growth of urban centres in Africa
- Identify the problems and solutions of urban centres in Africa

Skills

2

- Analyse the characteristics of urban centres in Africa
- Analyse the functions of urban centres in Africa.
- Explain the factors influencing the growth of urban centres in Africa.
- Apply the acquired knowledge to suggest solutions to control the problems of urbanisation in Africa

Attitudes and values

3

- Appreciate the importance of urbanisation in Africa
- Show concern for the challenges arising from uncontrolled rapid urban growth in Africa.
- Support initiatives aimed at reducing the impacts of rapid urban growth.

8.0. Introduction

This unit explores African urban centres, examining their characteristics, functions, growth factors, problems, and solutions, using case studies like Cairo and Johannesburg to understand urbanisation's impact on development.

Introductory activity

Mutesi and Kalisa visited Kigali, Cairo, and Johannesburg to explore urbanisation. They observed crowded streets, tall buildings, traffic jams, slums, and income disparities. They found that migration, industry, education, and transport influenced city growth, while unemployment, pollution, and inadequate housing created challenges. Their findings relate to the content of Unit 8. Try answering the questions below to demonstrate your understanding:

1. Fill in the blanks:
 - a) Population density in Kigali, Cairo, and Johannesburg shows _____ patterns due to urbanisation.
 - b) Major problems in African urban centres include _____ and _____.
2. Compare the factors influencing urban growth in Kigali versus Johannesburg.
3. Suggest solutions to manage rapid urbanisation effectively in Rwanda and other African cities.

8.1. Urban centers in Africa: Characteristics and functions

Learning activity 8.1

Read critically the story and answer the questions that follow:

Rukundo read a book about major urban centres in Africa. He learned that these cities feature crowded streets, slums, expensive housing, tall buildings, heavy traffic, and diverse human activities. They host businesses, schools, offices, industries, recreational facilities, and tourist attractions, with busy, well-connected roads lined with trees and electrical poles, unlike rural areas.

- a) Explain why traffic congestion occurs in major urban centres.
- b) Suggest two ways to improve mobility in crowded African cities.

❖ **Characteristics of major urban centres in Africa are the following:**

- They have large population and high population densities
- There are settlements ranging from slums to high-cost housing units
- Development of very tall buildings and high concentration of buildings
- The mobility of people in urban centres is very high.
- High traffic congestion occurs during peak rushing hours daily.
- Secondary and tertiary activities dominate urban economic life extensively.
- Residents experience extreme variations in income levels across cities.
- A high number of roads converge in city centres.
- People in urban centres perform many different daily functions.
- Roads are lined with trees and electrical poles throughout.

❖ Functions of urban centres in Africa include:

- Urban centres perform commercial functions because they host many businesses.
- Administrative functions exist due to offices located within urban centres.
- Cities provide transport and communication services for residents and visitors.
- Educational functions include schools, colleges, and universities within urban centres.
- Residential functions exist because people live in urban areas permanently.
- Industrial functions include manufacturing and production activities in urban centres.
- Recreational functions provide entertainment and leisure opportunities for city residents.
- Touristic functions attract visitors to urban centres for tourism activities.

Application activity 8.1

1. Which of the following is a type of settlement commonly found in African urban centres?
A. Rural villages B. Slums C. Forest clearings D. Nomadic camps
2. Which of these best describes the importance of industrial functions in city development?
A. Industries provide goods, employment, and contribute to economic growth
B. Industries only increase traffic congestion
C. Industries reduce urban population
D. Industries have no effect on urban development
3. Which of the following explains the importance of commercial functions in urban centres?
A. Commercial activities offer services, trade, and attract investments
B. Commercial activities reduce the number of roads
C. Commercial activities discourage urban migration
D. Commercial activities only exist in rural areas

8.2. Causes, effects and possible solutions of rapid urban growth in Africa

Learning activity 8.2

Urban centres in Africa face rapid growth, diverse functions, and various challenges. Think critically, and then reflect on your understanding to answer the following questions:

1. Which factor primarily contributes to rapid growth in African urban centres?
 - A. Rural-urban migration and natural population increase
 - B. Declining birth rates
 - C. Rural settlement schemes
 - D. Decrease in industrial activities
2. What is a major challenge caused by high population density in cities?
 - A. Improved road networks
 - B. Traffic congestion and pressure on housing
 - C. Expansion of green spaces
 - D. Increase in rural population
3. Reflect on how industrial, educational, and transport facilities influence urbanisation.
4. Reflect on how tourism and recreational functions contribute to city development.

❖ Causes responsible for rapid urban growth in Africa include:

- Rural-urban migration occurs as people search for better employment.
- Natural population increase happens due to improved health and nutrition.
- High demand for goods and services attracts more people.
- Industrial development draws job seekers into urban areas rapidly.
- Administrative activities encourage people to live near government offices.
- Transport and communication facilities attract residents and businesses efficiently.
- Mineral deposits in areas attract workers and economic activities.
- Availability of educational facilities encourages families to settle permanently.
- Agricultural activities in urban fringes support livelihoods and attract migration.

❖ Problems of urban centres in Africa include:

- Unemployment rises because many job seekers cannot find work.
- Inadequate social amenities in rural areas drive urban migration.
- Traffic congestion increases during peak hours daily in cities.
- Environmental degradation results from pollution, poor sanitation, and garbage.
- Poor planning creates insufficient space and substandard infrastructure problems.

- High crime rates threaten safety through banditry, drugs, terrorism.
- Street families emerge due to poverty and lack of housing.
- Inadequate housing leads to slum development in urban areas

❖ Possible solutions to urban centres in Africa

- Establish industries in rural areas to create more job opportunities.
- Develop education systems that generate employment for urban and rural residents.
- Improve social amenities in rural areas to reduce urban migration.
- Construct bypasses, tunnels, flyovers, or underpasses to ease traffic congestion.
- Expand road networks to improve traffic flow within urban centres.
- Provide dedicated paths for cyclists and pedestrians to enhance mobility.
- Enforce environmental laws to reduce pollution and manage urban waste.
- Recycle garbage for sustainability and future resource utilization in cities.
- Involve citizens in clean-up campaigns to maintain urban cleanliness.
- Restrict unplanned developments through strict urban planning regulations effectively.
- Build skyscrapers to solve space limitations in densely populated areas.
- Apply urban spatial master plans for proper city organization and management.

Application activity 8.2

1. Why do slums and inadequate housing develop in African urban centres despite development?
 - A. Poor urban planning and rapid population growth
 - B. Abundant green spaces and parks
 - C. Strict zoning laws
 - D. Low rural-urban migration
2. How can citizens actively help solve urban environmental problems?
 - A. By littering in public spaces
 - B. Participating in waste management, tree planting, and reporting pollution
 - C. Ignoring environmental regulations
 - D. Increasing traffic in congested areas
3. In your opinion, which solution is most effective for reducing unemployment in urban centres? Explain.
4. How do urban centres in Africa compare to rural areas in terms of opportunities and challenges?

8.3. Case studies: Cairo and Johannesburg

Learning activity 8.3

Mahoro and Ishimwe visited Cairo and Johannesburg to study urbanisation. They observed crowded streets, tall buildings, traffic congestion, industries, schools, hospitals, and cultural sites. Both cities faced challenges like high population density, pollution, unemployment, inadequate housing, and crime. They reflected on growth drivers, including rural-urban migration, economic opportunities, education, transport, and tourism, highlighting urbanisation's impact on African city development.

1. Explain how rural-urban migration contributed to the growth of both cities.
2. Suggest ways Johannesburg could reduce traffic congestion and improve mobility.

1. Cairo

It is the capital city and the largest city of Egypt. It is the 15th largest city in the world.

❖ Characteristics of Cairo

- It is the capital city and the largest city in Egypt
- It is located near River Nile and the Pyramids of Giza in the North of Egypt
- It is one of the largest cities in Africa with 528km²
- It is one of the most populated cities in the world with more than 23,074,000 in 2025.

❖ Functions of Cairo

- **Administrative centre:** Cairo is the political capital of Egypt. It is also the headquarters of the Arab League in Egypt.
- **Trade centre:** There are many trading activities carried out in Cairo. For example, trade in horticultural crops, textiles, livestock products and metal products.
- **Tourist centre:** The city has a number of tourist attractions. Examples are the Egyptian Museum, Cairo Tower, Old Cairo and Tahrir Square.
- **Industrial centre:** Major industries include food processing, iron and steel production and textile industries.
- **Transport and communication centre:** Cairo has an extensive road network, rail system, subway system and maritime services. The extensive road network connects Cairo to other major cities and villages in Egypt. Cairo also has many advanced communication facilities.
- **Medical centre:** Cairo offers high level of medical facilities at relatively low prices.

Cultural and educational center: It offers educational services for Egypt and the region. World's 2nd oldest institute of higher learning, Al-Azhar University, is located in Cairo.

❖ Factors, problems and solutions for the development of Cairo

Factors for growth of Cairo	Problems Cairo is facing	Solutions/measures
• Rural-urban migration • Availability of educational facilities • Natural increase in population • Availability of transport and communication facilities • Investment opportunities • Influx of tourists • Administrative activities.	• High levels of environmental pollution • Unemployment due to many people • Traffic congestion • Inadequate social amenities • High crime rates including terrorism • Inadequate housing facilities	• Enforce environmental protection laws • Create more job opportunities • Expand roads, reduce congestion • Improve social amenities • Increase security measures • Build affordable housing

2. Johannesburg city

Johannesburg is the largest city in South Africa and is one of the 50 largest urban areas in the world. Johannesburg, South Africa's biggest city and capital of Gauteng province, began as a 19th century gold-mining settlement.

❖ Characteristics of Johannesburg include the following:

- Johannesburg is the largest city in South Africa and a major urban, commercial, and industrial center.
- It is located on the Highveld plateau and is known for its modern infrastructure, high population density, and multicultural society.
- The city developed historically as a mining settlement after the discovery of gold and has evolved into a financial and service-oriented metropolis.

❖ Factors that contributed to its Growth include the following:

- Gold discovery (1886): The discovery of gold on the Witwatersrand attracted thousands of miners, investors, and workers, rapidly expanding the city.
- Industrialization: Growth of manufacturing, construction, and processing industries stimulated employment and urban expansion.
- Transport and infrastructure: Development of roads, rail networks, and later an international airport positioned Johannesburg as a major regional hub.
- Economic opportunities: Availability of jobs in finance, trade, telecommunications, and services attracted internal and international migrants.
- Urban modernization: Government and private investments in housing, education, and business districts supported continuous growth.

❖ Functions of Johannesburg include the following:

- Commercial and financial center: Hosts the Johannesburg Stock Exchange (JSE), banks, corporate headquarters, and major industries.
- Industrial hub: Produces steel, machinery, chemicals, textiles, and processed goods.
- Transportation hub: Acts as a major road, rail, and air link within South Africa and the wider region.
- Educational and research centre: Home to top universities and research institutions, such as the University of Johannesburg and Wits University.
- Tourism and cultural centre: Offers museums, historical sites, cultural events, and commercial attractions.

4. Challenges facing Johannesburg include the following:

- High crime rates: Social inequality and unemployment contribute to significant security concerns.
- Urban inequality and poverty: Large informal settlements lack adequate housing, services, and employment opportunities.
- Traffic congestion and transport inefficiencies: Rapid population growth strains transport networks.
- Environmental pollution: Air pollution, waste management issues, and water contamination affect public health.
- Service delivery pressures: Demand for electricity, clean water, housing, and sanitation often exceeds supply.

Application activity 8.3

Match the items in Column A with the correct description in Column B.

Column A	Column B
i. Cairo's tourist attractions	a) Discovery of gold on the Witwatersrand
ii. Factor that contributed to Johannesburg's growth in 1886	b) Air, steel, machinery, chemicals, and textile industries
iii. Major challenge faced by Cairo due to population growth	c) Egyptian Museum, Cairo Tower, Old Cairo, Tahrir Square
iv. Industrial function of Johannesburg	d) Traffic congestion and inadequate social amenities

End of unit assessment

❖ Knowledge and understanding:

1. Identify the main characteristics of urban centres in Africa, such as population size, economic activities, infrastructure, and social services.

❖ Skills:

2. Evaluate the major problems faced by urban centres in Africa.
3. Assess the impact of urban settlements on development in Africa.

❖ Attitudes and values

4. As a responsible citizen, suggest ways individuals, communities, and governments can support initiatives aimed at reducing the negative impacts of rapid urban growth in Africa.

UNIT 9

AGRICULTURAL SYSTEMS IN AFRICA



Key Unit Competence:

Evaluate the impacts of various agricultural systems on sustainable development in Africa.

Learning objectives

At the end of this unit you should be able to:

Knowledge and understanding

1

- Identify the different methods of subsistence farming in Africa
- Explain the methods of modern agriculture in Africa
- Outline the different types of livestock farming in Africa
- Identify the impact of agricultural activities in Africa.

Skills

2

- Examine the characteristics and practices of agricultural systems in Africa
- Investigate various modern agricultural methods currently used in Africa
- Examine the characteristics and methods of each type of livestock farming in Africa
- Evaluate the impact of agricultural activities in Africa

Attitudes and values

3

- Appreciate the importance of agricultural modernisation to the development of Africa.
- Show respect for strategies and innovations designed to solve farming problems

9.0. Introduction

Agriculture in Africa, involving crop growing and animal rearing, provides food, employment, and industrial raw materials. Practiced through traditional (subsistence) and modern farming, it uses different methods and tools, playing a key role in development and food security.

Introductory activity

Observe pictures showing subsistence farming, modern mechanised farming, and livestock rearing in different parts of Africa, then discuss differences in tools used, scale of production, and outputs and answer the questions that follow:



1. Identify the different types of crops and livestock shown in the images.
2. Explain the difference between traditional/subsistence farming and modern/commercial farming as illustrated in the photos.
3. Suggest how modern tools and machines (like the tractor in the photo) improve agricultural productivity.
4. Examine the benefits of mixed farming (crops and livestock) as shown in the images.

9.1. Subsistence farming in Africa

Learning activity 9.1

1. One of the main methods of subsistence farming in Africa is _____.
2. Pastoralism, ranching, and mixed farming are types of _____ farming.
3. Soil erosion, deforestation, and water pollution are examples of _____ impacts of agricultural activities.
4. Match the agricultural system with its description.

Column A	Column B
i. Subsistence farming	a) Combining crop cultivation and livestock rearing on the same farm
ii. Pastoralism	b) Farming primarily to feed the farmer's family, often on small plots
iii. Mixed farming	c) Raising livestock by moving herds to find pasture

1. Shifting cultivation/ slash and burn agriculture

This traditional subsistence farming, called shifting cultivation, uses fertile land temporarily, then moves to new virgin areas.



Examples of grown crops are: maize, yams, potatoes, beans, etc.

Characteristics	Advantages	Disadvantages
– Clearing of land using pangs, hoes and axes. – Burning of bushes to clear the land – Moving of farmers to the virgin land when the previous is exhausted – Mixture of crops in the same piece of land – Use of traditional techniques of farming.	– Source of food – Easy to control weeds through burning – It uses less labour – It generates jobs	– Destruction of organic matter in the soil due to burning – Destruction of environment through cutting and burning of trees and bushes – Degradation of vast areas of forested land

2. Small holder farming

This is the growing of crops on small plots of land.

Characteristics	Problems	Solutions
– Both subsistence and cash crops are grown – Land is subdivided into small uneconomical plots for farming – Simple tools are used – Mulching is a common practice and crop rotation is practiced to preserve soil fertility	– Pests and diseases destroy crops – Soil exhaustion due to its over cultivation – Decrease of production – Use of simple tools which may lead to low production – Shortage of land for agriculture due to the need of land for settlement as a result of population increase	– Application of pesticides to control pests and diseases – Rotational farming to let the land regain fertility – Using fertilizers to increase the productivity of the land – Land use consolidation and use of modern tools of farming – Setting aside settlement areas, while leaving out specified areas for farming

3. Rotational bush fallowing

Rotational bush fallowing is where an abandoned piece of land is left to regain its fertility before reverting back to it.

❖ Characteristics of rotational bush fallowing:

- It is practiced in areas with permanent settlements, where villages have their own land to till
- There is use of fire to clear the land and use of simple tools like hoes
- The land is never bare because each village has its own land which is cultivated according to a fixed rotation
- The time of cultivation of the land may vary from 3 to 5 years without rest.

Application activity 9.1

1. A major positive impact of agriculture in Africa includes:
a) Soil erosion b) Increased food supply
c) Water pollution d) Deforestation
2. Which type of livestock farming involves moving animals seasonally to find fresh pastures?
a) Ranching b) Pastoralism c) Mixed farming d) Plantation farming
3. One way to improve productivity in subsistence farming is to:
a) Reduce crop diversity b) Use improved seeds and fertilizers
c) Avoid irrigation d) Decrease farm size
4. Describe different methods of subsistence farming practiced in Africa and suggest at least two strategies to improve productivity

9.2. Modern agriculture in Africa

Learning activity 9.2

Choose the appropriate concepts or group of concepts from the following words in brackets (Plantation farming, Pastoralism, Mechanised farming and Mixed farming) and complete the statement below

1. _____ means raising livestock by moving herds to find pasture
2. _____ refers to the system using machinery to plough, plant, and harvest crops efficiently
3. ____ means combining crop cultivation and livestock rearing on the same farm
4. _____ means large-scale farming using commercial crops for export or sale

1. Modern agriculture in Africa

Modern methods of agriculture have been introduced in Africa by colonialists. They include plantation agriculture, green revolution and irrigation farming.

a) Plantation agriculture/extensive monoculture or large-scale farming

Plantation agriculture is the growing of one/single crop on a large scale for commercial purposes. The main crops grown are **coffee, tea, cocoa, rubber, oil palm, wheat, cotton and sugarcane**.

Characteristics/features of plantation	Factors favouring plantation
– Crops grown on very large-scale farms – Single crop grown using monoculture system – Requires many workers for farm operation – Needs large capital for investment and machinery – Owned by government, companies, cooperatives, or foreigners – Includes demonstration farms for farmer trainings – Uses scientific farming methods and fertilizers – Mainly designed for export markets – Produce processed before exportation – High production due to extensive cultivated land	– High and reliable rainfall – Presence of fertile and well-drained soils – Presence of water for irrigation – Presence of cheap labour – Presence of market for the crops grown – Favourable government policies – Presence of well-developed transport and communication networks. – Presence of large capital for investment

❖ Merits (advantages) and demerits of plantation agriculture

Merits (advantages)	Demerits (disadvantages)
– High production using modern farming methods – High quality produce from efficient techniques – Farmers gain skills through regular training – Provides employment opportunities to many people – Generates government revenue – Generates government revenue through taxation	– Prices fluctuate on world markets – Causes soil exhaustion through monoculture – Diseases can destroy large farm areas – Profits repatriated to foreign owners

- | | |
|--|---|
| <ul style="list-style-type: none"> – Promotes infrastructure development in surrounding areas – Earns foreign exchange through exports – Encourages development of processing industries – Minimizes wastage through recycling by-products – Supports out-growers with inputs | <ul style="list-style-type: none"> – Food crops neglected causing famine – Requires large capital investment – Crops take long to mature |
|--|---|

2. Green revolution

Green revolution is a type of agricultural technique aimed at increasing agricultural production.

❖ Main features (characteristics) of green revolution

- There is constant research on the improvement of the crops
- There is mixture of crops
- The land is always cultivated. It is not left for a certain time to regain fertility
- There is constant use of fertilizers and other chemicals
- There is use of pesticides and insecticides to control pests and insects
- There is constant training and supervision by experts on the farms
- There is high production due to use of scientific methods of farming
- There is use of improved seeds that can mature faster and yield more.
- There is use of irrigation farming
- There is use of farm machinery to increase in mass production.

❖ Merits (advantages) and demerits of green revolution

Merits (advantages)	Demerits (Disadvantages)
– There is increased production in quantity. – There is improved quality of the crop yields. – Self-sufficiency in food production – Sources of income to farmers – Improved standards of living of farmers – There is an increase of foreign exchange from exports – Development of food processing industries – Source of employment opportunities.	– Fall in world market price due to over-production – It is expensive to be done by small farmers – It is beneficial to rich people with large capital to invest – Unemployment due to use of mechanization – It is difficult to be carried out on small farms

3. Irrigation farming

Irrigation farming is an agricultural practice involving the artificial provisions of water for plant growth either permanently or temporary.

Types of irrigation

There are four main types of irrigation: **canal irrigation, Sprinkler irrigation, Drip trickle irrigation, and Bucket irrigation.**



- (i) **Canal irrigation:** This farming practice is carried out on gently sloping land where canals are constructed from rivers or lakes to supply water to farms. Its main advantage is effective control of water distribution, but it can cause waterlogging and soil salinity in clay soils and requires large amounts of water to operate efficiently
- (ii) **Sprinkler irrigation:** In sprinkler irrigation, water is pumped through pipes to sprinklers that spray it over crops like rainfall. Sprinklers may be fixed on or above the ground. This method allows farmers to control irrigated areas easily, but it is unsuitable for large plantations and during flowering of some crops.
- (iii) **Drip irrigation:** In this irrigation method, water is supplied through pipes with small holes that release drops directly at plant roots at intervals. It uses little water efficiently for many crops, but is costly to install, making it unsuitable for large farms
- (iv) **Bucket irrigation:** In bucket irrigation, water is carried using buckets or watering cans to irrigate each plant individually. This method is cheap and suitable for small-scale farmers, but it is very time-consuming and only practical for small gardens located near water sources.

Advantages of irrigation	Disadvantages of irrigation
– All types of crops can be grown throughout the year – Employment opportunities for many people – Source of foreign exchange when the production is exported – Improvement of infrastructures – Growth of market centres and collecting centres – Increase food security in country using it	– Increase of soil salinity due to high rate of evaporation – Water-logging which leads to destruction of crops in the farm – Breeding places for disease vectors such as mosquitoes – It may lead to soil and water pollution – It causes siltation of rivers

Application activity a.2

1. Explain the main methods of modern agriculture in Africa and discuss how these methods contribute to food security and economic growth
2. Analyse the impact of agricultural activities in Africa, discussing both positive effects and negative effects.
3. Evaluate how agricultural modernization can support Africa's development and suggest practical ways farmers and communities can adopt innovations to solve farming problems.

9.3. Livestock farming in Africa

Learning activity 9.3

Juma is a pastoralist from northern Tanzania who rears cows and sheep. Because rainfall is low and unreliable, he moves seasonally with his animals in search of water and fresh pasture. His livestock depend on natural grasslands and provide his family with milk, meat, and income. Juma faces challenges such as drought, diseases, overgrazing, and long movements, but pastoralism remains his cultural way of life.

1. What type of farming does Juma practice in northern Tanzania?
2. Why does Juma move with his cows and sheep from place to place?
3. Nomadic pastoralists mainly depend on _____ pastures for their livestock.
4. Livestock provides pastoralists with milk, meat, and _____
5. How does communal land ownership support nomadic pastoralism in Tanzania?

1. Nomadic pastoralism

This is a form of subsistence farming where cattle owners (pastoralists) walk long distances with their herds in search of water and fresh pastures.

Examples: Fulani of west Africa, Masai of East Africa, Turkana of North Kenya, Tuarg of the Sahara and Karamajong of North-Eastern Uganda.

a) Characteristics of nomadic pastoralism



Indigenous cattle breeds reared by pastoralists.

- They occupy areas of low and unreliable rainfall
- They keep cattle, sheep, goats and camels
- They keep traditional (indigenous) breeds of animals which give low and poor quality of production
- The animals depend on the natural pastures only
- Large number of animals are kept
- Animals are mainly kept for subsistence
- The livestock is owned by individuals; but the land belongs to the community
- Communal grazing is carried out
- Seasonal movements are necessary to ensure enough water and pasture
- The animals provide the basic food in the form of milk and blood.
- Overgrazing resulting from overstocking is done
- There is the burning of grass in the dry season to get fresh pastures.

b) Problems facing and their possible solutions the nomadic pastoralists

Physical Problems	Solutions
– Low rainfall causes water shortages, poor pastures. – Poor pasture quality deteriorates livestock health. – Long movements cause livestock weight loss, deaths.	– Construct boreholes to access underground water. – Build valley dams to trap rainwater. – Introduce ranching and paddocks for grazing.

– Animals affected by insects, pests, diseases. – Traditional breeds produce low-quality livestock products. – Livestock attacked by wild animal predators. – Overstocking causes overgrazing, soil erosion. – Water holes dry from overcrowded animals. – Communal grazing spreads livestock diseases quickly. – Burning grass deteriorates soil, harms livestock. – Neglect of facilities worsens pastoralist challenges. – Conservative culture limits adoption of improvements. – Lack of enough capital to improve their livestock practices	– Grow grain crops to supplement feed. – Establish milk collection centres for commercialization. – Control pests and diseases with vaccinations. – Quarantine livestock to prevent disease spread. – Educate pastoralists for improved livestock management. – Encourage adopting modern pastoralist farming practices
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c) Advantages (merits) and disadvantages (demerits)

Advantages (Merits)	Disadvantages/effects on environment
– Animals survive in harsh, low-rainfall areas – Seasonal movement ensures water and pasture access – Communal land allows free grazing and mobility – Use of local animal breeds adapted to climate – Local herbs used to treat animals naturally – Large herds ensure survival during droughts – Provides food: milk, blood, meat – Enables trade: butter, animal products – Source of heat and energy (cow dung)	– Overgrazing leads to soil erosion – Burning grasses causes low-nutrient, fire-resistant growth – Soil erosion worsens from burning practices – Communal grazing spreads diseases among livestock – Wells and boreholes dry due to overcongestion – Low production quality from local breeds

- Supplies clothing from skins and hides
- Contributes wealth, prestige, and social power
- Source of dowry and manure for crops

2. Ranching

This is the rearing of livestock on an extensive scale for commercial purposes. The reared animals include **cattle, sheep, goats, horses, or camels**.

a) Characteristics of Ranching in Africa

- Grazing animals in paddocks
- Animals are provided with veterinary services, water and salt
- Giving extra food to livestock during the dry season
- Ranches are owned by individuals
- Much of the pasture in the ranches is natural grasses.
- Rainfall is too low to support arable farming
- Livestock in ranches are of high quality
- Animals are reared for commercial purposes
- High quality of animals are reared
- There is regular disease control through spraying, dipping, etc.
- Limited movement of animals
- They are kept in permanent pastures.

3. Dairy farming in Africa

Dairy farming in Africa involves rearing cattle for milk, which can be consumed directly or processed into products like yoghurt, butter, and ghee; it is mainly practiced in South Africa and Kenya and is characterized by highly intensive production methods.

❖ Characteristics of dairy farming

- The main objective is to raise cattle for milk
- The animals are raised for commercial purposes
- Selected cattle breeds are kept yielding a lot of milk. Examples: Ayrshire, Jersey, Guernsey, Friesian, etc.
- Rich and nutritious grasses are grown to supplement the natural pastures
- There is the control of diseases and scientific management of farms
- Capital intensive techniques of production are used
- Returns are very high
- It requires well-developed transport since milk is perishable
- It is done in areas with cool and wet climatic conditions.
- It is mostly adapted in highland areas.

Application Activity 9.3

1. What is the meaning of nomadic pastoralism, livestock ranching, and dairy farming, giving one example for each in Africa.
2. Describe the main characteristics of nomadic pastoralism and explain why it is practiced in areas of low rainfall.
3. Assess the problems and environmental effects of nomadic pastoralism and propose sustainable solutions to improve livestock production in Africa.

9.4. The impact of various agricultural activities on Africa's development.

Learning activity 9.4

Musoni, a livestock farmer in Nyagatare, manages cows, goats, and sheep. He sells milk, meat, and skins in local markets, earning income that supports his family and farm workers. His animals' manure fertilizes his crops, improving harvests. The farm also provides raw materials for local industries and contributes to government revenue through taxes, while promoting trade across the region.

1. In what ways does livestock farming contribute to government revenue?
2. Explain how livestock farming creates employment opportunities in Africa.
3. Reflect on how livestock farming helps farmers save and invest for the future.

❖ Impact of various agricultural activities on Africa's development

- Source of food and livelihood from both crops grown and animals reared
- Source of government revenue through taxation
- Source of feeds for livestock
- Promotion of international trade (exports and imports)
- Creation of employment opportunities to many people
- Source of raw materials for processing industries
- Supports development of infrastructures
- Source of savings and investments
- Source of manure to be used in crop cultivation.

Application Activity 9.4

1. Why is livestock farming important for rural development in Africa?
2. How does integrating crops and livestock benefit a farm's productivity?

9.5. Case studies

Learning activity 9.5

Mohammad is a farmer in Egypt for diverse agricultural activities including cotton, cocoa, rubber, and dairy farming. In the Nile Valley, he grows cotton through irrigation, earning income and supplying raw materials to textile industries. He wishes to grow cocoa and rubber in West Africa to provide him with steady income, employment, and export opportunities. He also owned in Kenya, a dairy farm that produces a lot off milk for local consumption and processing industries. Despite challenges like pests, diseases, fluctuating prices, and poor infrastructure, Mohammad adopts modern techniques, veterinary care, and proper storage to increase productivity, support his family, and contribute to the regional economy.

1. How does cotton farming benefit farmers and the economy?
2. What conditions are necessary for successful cocoa and rubber cultivation?
3. Explain the main problems facing cash crop farmers in Africa.
4. How does dairy farming contribute to income and employment in Kenya?

1. Case study of cotton growing in the lower Nile valley

River Nile flows from Lake Victoria northwards through Uganda, South Sudan and Egypt into Mediterranean Sea.

Cotton is grown through irrigation in Egypt and Sudan. Egypt is the world's major producer of long staple cotton known as ***Sakellarides (Sakel)***



cotton grown in Nile River Valley of Egypt

a) Characteristics of cotton growing in the Nile Region

Cotton is usually grown in February, regularly watered and weeded, controlled by the government, produces high yields per unit area, and is harvested in August in Upper Egypt and September in Lower Egypt.

❖ Benefits (merits) of Cotton growing in the Lower Nile

Cotton growing provides employment and income to farmers, improves living standards and infrastructure, supplies raw materials to textile industries, earns foreign exchange for Egypt, and ensures productive use of land receiving little rainfall.

b) Problems of growing cotton in the Lower Nile

Pests and diseases destroy crops and increase control costs, while monoculture exhausts soil, excessive irrigation raises salinity, and the shortage of fertile soils limits cotton production and agricultural sustainability.

II. Cocoa growing in West Africa



Cocoa plantation, its fruits and beans

Cocoa originates from Brazil. It was firstly introduced in Africa on the Island of Sao Tome, then in Nigeria and finally in Ghana. The major countries growing Cocoa in Africa are Ivory Coast, Ghana, Nigeria, and Cameroun. Other countries in the other continents are shown on the map below:



- **Conditions:** Cocoa requires well-distributed heavy rainfall, high temperatures, fertile deep soils, high flood-free land, and tall trees for protection.
- **Uses:** Cocoa is used for beverages, chocolate, cocoa butter, oil, medicine, and cakes, providing versatile industrial and culinary products.

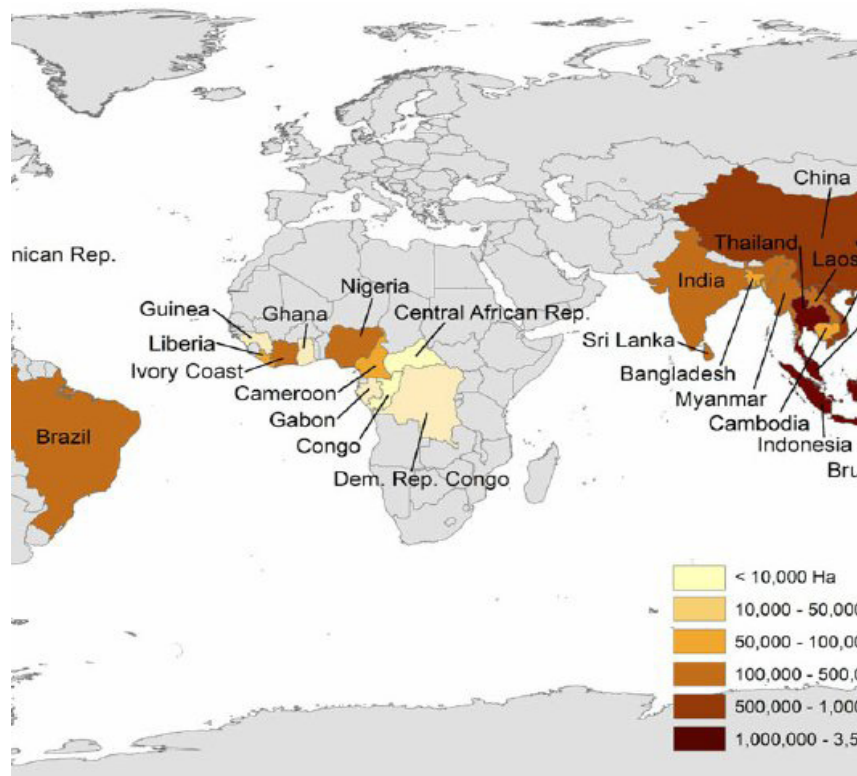
- **Problems:** Cocoa farming faces poor roads, tropical diseases, fluctuating market prices, and plant diseases like Swollen Shoot, reducing productivity.
- **Solutions:** Improving roads, farmer education, health facilities, government support, local use, and pest control can solve cocoa farming challenges.

III. Rubber growing in West Africa



Rubber plantation and its harvesting

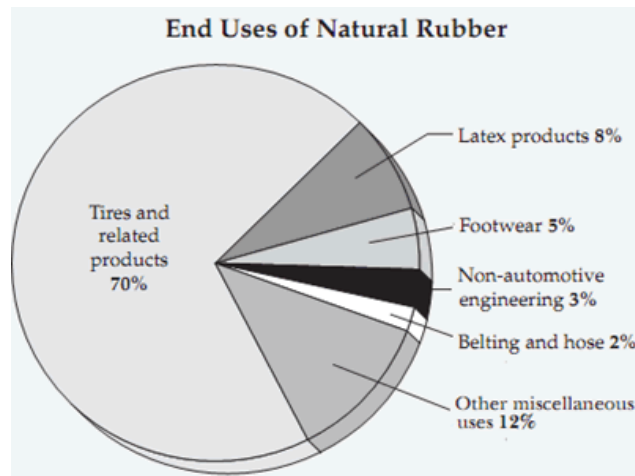
Rubber is a tree crop grown mainly in Liberia, Ivory Coast, Nigeria, Cameroun and DRC.



❖ The major producing countries of Rubber in Africa and the World

a) Uses of rubber:

Manufacture of erasers, soles of shoes, sandals, tires of locomotives and rubber cups and soap



❖ Uses of Rubber

b) Processing of rubber:

A latex tapper collects latex, takes it to factories, mixes it with water and acetic acid, rolls it into sheets, dries them in smoky sheds, producing crepe rubber.

c) Necessary conditions for growing rubber:

Rubber grows best in areas with heavy, well-distributed rainfall of about 1,750–2,500 mm per year, consistently high temperatures above 21 °C, and deep, fertile, well-drained soils.

d) Benefits of rubber to farmers:

Rubber farming provides employment opportunities and a steady income to farmers, which improves their standards of living and contributes to the development of infrastructure.

e) Problems facing rubber growing in west Africa:

Rubber production faces challenges such as competition from synthetic rubber, fluctuations in world market prices that discourage farmers, and the labor-intensive nature of harvesting, which is done manually.

f) Possible solutions to the problems of rubber growing in West Africa.

Challenges can be reduced by using modern farming methods, diversifying farmers' income sources, adopting improved harvesting techniques, and strengthening agricultural extension services to boost productivity, efficiency, and sustainable livelihoods.

IV. Case study of dairy farming in Kenya

There are two types of dairy farming in Kenya: *commercial dairy farming* and *domestic dairy farming*. The major areas with dairy farming in Kenya include: *Kikuyu, Nakuru, Kericho, Bungoma, Kakamega and Kiisi*.

a) Conditions and benefits favouring dairy farming in Kenya include:

Conditions	Benefits
– Adequate rainfall 1000–1500mm for pasture growth – Cool temperatures support exotic cattle survival – Fertile soils grow nutritious grasses – High humidity conditions – Exotic breeds and scientific livestock management – Availability of capital for investment – Skilled and unskilled manpower available – Demonstration farms teach better farming techniques – Experienced veterinarians present – High technology like electric milking machines – Large market for milk products – Well-developed transport routes	– Provides milk products to population – Source of income for farmers – Creates employment opportunities – Earns foreign exchange from exports – Develops milk processing industries

b) Problems and possible solutions of dairy farming in Kenya include:

Problems	Possible solutions
– Shortage of land – Occurrence of droughts – Poor transport during rainy season – Expensive cow maintenance inputs – Pests and disease – Inadequate farmer training	– Store hay and dry fodder for dry season – Provide veterinary services to farmers – Offer cooling facilities for market transport – Help control animal diseases and pests – Train farmers to improve production

Application activity 9.5

1. Suggest solutions farmers can use to overcome agricultural challenges.
2. How do modern techniques and veterinary care improve livestock productivity?
3. In what ways do these agricultural activities support exports and trade?
4. Reflect on the importance of infrastructure and transport in farming success

9.5. End of unit assessment

1. Project activity

1.1. Project Title: Improving subsistence farming in our community

1.2. Context

In many parts of Rwanda, including areas around Kigali, subsistence farming is the main source of food and income for families. However, farmers often face challenges such as poor soil fertility, small land size, unpredictable rainfall, and limited use of modern techniques. These challenges can lead to low crop production and food insecurity. By understanding these problems, students can help suggest and apply simple solutions to improve farming practices in their community.

1.3. Challenge

How can we take practical action to improve subsistence farming and increase food production in our community?

1.4. Task

- Investigate the main challenges faced by subsistence farmers in your area.
- Design a simple and practical solution to improve farming practices.
- Implement at least one real farming improvement activity.
- Share your results and encourage others to adopt better practices.

1.5. Project outputs (What you will produce)

A. Practical action (main output – required)

Carry out at least one real farming improvement activity, such as:

- Creating a small demonstration garden using improved practices.
- Applying organic manure or composting techniques.
- Practicing mulching, irrigation, or crop spacing.
- Introducing mixed cropping or crop rotation.
- Demonstrating simple water conservation methods.

B. Evidence of your work

- Photos, drawings, or before-and-after descriptions.
- A simple record of what you did (materials, steps, location).

C. Supporting materials

- A short report (1–2 pages) explaining the problem, action taken, and results.
- A simple sketch map or diagram of the farming area or demonstration plot.
- A poster or chart showing improved farming techniques

D. Communication

- A short oral presentation
- (Strongly encouraged) A community awareness product (e.g., advice to farmers, flyers, or demonstration session)

1.6. Constraints

- Use locally available and low-cost materials.
- Focus on small-scale but practical activities (school garden or nearby area).
- Ensure activities are safe and environmentally friendly
- Work within the given timeline.
- Solutions must be realistic and applicable in your community.

1.7. Guiding questions

- What challenges do subsistence farmers face in our community?
- How do these challenges affect food production?
- What simple and affordable methods can improve farming?
- How effective is our solution in improving farming practices?
- How can we encourage farmers to adopt these improvements?

2. Assessment questions

❖ Knowledge and understanding:

1. Identify a method of subsistence farming practiced in Africa:
A. Plantation agriculture
B. Shifting cultivation
C. Mechanised irrigation farming
D. Market gardening
2. Fill in the blanks (shifting cultivation, smallholder farming, rotational bush fallowing)
3. Three main methods of subsistence farming in Africa are __, ____, and __.

❖ Skills

4. Which of the following shows how modern agricultural methods improve food production?
A. Using simple hand tools only
B. Practising seasonal migration
C. Using tractors, irrigation, and improved seeds
D. Leaving fields fallow for long periods

5. Compare shifting cultivation and bush fallowing in terms of land use and productivity.

❖ **Attitudes & Values**

6. Suggest one strategy that farmers can adopt to reduce environmental damage while maintaining food production.
7. Explain why it is important for small-scale farmers to combine crops and livestock in their farms.



Key Unit Competence:

Examine the impact of forestry and forest exploitation on sustainable development in Africaa.

Learning objectives

At the end of this unit you should be able to:

Knowledge and understanding

1

- Outline the factors favouring forest growth and exploitation in Africa
- Identify the methods, factors, challenges and solutions of forest exploitation in Africa
- Explain the causes, effects and measures to control over-exploitation of forests in Africa

Skills

2

- Propose the possible solutions to problems affecting forest exploitation in Africa
- Explain the causes and effects of over-exploitation of forests in Africa
- Apply acquired knowledge to suggest measures to prevent over-exploitation of forests in Africa
- Evaluate the impact of forestry on sustainable development in Africa

Attitudes and values

3

- Appreciate the importance of forests and forest conservation in Africa.
- Show interest in conserving forest regions threatened by overexploitation

10.0. Introduction

Forestry involves planting, exploiting, and conserving forests, including silviculture and agroforestry, to ensure ecological balance, sustainable resource use, and environmental protection.

Introductory activity

During a field visit to Nyungwe Forest, students observe diverse trees, wildlife, and forest management practices. They learn how forests provide timber, medicine, and employment, while noting threats from over-exploitation, deforestation, and the importance of conservation for sustainable development.

1. List three factors favouring forest growth in Africa.
2. Explain the importance of forests to African communities.
3. Analyse problems affecting forest exploitation in Africa.

10.1. Forest exploitation in Africa: Methods and factors.

Learning activity 10.1

Rugwiro, a young entrepreneur in East Africa, runs a small timber business. He surveys forest areas before cutting trees using powered saws, removes branches, and trims trunks into logs. The logs are transported by trucks to riversides, then shipped to ports for export. Rugwiro relies on the abundant rainfall and warm climate that support dense forests, as well as nearby rivers for transporting logs. The high demand for timber allows him to earn a living with minimal specialized skills.

1. Fill in the blanks by extraction/harvesting, warm or rainfall
 - a) Forest exploitation refers to the _____ of forest resources such as timber for economic purposes.
 - b) In Equatorial Africa, forest exploitation is supported by abundant ____ and a ____ climate that favor dense forests.
2. Which of the following is the main method of forest exploitation described in the scenario?
 - A. Shifting cultivation
 - B. Selective cutting using powered saws
 - C. Mining under forest cover
 - D. Plantation agriculture
3. Using the case of Rugwiro's timber business, explain how physical and human factors contribute to the success of forest exploitation in Equatorial Africa.

1. Methods of forest exploitation in Africa

Forest exploitation is the making use of forest resources with a view of making money or earning a livelihood. It involves several key practices like **logging** (cutting trees for timber production), **lumbering** (processing of logs into usable wood), **agro-forestry** (combines crop farming with tree cultivation), **research** (study) and **silviculture** (managing forests to ensure sustainable growth).



❖ The methods of forest exploitation in Africa include:

1. Surveying the area: Identify trees and plan harvesting.
2. Establishing transportation routes: Determine paths for moving logs efficiently.
3. Cutting trees: Using powered saws to fell trees.
4. Removing branches and cutting trunks: Prepare logs into suitable lengths.
5. Transporting logs to collection centres: Gather logs before moving them further.
6. Moving logs to rivers or ports: Use tractors or trucks for export or processing

2. Factors favouring forest exploitation in Africa

- Luxuriant growth of trees due to heavy rainfall and high temperatures in Equatorial regions
- Presence of water bodies for transportation of logs
- High demand for forest products
- Forest exploitation requires little skills

Application activity 10.1

1. Fill in the blanks (warm, rainfall, market, income,)
 - a) High demand for timber encourages forest exploitation because it provides a ready ____ for forest products and increases ____ for producers.
 - b) A ____ climate with high ____ supports rapid forest growth, making continuous timber harvesting possible.
2. Why does forest exploitation generally require relatively little skill?
 - A. Trees grow naturally without human involvement
 - B. Forest exploitation is illegal in most regions
 - C. Basic tools and simple techniques are commonly used
 - D. Timber processing is fully automated
3. Referring to Rugwiro's timber business, suggest practical measures that can make forest exploitation sustainable and environmentally friendly.

10.2. Challenges and possible solutions of forest exploitation in Africa

Learning activity 10.2

Umutoni, a forest manager in West Africa, noticed that uncontrolled logging was causing soil erosion, flooding in river valleys, and landslides on steep slopes. Forests were shrinking, and local communities suffered from climate changes. To address these problems, he started reforestation projects, created forest reserves, promoted agro-forestry, and encouraged the use of fuel-efficient stoves and alternative energy. Laws were enforced to manage tree harvesting sustainably.

1. Fill in the blanks with the following concepts: land, landslides, planting and erosion
 - a) Uncontrolled forest exploitation can lead to environmental problems such as soil ____, flooding, and ____ on steep slopes.
 - b) Reforestation helps to manage forest problems by __ trees and restoring degraded __.
2. Why does deforestation increase soil erosion?
 - A. Tree roots absorb too much water from the soil
 - B. Removal of vegetation exposes soil to rain and runoff
 - C. Forest soils are naturally weak
 - D. Erosion only occurs in desert areas
3. Referring to Umutoni's actions, explain how sustainable forest management practices help reduce environmental and social problems in Africa.

❖ Problems resulting from forest exploitation in Africa

- Soil erosion because the land is exposed to agents of erosion such as wind and running water
- Climate change due to more temperatures which increase the rate of evaporation
- Flooding in river valleys
- Shrinking forest resources as trees and forests are used for various purposes
- Landslides in areas with steep slopes which are bare.

❖ Solutions to the problems of forest exploitation

- Reforestation in areas where trees have been destroyed
- Afforestation by planting trees where they have not been existed
- Enacting laws to effectively manage forests
- Creating forest reserves to protect some tree species
- Encouraging agro-forestry

- Encouraging use of alternative sources of energy to reduce overdependence on charcoal and firewood
- Using fuel efficient cooking stoves
- Encouraging use of efficient tree harvesting methods.

Application activity 10.2

1. Suggest measures to prevent over-exploitation of forests.
2. Evaluate the impact of forestry on sustainable development.

10.3. Over-exploitation of forests in Africa: Causes, effects and measures to control.

Learning activity 10.3

A young farmer, Atete, lives near a large forest in western Africa. She relies on it for firewood, food, and income from timber sales. Recently, she noticed fewer trees, drying rivers, and more soil erosion after heavy rains. Atete decided to plant trees, use solar energy for cooking, and educate her neighbors about sustainable forest practices. Slowly, the forest began to recover, and the community learned to balance their needs with forest conservation.

1. What problems did Atete notice in her forest?
2. Why were trees and rivers affected?
3. How did Asha help improve forest conditions?
4. Which alternative energy did Atete use?

❖ Causes, effects and measures of over-exploitation of forests in Africa

Causes	Effects	Measures
– Rapid population growth – Dependence on fuelwood – Commercial logging – Agricultural expansion – Weak forest governance	– Deforestation and forest degradation – Soil erosion and reduced fertility – Disruption of water cycle Contribution to climate change – Loss of forest-based livelihoods	– Strengthen forest laws and enforcement – Promote alternative energy sources – Encourage afforestation and reforestation – Implement sustainable forest management – Raise public awareness and involve communities

Application activity 10.3

1. Why is community involvement important in conservation?
2. How can forest over-exploitation affect livelihoods?

10.4. Impact of forestry on sustainable development in Africa.

Learning activity 10.4

Tuyizere, an environmentalist, visited a Umurinzi village where trees were cut for firewood, farms, and illegal logging. Rivers dried, and soil eroded. He helped villagers plant trees, use solar cookers, and prevent illegal logging. Gradually, forests recovered, water returned, soil improved, and villagers enjoyed better crop yields and cleaner water.

1. What were the main causes of forest degradation in Tuyizere's village?
2. How did over-exploitation of forests affect the soil and river?
3. What solutions did Tuyizere introduce to the villagers?

❖ Impact of forestry on sustainable development in Africa

- Provides employment and income to rural and urban populations
- Supply raw materials such as timber, fuelwood, and non-timber forest products.
- Supports industries like construction, furniture making, and eco-tourism.
- Contributes to national revenue and economic growth.
- Conserves biodiversity and wildlife habitats.
- Protects watersheds and regulates water flow.
- Reduces soil erosion and land degradation.
- Helps mitigate climate change through carbon sequestration.
- Improves livelihoods of forest-dependent communities.
- Enhances food security and availability of medicinal plants.
- Preserves cultural and traditional values linked to forests.
- Supports community development and social stability.
- Balances economic use of forest resources with conservation.
- Ensures long-term availability of forest resources for future generations.
- Promotes community participation and equitable benefit sharing.
- Supports national and regional sustainable development goals in Africa.

Application activity 10.4

1. Which human activities contributed most to the forest problems in the village?
 - A. Tree planting and wildlife protection
 - B. Use of solar energy and irrigation
 - C. Cutting trees for firewood, farming expansion, and illegal logging
 - D. Rainfall increase and soil fertilization
2. Why is involving local communities important in controlling forest exploitation and ensuring benefits from sustainable management?
 - A. Communities depend on forests, help protect them, and gain benefits such as better soils, water, and crop yields
 - B. Only governments are responsible for forest conservation
 - C. Community involvement increases illegal logging
 - D. Forest management does not affect local livelihoods.

End of unit assessment

❖ Knowledge and understanding:

1. Which of the following is a major effect of forest exploitation in Africa?
 - A. Increased soil fertility
 - B. Soil erosion and loss of biodiversity
 - C. Stable climate
 - D. Expansion of wetlands
2. Which factor favors forest growth in Africa?
 - A. Low rainfall and cold temperatures
 - B. High rainfall and warm climate
 - C. Frequent droughts
 - D. Sparse vegetation
3. Identify and explain two major effects of forest exploitation in Africa.

❖ Skills

4. Evaluate the impact of forestry and forest conservation on sustainable development in Africa.
5. Explain the causes and effects of desertification resulting from forest over-exploitation.

❖ Attitudes & Values

6. Suggest possible solutions to problems affecting forest exploitation in Africa.

UNIT 11

FISHING IN AFRICA



Key Unit Competence:

Assess the impact of fishing on sustainable development in Africa.

Learning objectives

At the end of this unit you should be able to:

Knowledge and understanding

1

- Identify the major fishing grounds in Africa.
- Identify the major fish species caught in Africa
- Explain the factors favouring the growth and development of fishing in Africa
- Identify the importance of fishing in Africa
- Outline the problems affecting fishing in Africa

Skills

2

- Draw a map of Africa to show major fishing grounds in Africa
- Classify fish species caught in Africa
- Analyse the factors responsible for the growth and development of the fishing industry in Africa
- Identify the importance of fishing in Africa
- Examine the problems affecting fishing in Africa and suggest solutions.

Attitudes and values

3

- Appreciate the importance of the fishing industry in Africa
- Show appreciation for fish as healthy food and advocate for its consumption in Africa.
- Support efforts to overcome problems affecting the development of fishing in Africa.

11.0. Introduction

Fishing is the extraction of fish from water bodies, supporting livelihoods, nutrition, trade, and economic development while requiring sustainable management and conservation.

Introductory activity

Kaliza lives near Lake Victoria, where her community depends on fishing for food and income. Recently, fish catches have declined due to overfishing, pollution, and poor equipment. Local leaders are discussing sustainable fishing practices to protect livelihoods and the lake ecosystem.

1. Using Kaliza's community as an example, show how fishing contributes to food supply and employment.
2. Analyse the problems affecting fishing in African inland waters such as Lake Victoria.
3. Propose sustainable fishing solutions that Kaliza's community could adopt to improve fish production and conserve resources

11.1. The major fishing grounds in Africa

Learning activity 11.1

One school holiday, Manzi and Isimbi traveled across Africa with their geography teacher to learn about fishing grounds. At Lake Victoria, they met fishermen using nets on calm freshwater. Along the Nile River, they saw boats carrying daily catches. Later, they visited Morocco's Atlantic coast, where cold ocean currents and wide continental shelves supported abundant marine fishing. The students realized how lakes, rivers, and seas provide livelihoods and food for millions across Africa.

1. Fill in the blanks with an appropriate concept/group of concepts (Ocean currents, Inland fishing grounds, Atlantic, Red Sea and Indian)
 - a) Lake Victoria and Lake Tanganyika are examples of major _____ in Africa.
 - b) Marine fishing grounds in Africa are mainly found along the _ and _ oceans
2. Which of the following rivers is an important inland fishing river in Africa? A.
A. Congo
B. Sahara
C. Kalahari
D. Atlas
3. Why is the Atlantic coast richer in fish than other African coasts?
A. It has warmer water throughout the year
B. It has fewer fishing activities
C. It is influenced by cold ocean currents causing nutrient-rich upwelling
D. It has shallow rivers entering the sea.

The major fishing grounds in Africa are classified into two main categories: inland (freshwater) fishing grounds and marine (saltwater) fishing grounds. Inland fishing is carried out in lakes, rivers, and dams, while marine fishing takes place in seas and oceans.

a) Inland or freshwater fishing grounds

The major inland fishing grounds in Africa are found:

- In **large natural lakes** such as Lake Victoria, Lake Kivu, Lake Tanganyika, Lake Albert, and Lake Malawi.
- In **human-made lakes** and reservoirs, including Lake Volta, Lake Kainji, Lake Nasser, and Lake Kariba.
- Along **major rivers** such as the Nile, Congo, Niger, Zambezi, and Orange Rivers.

b) Marine or Saltwater Fishing Grounds

- Marine fishing grounds in Africa are mainly located along the coasts of the Indian Ocean and the Atlantic Ocean, as well as in the Mediterranean Sea and the Red Sea.
- The richest marine fishing grounds are found along the Atlantic coast, particularly off the shores of Morocco, South Africa, Angola, and Nigeria, where cold ocean currents and continental shelves favor abundant fish stocks.

Application activity 11.1

1. Compare inland and marine fishing grounds in terms of location and water type.
2. Assess the role of cold ocean currents in supporting marine fishing.

11.2. Types of fish and fishing methods in Africa

Learning activity 11.2

During a school field trip, Mutesi and Manzi visited Lake Victoria and later the Indian Ocean coast. At the lake, fishermen showed them tilapia, Nile perch, dagaa, and catfish caught using baskets, hooks, lamps, and spears. At the coast, they saw tuna, sardines, mackerel, crabs, and lobsters harvested using trawlers, drifting nets, purse seines, and long hook lines. The students learned how traditional methods suit small inland waters, while modern methods support large-scale marine fishing, providing food, jobs, and income for Africa.

1. Fill in the Blanks (Using concepts from the above story)
 - a) Tilapia and Nile perch are examples of ____ fish species commonly found in African lakes.
 - b) Tuna and sardines are examples of ____ fish harvested from African ocean fishing grounds.
2. Which of the following fishing methods is mainly used in inland waters like Lake Victoria?
A. Trawlers B. Purse seines C. Baskets and hooks D. Long hook lines
3. Why are modern fishing methods mainly used in marine fishing?
 - A. They are cheaper and require no skills
 - B. They are suitable for shallow lakes
 - C. They support large-scale fishing in deep and open waters
 - D. They reduce the number of fish caught.

a) Types/species of fish in Africa

1. **Inland fish species:** The main fish species obtained from inland fishing grounds of Africa include Tilapia, Dagaa, Nile Perch, mud fish, cat fish, shell fish, and other freshwater species.



Tilapia



Dagaa



Catfish



Nile perch

2. **Marine fish species:** The main fish and seafood harvested from marine fishing grounds include Tuna, Sardines, Mackerel, Herring, Stockfish, Crabs, Lobsters, Shrimps, and other marine species.



Crab



lobster



Stockfish



Shrimp

b) Major fishing methods

The fishing methods used in Africa are both traditional and modern.

1. Traditional fishing methods

- **Spearing method:** Fish are caught in shallow water using a spear or sharp stick.
- **Hook and line method:** A baited hook on a line captures fish pulled from water manually.
- **Basket method:** Fish swim into baited baskets with narrow exits and get trapped.
- **Gill-net method:** Vertically held nets trap fish by gills as they try to pass through.
- **Use of herbs:** Crushed herbs stun fish in calm waters, allowing easy hand collection.
- **Trapping method:** Fish are caught using fixed traps in rivers, ponds, or streams.
- **Lamp attraction method:** Fish are attracted to lights at night and then captured.



Spearing method



Hook and line method



Basket Method

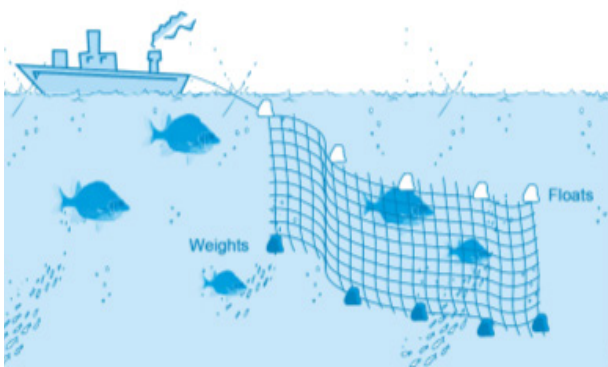


Gill net Method

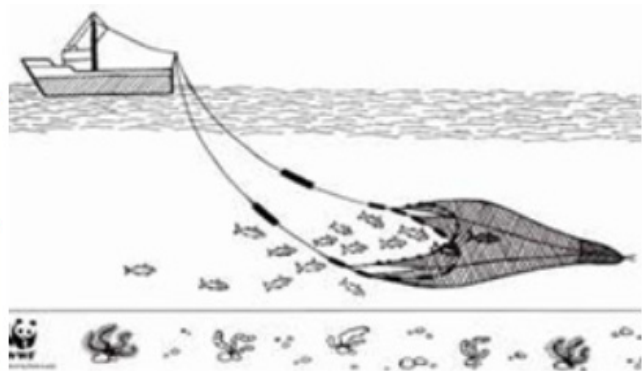
2. Modern fishing methods

These are also called **commercial fishing methods**. They are used in the major fishing grounds. They include:

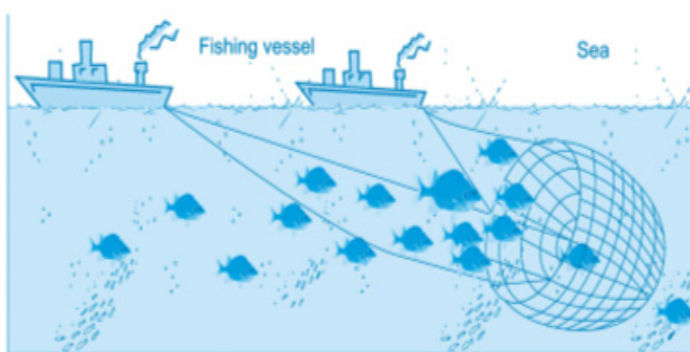
- **Drifting method:** A vertically held net is dragged by a boat, trapping fish by gills.
- **Trawling method:** A bag-shaped net is dragged along the seabed by a trawler.
- **Seining method (general):** Fish are surrounded with a large net and then pulled together.
- **Purse seining method:** A net surrounds fish and is closed at the bottom to trap them.
- **Long hook lining method:** A long line with baited hooks is used to catch fish from boats



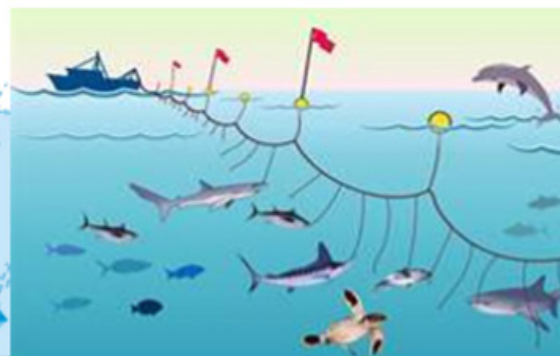
Drifting method



Trawling method



Purse seining method



long hook lining method

Application activity 11.2

1. Explain how the lamp attraction method helps fishermen at night.
2. In your opinion, which fishing method is more sustainable? Give reasons.

11.3. Factors, importance, problems, and possible solutions

Learning activity 11.3

During their geography excursion, Kaneza and Izere visited Lake Tanganyika and later the Atlantic coast of West Africa. They observed how cold ocean currents, wide continental shelves, rivers, and lakes supported rich fishing activities. Fishermen explained how fishing provides jobs, food, foreign exchange, and supports industries. However, they also saw problems like overfishing, pollution, poor storage, and foreign vessels reducing local catches. Community leaders showed them solutions such as fish farming, seasonal fishing laws, cooperative societies, better transport, storage facilities, and navy patrols. The students understood that fishing is vital but must be well managed.

1. Cold ocean currents and wide continental shelves are examples of:
 - A. Problems affecting fishing in Africa
 - B. Factors that favour rich fishing activities in Africa
 - C. Fish preservation methods
 - D. Foreign vessels reducing local catches
2. Fish farming and seasonal fishing laws are examples of:
 - A. Problems affecting inland fishing
 - B. Factors favouring overfishing
 - C. Solutions adopted to manage fishing sustainably and conserve resources
 - D. Marine fishing methods
3. Which of the following problems was observed affecting fishing activities in Africa?
 - A. Overfishing and pollution
 - B. Excessive rainfall increasing fish stocks
 - C. Abundant fish causing low market prices
 - D. Efficient storage and transport systems

a) Factors and importance of fishing in Africa

Factors Favoring Fishing in Africa	Importance of Fishing in Africa
– Cold ocean currents supply nutrients – Wide, shallow continental shelves – Convergence of warm and cold currents – Long and indented coastlines – Estuaries, lagoons, river mouths (providing breeding grounds) – Mild climate and calm seas (favouring year-round fishing) – Numerous lakes, rivers, and ponds	– Creates jobs, improving living standards – Fish is a source of protein (improving nutrition) – Development of related industries (e.g., canning, net making) – Foreign exchange from exported fish – Fish wastes used for fertilisers, glue, and other by-products – Fish oil used in medicine production

– Cold ocean currents supply nutrients – Wide, shallow continental shelves – Convergence of warm and cold currents – Long and indented coastlines – Estuaries, lagoons, river mouths (providing breeding grounds) – Mild climate and calm seas (favouring year-round fishing) – Numerous lakes, rivers, and ponds (supporting inland fishing)	– Creates jobs, improving living standards – Fish is a source of protein (improving nutrition) – Development of related industries (e.g., canning, net making) – Foreign exchange from exported fish – Fish wastes used for fertilisers, glue, and other by-products – Fish oil used in medicine production
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b) Problems affecting fishing in Africa and their possible solutions

Problems affecting fishing in Africa	Possible Solutions in Africa
– Straight coastlines limit port locations – Poor fishing methods catch immature fish – Inadequate storage spoils fish quickly – Water pollution kills fish – Poverty limits local fish markets – Water hyacinth and weeds cover lakes – Dangerous aquatic animals make fishing risky – Rapids and waterfalls hinder fishing – Lack of funds forces inefficient methods – Overfishing occurs where control is weak – Poor transport infrastructure causes spoilage – Foreign countries' fishing reduces local fish – Limited local fish supply	– Some natural problems cannot be solved – Restrict fishing to specific seasons – Introduce storage facilities near fishing grounds – Enact and enforce anti-pollution laws – Encourage cooperative societies for financial support – Clear water weeds to enable fishing – Educate locals on safe fishing practices – Build alternative access or use safer routes – Provide financial support for modern equipment – Issue fishing licenses to control numbers – Improve transport facilities for quick delivery – Use navy patrols to control territorial waters – Encourage fish farming and restocking

Application activity 11.3

1. How does water pollution affect fishing?
2. Suggest two solutions to reduce overfishing in Africa.
3. Why is fish farming important for African population?

End of unit assessment

❖ Knowledge and understanding:

1. Which of the following is a major inland fishing ground in Africa?
A. Sahara Desert
B. Lake Victoria
C. Kalahari
D. Atlas Mountains
2. Which of the following is a traditional fishing method in Africa?
A. Purse seining
B. Gill net
C. Basket traps
D. Trawlers
3. Identify four major fishing grounds and two major fish species caught in Africa.

❖ Skills

4. Draw a map of Africa showing the major fishing grounds accurately.

❖ Attitudes & Values

5. Evaluate the importance of fishing in Africa, examine the main problems, and suggest possible solutions to overcome them.

UNIT 12

MINING IN AFRICA



Key Unit Competence:

Analyze the impact of mining on sustainable development in Africa.

Learning objectives

At the end of this unit you should be able to:

Knowledge and understanding

1

- Identify the major types of minerals and mining areas of Africa.
- Outline different methods of mining in Africa.
- Explain the factors favouring mineral exploitation in Africa.
- Identify the problems affecting mining in Africa.
- Show the impact of mining on sustainable development of Africa.

Skills

2

- Locate the major minerals and mining areas of Africa.
- Describe the different methods of mining in Africa.
- Analyse the factors favouring the exploitation of minerals in Africa.
- Analyse the problems affecting mining in Africa
- Propose possible solutions of the problems affecting mining in Africa
- Evaluate the impact of mining on sustainable development in Africa.

Attitudes and values

3

- Appreciate the importance of various types of minerals found in Africa.
- Show continual desire to protect areas under over exploitation of minerals.
- Show concern for the effects of mining in Africa and suggest solutions.
- Appreciate the importance of mining on sustainable development of Africa

12.0. Introduction

Mining is the extraction of valuable minerals from the earth, supporting Africa's economic growth, industrial development, and sustainable use of natural resources.

Introductory activity

Students visited a mining area in Africa and observed open pits, heavy machinery, and workers extracting gold, diamonds, copper, cobalt, iron ore, bauxite, and coltan. They learned mining supports employment, industries, and government revenue, but also causes environmental degradation, pollution, and health risks. Teachers emphasized responsible mining, environmental conservation, land rehabilitation, and safety measures for sustainable development.

1. Identify major minerals mined in Africa and their mining areas.
2. Outline the different methods of mining used in Africa.
3. Propose sustainable mining practices to reduce environmental damage and support development.

12.1. Types of minerals and their mining areas in Africa

Learning activity 12.1

Dr. Mwizerwa advises the African Union (AU) on mineral investments. He must choose between promoting continental factories using metallic minerals (like Copper and Iron) or securing future power from Energy minerals (like Oil and Gas). His report covers major producers: South Africa (Gold, Iron), the Copper Belt, and Nigeria/Libya (Oil).

1. Name the three specific minerals listed in the text that are classified as Ferrous metallic minerals.
2. What is the main difference between a metallic mineral (like Gold) and a non-metallic mineral (like Salt)?

❖ Minerals can be classified into:

1. Metallic minerals:

Metallic minerals contain extractable metals used for economic and industrial purposes. They are usually shiny, dense, good conductors, occur as compounds, and are classified into ferrous (iron-containing such as Haematite, Limonite and Siderite) and non-ferrous minerals (no iron containing like Gold, Copper, Aluminium, Tin, Lead and Zinc.).

2. Non-metallic minerals:

Non-metallic minerals are widely found in Africa and mainly used in construction, agriculture, and chemical industries, including sulphur, asbestos, salt, phosphates, graphite, and building stones.

3. Energy metallic minerals

They include metals or organic materials that serve as sources of energy. They provide fuel or radioactive energy for electricity generation or industrial use, with common examples being petroleum, natural gas, coal, and uranium.

Some of the major mining areas in Africa include South Africa, Copper Belt in Democratic Republic of Congo and Zambia, Nigeria, Angola and Libya.

4. Major mining areas in Africa

Some of the major mining areas in Africa include South Africa, Copper Belt from DRC to Zambia, Nigeria, Angola and Libya.

a) Mining in South Africa

South Africa has valuable minerals including gold, uranium and diamonds (mined in Kimberley), iron, phosphates, lead, zinc, copper, tin, chrome, Coal (mined in Transvaal, Orange Free State and Natal) and asbestos. Gold, a shiny metal used for Jewellery, is mined underground at Witwatersrand near Johannesburg, making the country a leading producer.

Gold mining in the Johannesburg (Witwatersrand) area was influenced by abundant gold deposits, availability of power and coal, good transport networks, modern machinery and technology, cheap labour, and large local and international markets.

b) The Copper Belt of DRC and Zambia

The Copper Belt stretches from Ndola in Zambia through Lubumbashi to Kolwezi in southern Katanga, and Zambia is the world's second largest copper exporter after Chile, mining copper by opencast and underground methods

c) Mining areas in Nigeria

Nigeria is Africa's leading oil exporter, with oil drilled mainly in the Niger Delta, largely exported as crude, alongside other minerals such as coal, tin, iron, limestone, and phosphate

d) Mineral areas in Angola

Angola's major minerals include crude oil, diamonds, and natural gas, with oil mined along the coast and diamonds extracted from riverbeds of the Kwango, Kasai, and Chicopa rivers.

e) Major minerals in Libya

Libya's main mineral is sulphur-free oil, refined into products like petrol, diesel, paraffin, and tar for use. It is mined around the coast of Mediterranean Sea.

Application activity 12.1

1. Which two factors that supported high gold output at Witwatersrand would also help the Copper Belt (DRC/Zambia) become a major copper exporter?
 - A. High rainfall and fertile soils
 - B. Cheap labour and good transport networks
 - C. Tourism and political stability
 - D. Dense forests and rivers
2. Which option is better for Africa's future economic stability?
 - A. Exporting crude oil only without processing
 - B. Depending entirely on foreign mining companies
 - C. Investing in factories to process copper and iron locally
 - D. Reducing mineral production
3. Which of the following is a quick way Africa can earn money from non-metallic minerals?
 - A. Leaving them unexploited
 - B. Using them only for local housing
 - C. Processing building stones and salt for construction and industry
 - D. Exporting them without any value addition.

12.2. Methods and factors for mineral exploitation in Africa.

Learning activity 12.2

You are part of two dynamic teams exploring a resource-rich region of Africa, tasked with deciding the best methods for extracting valuable resources while prioritizing safety, efficiency, and environmental impact. Team **A** has discovered iron ore just beneath the soil and diamonds in a nearby riverbed, while Team **B** must navigate the complexities of accessing deep gold deposits and ensuring the safe extraction of liquid natural gas from below the surface. As you deliberate on the extraction methods, conduct research on the methods of mining. Guided by the below questions, prepare a presentation for your class.

1. What mining method would you recommend for Team **A**'s iron ore extraction, and why is it the most suitable choice?
2. For the diamonds found in the riverbed, what extraction technique should Team **B** use? How does this method ensure minimal environmental disruption?

❖ Mining methods include:

1. Surface mining

Methods for mining coal, iron, bauxite, limestone, clay, rocks, and phosphate either by opencast or alluvial methods:

a) Opencast or stripping mining: Extracts minerals near the surface by removing overlying soil and rock layers.

b) Alluvial or placer mining: Recovers minerals from riverbeds, sand, or gravel using water and manual or mechanical methods



A limestone quarry



Coal mine



Panning method



Open cast method

3. Underground mining methods to mine petroleum, natural gas, sulphur, potash, salts, and coal are extracted using drift, slope, shaft, solution, and drilling methods.

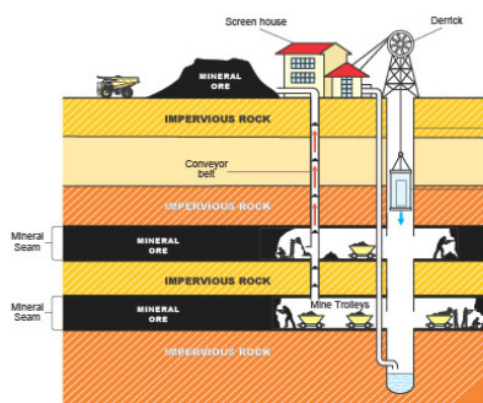
a) Drift mining method: Drift mining is a type of underground mining where miners dig a horizontal tunnel into the side of a hill or mountain to reach minerals

b) Slope method: Uses inclined tunnels or cable systems to reach steeply tilted mineral deposits.

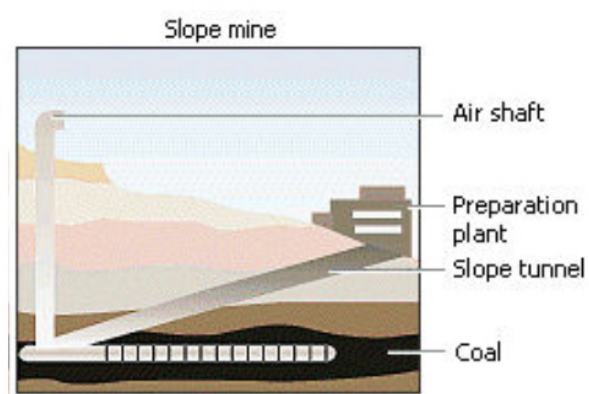
c) Shaft method: Excavates vertical or near-vertical tunnels to access deep mineral deposits.

d) Solution method: Dissolves and extracts soluble minerals like sulphur, potash, and salts from underground.

e) Drilling method: Extracts petroleum, natural gas, or deep minerals by penetrating rock layers with drills.



Shaft tunnel



Slope shaft

Application activity 12.2

1. To access deep gold deposits. What underground mining methods could be used, and what are the safety considerations involved?
2. Liquid natural gas extraction can be hazardous. What safety measures and technologies should be employed to minimize risk during extraction?

Learning activity 12.3

The Iron Ore Mine of X region is closing. Problems included low-quality ore and political disputes. The closure caused massive job loss (unemployment) and left behind large toxic waste heaps, causing land degradation. The government must now implement a new national strategy to ensure that future mining is truly sustainable.

1. According to you, name two specific problems that can cause people to leave a mining area, related to environmental damage.
2. List two ways the government can enhance sustainable mining that directly addresses the problem of **"shortage of skilled labour"** and **"Inadequate capital for investment."**

a) Problems affecting mining in Africa:

- Environmental protection laws restricting mining activities: Laws to conserve the environment may limit mining locations or methods.
- Small size of mineral deposits in some countries: Limited mineral quantities reduce profitability and discourage large-scale exploitation.
- Low value of minerals: Minerals with low market value generate little revenue, affecting mining investment.
- Low quality of mineral ores: Poor ore quality requires more processing, increasing costs and reducing efficiency.
- Poor methods of extraction: Outdated or inefficient mining techniques limit production and can damage the environment.
- Limited market for minerals: Lack of local or international buyers reduces demand and profitability.
- Inadequate capital for investment: Insufficient funds prevent purchase of modern equipment and technology for mining.
- Shortage of skilled labour: Lack of trained workers reduces mining efficiency and technological adoption.
- Political disputes over border minerals: Conflicts between countries can halt or restrict mineral exploitation in border areas
- Overdependence on mining, neglecting other economic activities
- Environmental degradation, loss of vegetation, and farmland destruction:
- Unemployment and industrial decline due to mineral exhaustion
- Migration caused by environmental pollution.
- Underground mining can cause land to sink, damaging buildings and infrastructure.
- Mining waste may contain harmful chemicals affecting health and soil.
- Water in abandoned quarries can breed mosquitoes, spreading malaria and other diseases.

- Radioactive mining can contaminate underground and surface water sources.
- Mining and coal use release greenhouse gases (like CO₂), contributing to air pollution and climate change

b) Ways to enhance sustainable mining in Africa

- Enforce sustainable mining and environmental management practices.
- Invest in modern mining technologies and efficient extraction methods.
- Improve transport, power, and communication infrastructure.
- Train skilled labour for mining industries.
- Mobilize capital through public-private partnerships.
- Promote regional cooperation for cross-border mineral management.
- Diversify the economy to reduce overdependence on mining.
- Implement proper waste management to protect health and environment.
- Expand local and international markets for minerals.

Application activity 12.3

1. Compare the environmental impacts of the extraction methods chosen by both teams and explain how can you implement sustainability practices in their operations?
2. Explain the concept of **"overdependence on mining."** What two other economic activities might be neglected when a country relies too much on mining?

12.4. Impact of mining on sustainable development of Africa

Learning activity 12.4

The newly opened **'Prosperity Mine'** in Ghana is boosting the region. It creates jobs, pays taxes to the government, and builds new roads and hospitals. Its mineral exports bring in foreign money, strengthening the national economy and encouraging trade with other nations.

1. Name two specific types of infrastructure or social amenities (like a road) that mining activities often support the establishment of.
2. What three types of minerals or materials listed in the text supply energy for industrial activities and transportation?

❖ The impact of Mining on the Sustainable Development of Africa

- Minerals are essential for industrial production and manufacturing growth.
- Petroleum, coal, and natural gas supply energy for industrial activities.
- Mining creates employment opportunities for skilled and unskilled workers.
- Governments earn revenue through taxes from mining activities

- Mineral exports generate foreign exchange that supports national economy.
- Mining promotes urban development and construction of infrastructure
- Mining supports establishment of social amenities like schools, hospitals, and services.
- Mining areas attract tourists interested in natural resources and mining.
- Mining stimulates growth of other economic sectors and related industries.
- Minerals like coal and petroleum provide fuel for locomotives and transportation.
- Mining encourages international cooperation through trade and cross-border agreements.

Application activity 12.4

1. Mining creates jobs and also stimulates the growth of other economic sectors. Give two examples of those other economic sectors or related industries that would grow because a new mine opened.
2. Discuss problems that affect the mining industry in Africa.
3. Briefly discuss the impact of mining on sustainable development in Africa.

End of unit assessment

❖ Knowledge and understanding:

1. Which of the following is an energy mineral found in Africa?
A. Gold B. Coal C. Salt D. Copper
2. Which mining method is most suitable for extracting iron ore near the surface?
A. Shaft mining B. Drift mining C. Opencast mining D. Solution mining
3. Name two metallic minerals and one non-metallic mineral found in Africa, giving at least one mining area for each.

❖ Skills

4. Explain why over-reliance on mining can be harmful to Africa's economy and suggest one other economic activity that should also be developed.
5. Suggest two ways Africa can ensure sustainable mining while maintaining economic growth.

❖ Attitudes & Values

6. Recommend a mining method for extracting diamonds from a riverbed and explain why it minimizes environmental damage.

UNIT 13

POWER AND ENERGY IN AFRICA



Key Unit Competence:

Evaluate the impact of power and energy production on sustainable development in Africa.

Learning objectives

At the end of this unit you should be able to:

Knowledge and understanding

1

- Outline the factors favouring power and energy production in Africa.
- Explain the importance of power and energy in Africa.
- Identify the problems hindering the effective production of power and energy in Africa.

Skills

2

- Analyse the factors favouring power and energy production in Africa
- Evaluate the importance of power and energy in Africa
- Analyse the problems hindering the effective production of power and energy in Africa and possible solutions.

Attitudes and values

3

- Appreciate the importance of power and energy for the development of Africa
- Show concern to propose the solutions to the problems affecting energy and power production in Africa
- Appreciate the efficient use of power and energy in Africa.]

13.0. Introduction

Power and energy are essential for Africa's development, supporting production, industrial growth, and daily life through sustainable and efficient resource use.

Introductory activity

Students study hydroelectric dams, solar farms, and wind turbines in Africa. They learn energy supports factories, homes, and schools. Teachers explain challenges like frequent power shortages, limited infrastructure, and fuel dependency, and discuss solutions such as renewable energy and efficient distribution systems.

1. Explain the importance of energy in Africa.
2. Analyse factors favouring power and energy production
3. Describe problems hindering effective energy production.
4. Suggest solutions to improve energy production.

13.1. Factors favouring power and energy production in Africa

Learning activity 13.1

A Rwandan energy student, Kamanzi, is studying energy production across Africa. She observes that countries like Kenya are using geothermal energy from the Rift Valley, South Africa relies on coal, Nigeria exploits oil and gas, and many regions are developing solar power due to abundant sunshine. She notes that growing populations and infrastructure expansion have increased energy demand, creating a need for sustainable solutions.

1. Identify the main renewable and non-renewable energy resources found in Africa.
2. Explain how population growth and infrastructure development influence energy demand in Africa.

Africa's energy sector is shaped by both natural resources and socio-economic factors. The continent's development, population growth, and infrastructure improvements have all influenced energy demand and production.

- As Africa continues to develop, energy demands and production also grow.
- The increasing need for power has driven investments in various energy sectors.
- Improved infrastructure has made it possible to develop several power stations across the continent, supporting both industrial and domestic energy needs.
- The presence of huge renewable energy resources across Africa contributes significantly to energy production. These resources are evenly distributed and abundant, including geothermal energy in the Rift Valley and extensive water resources.
- Fossil energy resources also play a major role. Examples include oil and gas reserves in northern, western, and southern Africa, as well as coal deposits in western and southern regions.
- The availability of sunshine in almost all African countries throughout the year

has facilitated the production of solar energy, taking advantage of the continent's favorable climate.

- Population increase has further intensified the demand for power and energy, highlighting the need for sustainable energy solutions to support growing communities and industries.

Application activity 13.1

Using relevant examples from the Rwanda or any other country in Africa, describe the factors that explain how African countries utilize their energy resources.

13.2. Importance of power and energy in Africa

Learning activity 13.2

Manzi, a Rwandan geography student, visits South Africa and Nigeria to study how power and energy influence economic development. He observes that industries have grown, urban centres have expanded, farms are more modern, transport networks have improved, and employment has increased. He also notes that governments earn revenue and foreign exchange from energy sectors.

1. List the ways in which power and energy contribute to economic development in Africa.
2. Explain how energy availability improves agricultural production and industrial growth.
3. Using examples from the scenario, describe the relationship between energy production and urban growth

Power and energy play a very important role in the economic development of countries in Africa. The importance of power and energy in Africa include the following:

- *Increased employment opportunities:* People are directly employed in the power and energy sectors or in other sectors that use power and energy such as industries. This has raised people's standards of living.
- *Development of industries:* Availability of power and energy lead to development and expansion of industries.
- *Growth of urban centres:* Many urban centres have come up and others have expanded due to availability of power and energy.
- *Improvement in agricultural production:* Availability of power and energy resources has led to modernisation and expansion of the agricultural sector. This has in turn increased agricultural production.
- *Provision of energy for domestic use:* Energy is required for cooking, lighting, heating and providing power to some domestic appliances.
- *Expansion of the transport sector:* The transport sector is the largest consumer of petroleum. As the rate of production of petroleum increases, it causes an expansion in the transport sector.

- *Source of revenue for governments:* Taxes generated from those employed directly or indirectly in the power and energy sectors earn revenue for the countries with these sectors.
- *Generation of foreign exchange:* Some of the multi-purpose river projects are tourist attractions. This generates foreign exchange for countries with these projects.

Application activity 13.2

1. Analyse the impact of energy sectors on employment and government revenue in African countries.
2. Suggest ways African countries can further use power and energy to promote economic development.
3. Evaluate the role of energy-related tourism projects in generating foreign exchange and supporting local economies.

13.3. Problems and solutions of power and energy production in Africa

Learning activity 13.3

Keza, a Rwandan energy officer at REG, is conducting research on power production in Africa. She observes that many countries face challenges such as high production costs, inadequate capital, shortage of skilled workers, political instability, fluctuating river volumes, and low industrialisation. She also notes that initiatives like small rural power projects, renewable energy development, regional cooperation, and affordable tariffs are helping to address these challenges in Rwanda and they can be diversified to other African countries.

1. Identify the main problems affecting power and energy production in Africa.
2. Using examples from the scenario, describe how fluctuations in river volumes affect hydroelectric power production.
3. Suggest practical measures that African countries can implement to address the shortage of skilled manpower in the energy sector.

d) Problems of power and energy production in Africa

Africa has large amounts of power and energy resources. These have not been fully exploited due to the following problems:

- iv. Inadequate capital:** Production of power requires a lot of capital. Most African countries cannot afford to invest in power projects due to limited financial resources. Inadequate capital leads to insufficient maintenance of equipment and lack of modernisation.
- v. Shortage of skilled manpower:** There is shortage of skilled manpower which makes African countries hire experts from developed countries. This makes the

production of power and energy expensive.

vi. Political instability: There has been instability in some African countries. This has discouraged potential investors in the power and energy sectors.

vii. High cost of power: Due to low-income levels, majority of people cannot afford to pay for power. This tends to limit production of power in some areas due to low returns.

viii. Fluctuation of volume of water in rivers: Due to changes in climatic changes, water volume in rivers fluctuates. This at times may lead to interruptions in the production of hydroelectric power.

ix. Low level of industrialization: In many countries of Africa, industrialisation is not fully developed to necessitate high energy production.

b) Solutions of power and energy production in Africa

The following are some possible solutions to the problems of power and energy production in Africa:

- Intensifying regional and interregional cooperation on renewable energy production and transmission.
- Improving efficiency in production, distribution and use of energy.
- Ensuring there is peaceful political situation in order to attract investors in the energy sector.
- Encouraging development of small power projects especially in the rural areas.
- Lowering the pricing of power to make it affordable to people with low income.
- Strengthening regional cooperation on research and training of skilled personnel required in the energy sector.
- Fully utilizing geothermal, wind and solar resources for renewable energy production.

Application activity 13.3

1. Suggest sustainable strategies African countries could adopt to meet increasing energy demands.
2. Evaluate the impact of energy production on both economic growth and the environment in Africa

13.4. Case studies: Hydro -electric power stations/projects in Africa: Aswan (Nile) in Egypt, and Akosombo (Volta) in Ghana.

Learning activity 13.4

A group of Rwandan geography students, led by Jean, is studying major hydroelectric dams in Africa. They visit the **Aswan High Dam** in Egypt and **Akosombo Dam** in Ghana. They observe that both dams provide electricity, support irrigation, create jobs, promote fishing and tourism, and improve living standards. However, they also learn about challenges such as siltation, high maintenance costs, water pollution, droughts, and displacement of people. Students discuss solutions like silt removal, mechanical control of weeds, training skilled personnel, and building small supplementary dams.

1. Identify the main purposes for establishing the Aswan High Dam and Akosombo Dam.
2. Explain the importance of these dams to local communities and national economies.
3. Describe the problems experienced in power production at both dams.

13.4.1. Aswan High Dam

Aswan High Dam is in the southern part of Egypt. It was built across River Nile which is the longest river in the world. It was initiated by Gamal Abdel Nasser who was the president of Egypt at that time. Construction started in 1960 and was completed in 1970. During the construction, about 100,000 people were relocated. This was to provide space for the dam. The reservoir created by the dam forms Lake Nasser. It is about 500 kilometres long with part of it extending to Sudan.



d) Reasons for the establishment of Aswan High Dam and its importance

Reasons	Importance
- To control the floods along river Nile - To provide water for irrigation since the largest part of Egypt is within Sahara Desert - To generate hydro-electric power - Providing water for both industrial and domestic use - To create employment opportunities for many people - To control siltation of river Nile	- Source of electricity for industrial and domestic use - Provision of water for irrigation from Lake Nasser - Promotion of fishing from lake Nasser - Attraction of tourists which give foreign exchanges - It has helped to control floods in the lower sections of River Nile - Promotion of water transport along Lake Nasser - Creation of employment opportunities for many people - Industrial development due to electricity generated - Source of government revenue due to taxation of workers' income - Improvement of standards of living of people due to increased income

e) Problems experienced in the production of power and their possible solution at the Aswan High Dam

Problems	Possible solutions
- Deposition of silt in the reservoir which reduced the amount of water available for use - Development of seaweeds which tend to limit navigation - High evaporation rates leading to loss of water - Water pollution leads to high cost of treatment of drinking water - The excessive utilization of water resources affects the volume of water required - High cost of maintaining equipment involved in the power production	- Regular removal of silt from the reservoir - Mechanical harvesting and biological control of seaweed - Negotiate with countries in the upper course of Nile River on proper utilization of the Nile water - Regular maintenance of power plants - Avoiding water pollution by removing algae alongside the weeds

13.4.2. The Akosombo Dam (Volta River project)

The Akosombo Dam (1961–1965), initiated by Dr. Kwame Nkrumah on Ghana's River Volta, is a Hydro-electric Project. It created Lake Volta, the world's largest human-made lake by surface area (8,502 sq km = 3.6% of Ghanaian area), providing vital power and water resources



f) Reasons for the establishment of Akosombo Dam and its importance

Reasons for the establishment of Akosombo Dam	Importance of Akosombo Dam
– To generate hydro-electric power to use in mining and refining of bauxite into Aluminium – To regulate the flow of River Volta by fighting against flooding – To develop manufacturing industries – To create employment opportunities for many people – To provide water for irrigation in order to increase food production – To improve the standards of living of people through various income generating activities.	– Source of electricity to be used in homes, in mining, and in industries – To get enough water for irrigation so as to increase food production – Source of fish from Volta Lake – Attraction of tourists which give foreign exchanges – Promotion of water transport (Navigation) along Lake Volta – Sources of employment opportunities to many people – Source of water for industrial and domestic uses.

g) Shortcomings (problems) experienced in the production of power at the Akosombo Dam and their possible solutions

Shortcomings (problems)	Possible solutions
– Occasional droughts which led to reduction in water levels in Lake Volta. – Inadequate capital to invest in expanding and maintaining the project – Low level of technology which led to reliance on foreign experts – Siltation of the Lake due to deposition of silt in the rivers. – Spread of diseases during the construction of the Dam – Displacement of people due to the need for large land for the establishment of the Dam and Lake Volta.	– Strengthening technical education and training in order to get skilled personnel in energy sector – Encouraging private investors to invest in the energy sector – Regular removal of silt from the Lake – Building smaller dams to supplement the energy generated by the Akosombo Dam. – Reduce overutilization of the dam by encouraging people to engage in alternative sources of income.

Application activity 13.4

1. The **Aswan High Dam** was constructed across the _____ River to control floods and provide water for irrigation.
2. Lake _____ covers an area of _____ km², making it the largest man-made lake in the world by surface area.
3. One of the major problems affecting power production at both dams is _____, which reduces water availability and affects electricity generation.
4. Explain the importance of the Aswan High Dam and the Akosombo Dam to the economic development of Egypt and Ghana, highlighting at least four benefits.
5. Discuss the challenges experienced in power production at these dams and suggest possible solutions to overcome these problems.

End of unit assessment

❖ Knowledge and understanding:

1. Which of the following is a major renewable energy source in Africa?
 - a) Coal
 - b) Petroleum
 - c) Solar energy
 - d) Natural gas
2. Which strategy can help African countries improve energy production?
 - a) Ignoring renewable energy
 - b) Encouraging energy efficiency
 - c) Limiting infrastructure development
 - d) Relying only on imported fuels
3. Outline three factors favouring power and energy production in Africa.

❖ Skills

4. Analyse how energy from different sources (wind, water, solar, petroleum) contributes to Africa's power supply.
5. Identify and analyse the main problems hindering effective power and energy production in Africa

❖ Attitudes & Values

6. Evaluate the importance of power and energy for economic development and daily life in Africa
7. Suggest five possible strategies that African countries can use to enhance power and energy production and ensure their efficient utilization

UNIT 14

INDUSTRIAL DEVELOPMENT IN AFRICA



Key Unit Competence:

Analyze the impact of industrialization on sustainable development in Africa.

Learning objectives: At the end of this unit you should be able to:

Knowledge and understanding

1

- State the types of industries in Africa
- Identify the major industrial areas in Africa
- Identify the factors influencing industrialisation in Africa.
- Explain the importance of industries to the economy of Africa.
- Identify the problems affecting industrial growth in Africa and their possible solutions
- Outline the prospects for industrial development.

Skills

2

- Classify the major types of industries in Africa
- Locate the major industrial areas of Africa.
- Analyse the factors influencing industrialisation in Africa.
- Analyse the effects of industrialisation on the economy of Africa
- Examine the problems and prospects for industrial development in Africa

Attitudes and values

3

- Show a concern to minimize the industrial effects on the environment
- Appreciate the importance of industries for the sustainable development of Rwanda

14.0. Introduction

Industries produce goods and services for consumption or sale; industrialization concentrates production, promotes economic growth, and involves factories where machines facilitate large-scale production within specific regions in Africa.

Introductory activity

Scenario

The government of Rwanda is considering building a new cement factory to support construction and regional trade. Learners are asked to advise on the best location by thinking about resources, people, and the environment.

Questions:

1. What raw materials and energy sources are needed for a cement factory, and where can they be found?
2. How do transport and communication networks influence the choice of location?
3. In what ways do population distribution and settlement patterns affect labor supply and markets?
4. What environmental concerns should be considered before establishing the factory?

14.1. Types of industries and major industrial areas of Africa.

Learning activity 14.1

A group of Rwandan staff in charge of industry travelled to South Africa and Egypt to acquire skills on industrial development in Africa. They compare South Africa and Egypt, observing that South Africa has highly developed industries in regions like the Rand and Western Cape, producing metals, chemicals, cars, and food products. Egypt's industries are concentrated in Cairo, Alexandria, and Helwan, producing electronics, textiles, chemicals, and sugar. They noted that factors such as raw materials, skilled labor, power, infrastructure, markets, and government policies have favored industrial growth in both countries. Back to they are using the gained skills to speed up the industrial sector in Rwanda.

1. Identify the main industrial regions in South Africa and Egypt and the types of industries found there.
2. Explain the factors that have favored industrial development in South Africa and Egypt.

1. Types of industries in Africa

Industries are classified into 3 categories/types:

- a) Primary /extractive industry:** industry that deals with extraction of natural resources and raw materials like mining, fishing, agriculture and forestry.
- b) Secondary (Manufacturing) Industry:** This industry involves the processing of raw materials into finished or semi-finished consumer goods. It is divided into **heavy industries** (producing large, bulky goods using iron, steel, cars,

ships) and **light industries** (producing small, lightweight goods like textiles, food, cosmetics, plastics).

c) Tertiary / service industry: these are industries that provide services such as banking, insurance, transport and recreation (tourism). They are activities that make it easy for the primary and secondary activities to be carried out.

Note that: Depending on size, industries are **large-scale** or **small-scale**. They are also classified as **craft**, **art**, and **modern industries**. Craft industries use hand tools or small machines, art industries produce artistic objects, and modern industries involve manufacturing.

2. Industrial development in South Africa

South Africa has a highly developed industrial sector compared to other African countries. This is because it was in contact with the developed countries for a long time.

- The four major industrial areas in South Africa are: The Rand, Western Cape centred on Cape town, Eastern Cape centred on Port Elizabeth and East London and Durban-Pinestown region of Natal.
- Some types of industries in South Africa include metal working and engineering, food processing industries, electrical machinery and equipment factories, chemical industries, oil refineries industries, paper manufacturing industries and car assembly industries

❖ Factors that have favoured industrial development in South Africa

- Well-developed transport and communication networks in form of roads, railways, airports and seaports among others.
- This factor has favored easy transportation of raw materials and finished products.
- Availability of raw materials in form of minerals such as gold, iron, lead, silt, coal, diamonds and so on.
- There are also agricultural raw materials like sugarcane, fruits, maize, hides and skins, and tobacco.
- Availability of local and foreign markets for finished goods
- Availability of power in form of coal and electricity
- Availability of cheap skilled and unskilled labour local
- Availability of capital to invest in developing industries. This capital comes from the selling of precious minerals.
- Availability of water from river Vaal is to be used in industries of the Rand region.
- Foreign investment due to cheap labor, presence of markets and raw materials.

- Government policy encouraging home production and restricting imports of products which are locally produced.

15.2. Industrial development in Egypt

Egypt is the second most industrialized country in Africa after South Africa. Most of the industrial developments are in Lower Egypt.

The main industrial centres are **Cairo, Alexandria and Helwan**. Other centres are **the Suez and Nile Delta**.

❖ The main industries found in Egypt with their locations and products:

Industry type	Location(s)	Main products
Electronics industries	Cairo, Alexandria, 10th of Ramadan	TVs, radios, computers, mobile devices
Chemical industries	Cairo, Suez, Alexandria	Fertilizers, plastics, detergents, paints
Petro-chemical industries	Suez, Alexandria, Port Said	Plastic products, synthetic fibers, chemicals
Textile and spinning industries	Cairo, Alexandria, Mansoura, Damietta	Cotton fabrics, yarns, garments
Engineering industries	Cairo, Alexandria, Helwan	Machinery, tools, industrial equipment
Sugar refining industries	Upper Egypt (Assiut, Sohag, Qena)	Refined sugar
Food processing industries	Cairo, Alexandria, Tanta	Canned foods, dairy, and beverages
Home appliance industries	Cairo, 10th of Ramadan	Refrigerators, washing machines, ovens
Automobile industries	Cairo, 6th of October City, Mansoura	Cars, trucks, buses
Oil refineries	Alexandria, Suez, Cairo	Petrol, diesel, kerosene, and lubricants

❖ Factors that have favoured industrial development in Egypt

- **Availability of power:** Egypt has hydroelectric power from the Aswan High Dam and oil used to produce electricity, which is highly used in industrial operations.
- **Availability of raw materials:** Agricultural raw materials such as cotton and sugarcane, and minerals like petroleum, iron ore, limestone, and gold support industries.
- **Availability of markets:** Industries have both local and foreign markets for their products.
- **Good transport and communication:** Egypt has well-developed transport

systems, such as rivers, roads, and railways, that help move raw materials and finished goods.

- **Availability of labour and technology:** There are skilled workers and improved technology for industrial production.
- **Supportive government policies:** The government encourages industrial development through favorable policies.

Application activity 14.1

1. Complete the missing words:
 - a) The _____ industry in Africa is based mainly on the extraction of minerals such as gold, copper, and diamond.
 - b) Industries that produce goods for direct consumption, such as food and clothing, are known as _____ industries
2. Egypt's textile industry is mainly supported by:
 - A) Cotton production
 - B) Oil reserves
 - C) Diamond mining
 - D) Hydropower
3. Compare and contrast industrial development in South Africa and Egypt, considering raw materials, technology, and government policies.

14.2. Factors and importance of industrial development in Africa

Learning activity 14.2

The government of Kenya is planning to expand its industrial sector by building a new food processing plant near Nairobi. The aim is to reduce imports, create jobs, and supply both local and regional markets.

1. What two main factors should the government consider when deciding the location of this plant, and how will the plant contribute to Kenya's economic development?
2. How does industrial development contribute to employment creation and urbanization across African countries?

(a) Factors influencing industrial development and its importance

Factors and Determinants for Industrial Growth	Importance/benefits of Industrial development
– Availability of raw materials (minerals, agricultural produce). – Capital investment from local or foreign sources. – Availability of skilled and unskilled labor.	– Provides employment opportunities, reducing unemployment. – Increases people's income and improves standards of living. – Provides essential consumer goods (sugar, textiles, shoes).

– Well-developed infrastructure (roads, railways, energy) and government policies. – Market access (local and foreign) and access to modern technology. – Peace and political stability. – Proximity to raw materials. – Proximity to the market. – Availability of a reliable source of power. – Supportive government policy.	– Diversifies the national economy. – Promotes self-sufficiency and reduces reliance on imports. – Government gains revenue through taxation. – Foreign exchange is earned after commodities are exported. – Development of urban centers. – Development of infrastructure (transport, hospitals, schools). – Improvement of international trade relations.
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Application activity 14.2

Explain how supportive government policy contribute to the development of industrial sector in Africa

14.3. Problems, solutions, and prospects for industrial development in Africa

Learning activity 14.3

A group of young entrepreneurs in Nigeria start a food-processing factory. They struggle with high electricity costs, poor roads, limited capital, and competition from imported goods. The factory creates jobs but causes pollution, attracts rural migrants, and affects nearby farms. Government proposes reforms to support sustainable industrial growth.

1. Identify three challenges faced by the factory related to capital, energy, and markets.
2. Why does competition from developed countries threaten local industries?
3. Mention two environmental problems caused by the factory.
4. Suggest three solutions the government can use to support this factory.

14.3.1. Problems affecting and resulting from Industrial development in Africa

This table clearly distinguishes between the challenges hindering industrial growth and the negative consequences that industrialization itself can cause.

Problems/challenges affecting industrial growth in Africa	Problems resulting from industrial development (Negative impacts)
– Limited market due to low purchasing power – Competition from manufactured goods from developed countries; competition due to similar local products. – Shortage of capital to invest; limited access to credit and high interest rates. – High operational costs due to expensive equipment and energy. – Limited raw materials (minerals) due to exhaustion of some deposits. – Poor and inefficient transport networks; limited and expensive energy for industrial consumption. – Political instability discouraging foreign investors; low level of technology in industrial development and management.	– Pollution of air (harmful gases), water, and land. – Global warming and pollution of surrounding areas due to large quantities of heat (boilers/turbines). – Destruction of the environment during the location/setting up of industries. – Rural-urban migration, as most industries are concentrated in urban centers. – Neglect of other sectors of the economy, especially agriculture, due to employment opportunities in mining/industry.

14.3.2. Solutions and prospects for industrial development in Africa

The following are some ways of improving and expanding the industrial sector in Africa:

- Providing adequate and efficient power supplies by investing in new power plants
- Improving the quality of infrastructure in order to promote transportation of raw materials and final products.
- Developing information and communication technologies in order to reduce transaction costs in the industrial sector.
- Widening the market base by expanding regional markets and international markets.
- Adding value to the quality of products manufactured to make them marketable and competitive.
- Providing training programmes in managerial, technical and entrepreneurial skills.
- Promoting a conducive business environment.
- Developing modern transport network to ease transportation of goods and services
- Encouraging industries that are more efficient in the use of resources for example, those that generate less pollutants and waste.

Application activity 14.3

1. Explain how improved energy supply and ICT can reduce production costs.
2. Why is value addition important for African industrial products?
3. Propose one sustainable industrial practice that reduces pollution and waste.

End of unit assessment

❖ Knowledge and understanding:

1. Fill in the blank:
 - One of the most important factors that attract industries to Africa is the availability of _____, which ensures continuous production.
2. Short answer
 - Explain two reasons why industrial development is important for African countries.

❖ Skills

3. Scenario-based: A Nigerian food-processing factory struggles with high electricity costs, poor roads, and competition from imports.
Question: Suggest two practical solutions the government could implement to help the factory thrive.
4. A cement factory is built near limestone but far from the market. What is the biggest drawback of this location?
A. High cost of raw materials C. High transport cost for finished products
B. Lack of skilled labor D. Excessive government taxes
5. A textile factory in Egypt causes air and water pollution but provides jobs. Suggest one strategy that balances employment and environmental protection.

❖ Attitudes and values

6. Why should African youths support local industries even when imported goods are cheaper?
7. A factory is being planned near a river, which provides water for production. Local communities are concerned that the factory may pollute the river and harm fishing and farming. As a responsible citizen, what two actions would you take to ensure industrial development is both profitable and environmentally friendly?

UNIT 15

TRANSPORT, COMMUNICATION AND TRADE IN AFRICA



Key Unit Competence:

Evaluate the impact of transport, communication and trade on sustainable development in Africa.

Learning objectives: At the end of this unit you should be able to:

Knowledge and understanding

1

- Identify the types of transport in Africa
- Identify the factors influencing the development of transport in Africa
- Outline the importance of various types of transport.
- State the problems affecting transport
- Identify types of communication
- Outline the factors influencing the development of communications in Africa
- State the importance of various types of communications in Africa
- State the problems affecting communication in Africa
- Identify the types of trade in Africa
- Identify the importance of trade in Africa.
- Explain the problems hindering trade in Africa.
- Identify the regional integrations in Africa, and internal and international trade.

Skills

2

- Compare different types of transport and establish the most suitable for Africa
- Analyse the factors influencing the development of transport in Africa
- Evaluate the importance of various types of transport
- Analyse the problems of transport in Africa and propose solutions.
- Categorize different types of communication systems used in Africa.
- Analyse the factors influencing the development of communication in Africa.
- Evaluate the importance of various types of communication.
- Analyse the problems affecting communication in Africa and propose solutions

3

Attitudes and values

- Appreciate the importance of transport in the development of African countries
- Appreciate the importance of communication in the development of African countries
- Appreciate the importance of trade in supporting the development of other economic activities in Africa

15.0. Introduction

Transport, communication, and trade link people and markets, support economic activities, promote regional integration, and enhance development by enabling movement, information exchange, and exchange of goods across Africa.

Introductory activity

Read the following story then answer the questions that follow:

Akaliza, a sales lady in Kicukiro, owns a boutique selling businesswomen's clothes and offers mobile money transfer services. She received a call from her mother in Nyamata about their father's annual commemoration and to buy cheaper city rice and wheat flour. She travelled by bus, preferring Rwanda's good roads, eager to reunite with her mother after being apart for a long time.

Using the story above:

1. Identify the various things that point towards transport, communication and trade.
2. Give a brief explanation why you think these three services are important to the people in Rwanda.

15.1. Transport in Africa

15.1.1. Types and factors influencing transport development in Africa

Learning activity 15.1.1

Akayesu studies transport in Africa, which moves goods and people, linking resources and markets. She found that major types of transport include land, water, and air. Roads like the Trans-African Highway and railways like TAZARA promote trade, industry, and urban growth. Waterways and the Suez Canal facilitate cheap transport, while air transport is fastest.

1. List the major types of transport in Africa.
2. Identify at least two major roads and railways in Africa.
3. Explain the factors that are affecting the development of transport in Africa.

❖ Major types of transport development

Transport refers to the physical movement of goods and people from one place to another. It is one of the aids to trade since it facilitates movement of goods from the source to the destination. The major types of transport in Africa are Land transport, Water transport and Air transport.

Means and types of transport

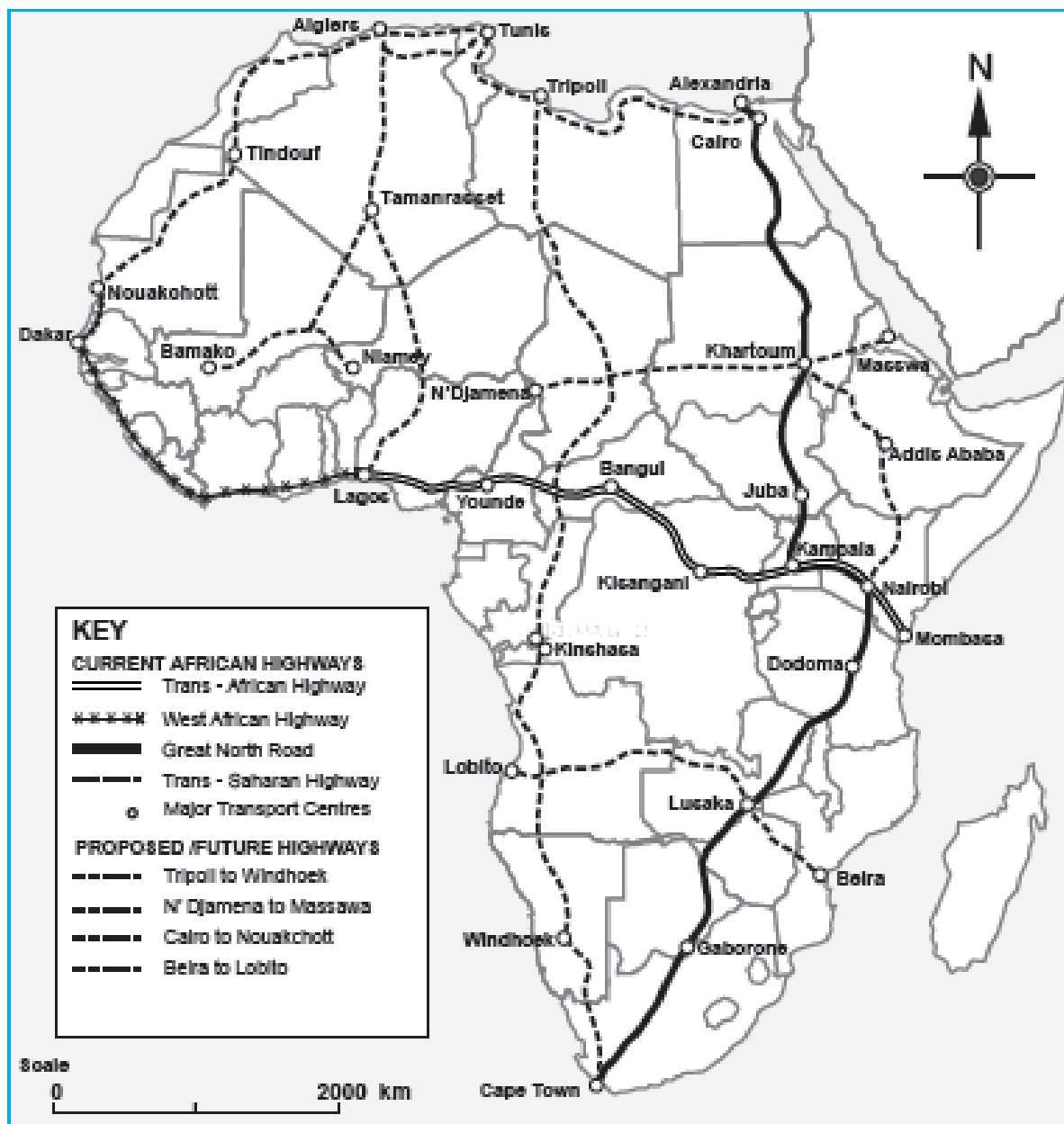


A. Land transport

It is a transport that takes place on land. It includes road transport, railways transport, pipeline transport, animal transport (in desert areas), human portage and transmission lines (movement of power /Electricity).

- (i) **Road transport** involves the use of motor vehicles, bicycles and motorcycles on **trunk roads/highways** (tarmac roads) and **feeder roads** (loose surface roads).

The main African roads are the **trans-African Highway** and the **Great North Road or Pan-African Highway** which is linking the important cities like Cape Town, Johannesburg, Pretoria, Harare, Lusaka, Arusha, Nairobi, Khartoum and Cairo.



❖ Advantages and disadvantages of road transport

Advantages	Disadvantages
– They are flexible: there is accessibility to various places. – They are reliable since they can be used at one's convenience due to the fact that there is no time schedule. – Link people, resources and activities.	– Easily affected by weather (heavy rain, floods). – Environmental problems: Noise and air pollution. – Not suitable for transport of bulky goods; it is costly and road get damaged.

- Cheaper and faster to use over short distances. - Suitable for movement of light goods. - Appropriate in hilly areas compared to other means such as railways. - They link other means of transport like ports, railway stations and airports. - They lead to establishment of market centres in rural areas.	- Roads frequently get congested (traffic jam) leading to delays of movement of people and goods. - There are many road accidents due to poor visibility. - Topographical features (mountains, hills) make roads construction difficult.
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❖ A case study of the Trans-African Highway

Trans-African Highway starts from Mombasa (Kenya) through Uganda, DRC, Central Africa, Cameroun and ends in Lagos. It links the east coast of Africa to the West coast.

❖ Advantages of the Trans-African Highway

- It has increased trade between West Africa and east Africa.
- Landlocked countries such as Uganda and the Central African Republic have got an easy access to the sea.
- It has stimulated the exploitation of natural resources
- It has encouraged the growth of trading centers and towns such as Nairobi, Kampala, and others.
- It has stimulated the growth of agricultural activities along its length.
- It has led to industrial development.
- It has encouraged political and economic cooperation among the states of Kenya, Uganda, DRC, Cameroun and others through which it passes.
- It has encouraged tourism
- It has encouraged the spread of new ideas in different sectors of economy.

❖ Obstacles Faced by the Trans-African Highway

- There are political differences between countries that reduce the effective use of the road.
- Some sections of the road are difficult to use in rainy seasons because they are not under tarmac.
- There are high costs of maintenance by individual countries.

(ii) Railway transport:

It involves trains moving on constructed track routes known as railways.

This form of transport is not well developed in many parts of the continent.

Reasons for poor development of railway in Africa

- Insufficient capital for the construction of railways
- Large areas of Africa are still uninhabited making it very uneconomical to construct railway lines through them
- Political factions prevent the economic and sensible planning of railway networks
- Hindrances of physical factors such as highlands, dense forests and arid lands
- Limited trade among the African countries
- Lack of adequate relevant technology in the construction and maintenance of railway lines
- Limited skilled manpower needed in the construction of railway lines

❖ Advantages and disadvantages of railway transport

Advantages	Disadvantages
– Suitable for transporting heavy and bulky goods over a long distance. – Carries many people for a long distance – Are time bound, making them to be suitable to passengers and traders who work with tight time schedules – They are not affected by congestion – Trains are less prone to accidents compared to motor vehicles – It is very cheap to maintain railways once they are constructed. – Less affected by weather	– Lacks flexibility can only go where rail lines are constructed. – Construction costs are high. – Physical landscapes (mountains, hills) are obstacles to construction. – Not suitable for perishable products like flowers, fruits, vegetables or unrefrigerated milk due to low speed. – They cannot be used during the time of construction.

❖ Case study of Railway Transport in South Africa

South Africa has the most developed railway network on the African continent. Some railway lines are electrified. E.g.: Durban to Johannesburg is one of the longest electrified lines in the world.

❖ Importance and problems (shortcomings) of railway transport in South Africa

Importance	Problems (shortcomings)
– It has facilitated the exploitation of minerals – It has facilitated the exportation of goods produced in south Africa and importation of goods from the foreign countries – Railway lines link the interior to the coast; thus facilitating the importation and exportation of products – Promotion of agriculture – They have stimulated the development of manufacturing industries – Development of urban centres – Development of tourist industry in south Africa – Generation of employment opportunities to many people	– It is not effective for the transportation of perishable commodities because it is slow – It is limited by the terrain of steep gradients – Double costs have to be incurred of loading and unloading at the railway terminals.

❖ A case study of Tanzam or Tazara Railway

The Uhuru (TAZARA) Railway was built between 1970 and 1975 by the Tanzania-Zambia Railway Authority. It runs 1,870 km from Kapiri-Mposhi in Zambia to Dar es Salaam in Tanzania. Constructed with Chinese finance and technical assistance, it was commissioned in 1975 to provide Zambia an alternative sea route and open up productive but remote areas of southern Tanzania.

❖ Benefits and problems (shortcomings) of the Tanzam railway

Benefits	Problems (shortcomings)
– Provided landlocked Zambia access to the sea route. – Increased productivity in remote parts of Tanzania – Strengthened political and economic cooperation between Zambia and Tanzania. – Stimulated mineral exploitation, copper Zambia, coal iron southern Tanzania. – Facilitated transport of heavy machinery and oil imports inland. – Stimulated industrial growth in Zambia and Tanzania. – Provided employment for drivers, technicians, and others	– It has suffered from competition from road transport – Political changes in southern Africa have lessened the need to use northern routes – Handling facilities are largely inadequate due to periodic congestion of shipping at Dar-es-salaam. – It is unsuitable for the transportation of perishable commodities due to serious delays.

- Contributed to growth and expansion of Dar-es-Salaam port facilities.
- Stimulated growth of trading centres and towns along railway

iii. Human portage and animal transport in Africa

This is the oldest method of transport. Animals like oxen, camels, donkeys and horses are used to carry heavier loads as compared to what people can carry. The chief advantage of human and animal transport is flexibility.

Addition information on advantages and disadvantages were inserted

iv. Pipeline transport in Africa

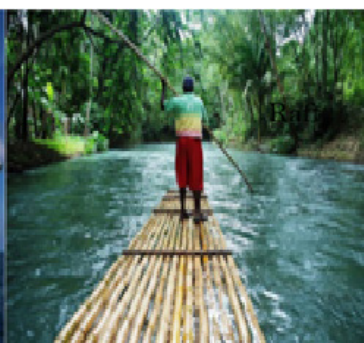
It is the transportation of liquids or fluids. The main substances which are transported are oil, water and gas. In Africa, it is highly used in Saharan countries; where oil is mined and is used to transport oil towards refineries located along the coastline.

❖ Advantages and disadvantages of pipeline transport

Advantages	Disadvantages
– It is never affected by weather – There is no congestion or delay in transportation of fluids – It does not pollute the environment – It can be very flexible, mostly for water transport – It is cheap to maintain after construction – It can transport a large quantity of commodities to its destination without interruptions.	– It can only transport one type of commodity – In case of damage, it can result in excess wastage and pollution depending on the type of fluid. – It is expensive to construct a pipeline. I added other.

B. Water transport

This is a movement of goods and passengers on water by use of ships, ferries, canoes, and boats. It is carried out on lakes, rivers, oceans and seas. Rivers and lakes are referred to as **inland water ways** while oceans and seas are referred to as **sea ways**. The two greatest advantages of water transport are that it uses existing routes and it is the cheapest form of transport for many commodities.



Generally, in Africa water transport has not been well developed. This is due to the following reasons:

- Rivers interrupted by waterfalls, rapids, and floating vegetation.
- River flows vary seasonally due to uneven rainfall patterns.
- Many rivers are short, shallow, limiting navigation use.
- Some rivers cross uninhabited, inhospitable areas with little value.
- Lack of enough capital to develop port facilities
- Political differences which limits the use of the rivers
- Low level of technology to develop ports and related facilities.

❖ River transport in Africa

Rivers in Africa are navigable for short distances especially where the speed of flow of the river is low. It is often done on some sections of rivers such as Nile, Congo, Senegal, Niger, and Volta.

❖ Lake Transport in Africa

Most of the navigable lakes in Africa are found in east Africa. The major ones are the following:

- **Lake Victoria:** it has facilitated trade between Tanzania, Kenya and Uganda. Some of the ports on this lake are Kisumu and Homa Bay in Kenya, Jinja Port Bell, Entebbe in Uganda, Bukoba, Musoma and Mwanza in Tanzania.
- **Lake Tanganyika:** it facilitates waterway linking Tanzania, Burundi, DRC and Zambia. The main ports on it are Kasanga and Kigoma in Tanzania, Bujumbura in Burundi, Mpulungu in Zambia and Kalemie in DRC.
- **Lake Malawi:** This acts as a water transport route for Malawi, Tanzania and Mozambique.
- **Lake Albert:** this links DRC and Uganda.
- **Lake Chad** in West Africa links Chad, Nigeria, Cameroun and Niger.
- **Ocean transport in Africa:** It is carried out on Atlantic Ocean, Indian Ocean, Mediterranean Sea and Red Sea linking Africa to the other continents.

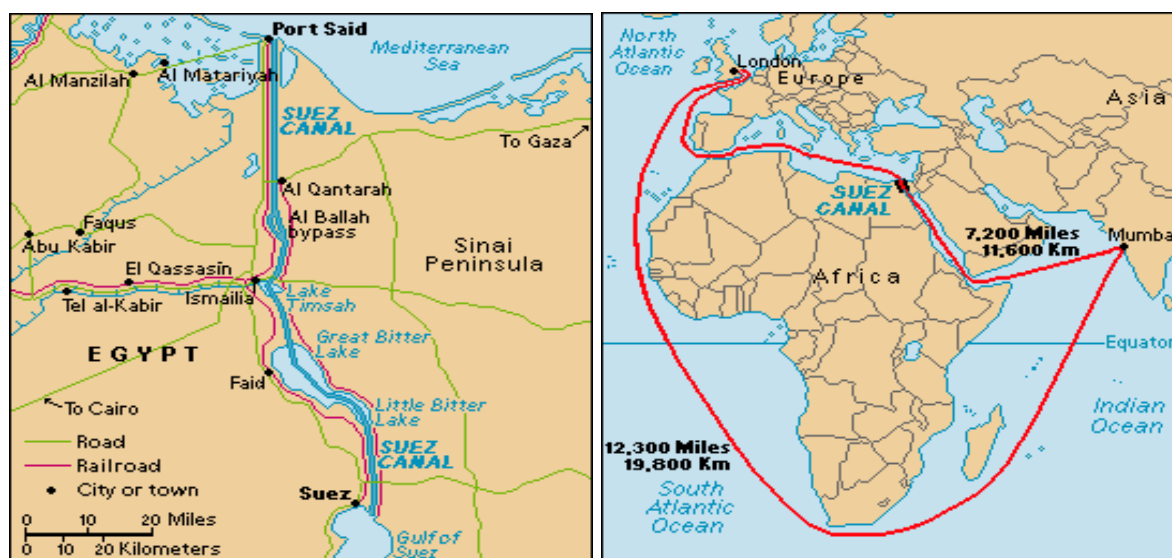
❖ Advantages and disadvantages of water transport or seaways.

Advantages	Disadvantages
– Cheap and convenient for transporting bulky goods and passengers – Less affected by robbery – Does not need any construction of routes	– Port cost are very high – It is slow thus not suitable for urgent and perishable goods – Affected by weather changes such as storms. Dry season may also reduce water volume and limit use of canals.

- | | |
|---|--|
| <ul style="list-style-type: none"> – Suitable for transportation of fragile commodities such as glass since shaking is limited. – The routes are natural and free – Waterways are smooth and do not need repair. | <ul style="list-style-type: none"> – There may be risks of accidents leading to losses of goods and human lives in case of typhoons and tsunamis. – Limited to areas with navigable water – Construction of seaport terminals is very expensive. – Sea vessels are expensive to buy, to maintain and to repair. – Limited experts to operate sea vessels. |
|---|--|

❖ A case study of the suez canal

This is an artificial waterway running north to south from the Gulf of Suez to Port Said at Mediterranean Sea. It was constructed in 1869 and it is linking the Far East with Europe. It provides a shortcut for ships operating between both European and American ports and ports located in southern Asia, eastern Africa and Oceania.



The Suez Canal on River Nile. Fig.16.4: A Shortcut for Ships provided by the construction of the Suez Canal

❖ Importance of the Suez Canal

- Promotion of trade between Europe and the Far East commodities such as cotton, tea, coffee, sugar, etc
- It reduces transport costs because of reduced distance
- It has opened up a large market for manufactured commodities from Europe and a large market for raw materials from the Far East.
- It has provided cheap water transport
- The development of industries due to access to raw materials and markets of manufactured products
- It has encouraged the development of ports

- Increase of revenue to countries such as Egypt
- It has facilitated the exploitation of natural resources

Shortcomings of the Suez Canal

- The depth and the width of the canal limits the size of vessels used on the route
- It is affected by political insecurity in the Middle East as a whole. New points were added.

C. Air transport

Air transport began with the invention of the aero plane in 1903 by the Wright brothers. It takes various forms: airstrips, commercial planes, military air crafts, helicopters, jumbo jets and gliders. The terminals for air lights are known as **airports**.



❖ Advantages and disadvantages of air transport

Advantages	Disadvantages
– It is the fastest means of transport; hence great saving of time – Suitable to carry passengers and perishable goods for long distances – Link Africa to other regions – It does not require any route construction – It is relatively independent of physical barriers – They link up the remote and inaccessible areas with developed and settled parts	– Causes air and noise pollution – Construction of airports and strips is very expensive – Aircraft consumes large quantities of fuel – It is not suitable for bulky commodities. – Competition with well-established airlines from developed countries in other continents. – Sometimes there are interruptions caused by bad weather. It is very expensive to buy airplanes.

❖ Factors influencing the development of transport in Africa

- The development of transport in Africa has been influenced by:
- Transport improved through colonial influence and indigenous routes.
- Hills and mountains make constructing routes expensive, difficult.
- Rivers, lakes aid transport; waterfalls hinder navigation.
- Dense forests make building transport networks costly, difficult.
- Heavy rainfall softens ground, making roads slippery.
- Capital enables roads, railways, airports linking economic areas
- Transport networks support trade among landlocked countries.
- Government policies influence transport network distribution efficiently.
- Advanced technology allows development of modern transport systems.
- Political instability discourages transport development in Africa.
- Skilled manpower availability mostly depends on foreign companies.

Application activity 15.1.1

1. Explain the benefits of the Suez Canal to international trade.
2. Compare land, water, and air transport in terms of speed, cost, and accessibility.

15.1.2. Importance, problems, and possible solutions for transport in Africa.

Learning activity 15.1.2

Atete, a young geography enthusiast, traveled across Africa to study transport. She discovered that mountains, dense forests, heavy rains, political instability, and lack of funds slowed the movement of goods and people. Yet, she also saw hope: with modern technology, skilled workers, smart investments, and supportive government policies, Africa's roads, railways, ports, and airports could become efficient, safe, and modern networks connecting the continent.

1. Atete observed that transport in Africa is slowed by _____, _____, heavy rains, political instability, and lack of funds.
2. According to Atete, Africa's transport networks could improve with _____, skilled workers, smart investments, and supportive government policies.
3. Discuss the challenges hindering transport development in Africa and suggest possible solutions, highlighting the role of technology, capital, and government policies.

a) Importance of transport to trade and development

- It has facilitated the extraction of natural resources.
- Transport is important to transport workers to work places like plantations, produce to processing factories and to market centers.
- Promotion of Tourism by linking tourists to the interesting areas having tourist attractions.
- Sources of government revenues as the transport companies pay taxes to the states
- It has led to urban development as many urban centers have developed where transport network converge.
- Has facilitated the mobility of passengers because people move from one place to another with the aid the transport
- Has led to industrial development as it moves raw materials to industries for processing and finished products from industries to markets.
- Has created employment opportunities to many people as a large number of people are employed in the transport sector
- Has promoted trade because it facilitates both internal and external trade as commodities are transported from one place to another.
- Improving agriculture by linking areas of agriculture such as plantation and ranches with the markets and collection centres.

b) Problems and solutions of the development of transport sector in Africa

Problems	Possible solutions
– Transport causes air, water, land pollution. – Accidents cause loss of life, property. – Construction displaces people and damages property. – Transport development encourages smuggling and illegal trade. – Developing countries depend on external transport systems. – Facilitates rural-urban migration affecting population distribution. – Steep terrain makes construction difficult and expensive. – High maintenance reduces transport efficiency, operations. – Harsh climate hinders transport during rainy seasons. – Urban traffic congestion delays people and goods. – Inadequate ports and waterways hinder transport. – Terrorism, insecurity, slow operations affect safety.	– Enforce environmental laws to reduce transport-related pollution. – Compensate and resettle people displaced by construction projects. – Strengthen border controls to curb smuggling and illegal trade. – Promote local industries and skills to reduce dependence. – Improve health screening to prevent disease spread via transport. – Enhance rural services to reduce rural-urban migration pressures. – Ensure equitable infrastructure to reduce regional and gender imbalances. – Encourage partnerships to address inadequate capital for development. – Construct all-weather routes and bridges for difficult terrain. – Modernize ports and integrate transport systems for efficiency.

Application activity 15.1.2

1. Analyze the challenges that hinder transport development in Africa.
2. Suggest solutions to improve transport networks across Africa.

15.2. Communication in Africa

15.2.1. Means of Communication and factors influencing communication in Africa.

Learning activity 15.2.1

Mukunzi, a geography student, observed how people in Africa communicate. He saw written methods like newspapers and emails, audio methods like radios and phones, and audio-visual methods like television and video calls. Communication was faster in cities with electricity, ICT, and government support, but slower in rural areas due to low literacy, poor infrastructure, and difficult terrain.

Read the text above and fill in the blanks using the following words: communication, audio, television, rural.

1. The process of sharing information from one person to another is called -----
2. Radios and phones are examples of -----methods of communication.
3. -----and video calls are examples of audio-visual communication.
4. Communication is slower in ----- areas due to poor infrastructure and low literacy.

a) Forms and means of Communication

Communication in Africa refers to the methods used to pass messages from one person to another. It plays an important role in education, trade, health, governance, and social life. Effective communication helps people share information quickly and supports social and economic development across the continent.

Forms of communication	Means of communication
1. Written communication	Letters, newspapers, printed materials, handwritten materials, billboards, emails.
2. Audio (verbal) communication	Face -to-face conversation, telephones, mobile phones, radio.
3. Audio-visual communication	Television, videos, video calls, online learning platforms.
4. Electronic communication	Internet, e-learning platforms, satellites, fibre optics, emails, mobile phones

b) Factors influencing the development of communication in Africa

- **Availability of capital:** Capital enables construction and purchase of communication facilities across Africa.
- **Literacy levels:** Educated people use communication facilities more than illiterate populations.
- **Availability of electricity:** Electricity supply is essential for operating communication tools effectively.

- **Mountainous relief:** High mountains influence placement and construction of communication transmitters.
- **Climate:** Clear weather improves communication, while heavy rain hinders operations.
- **Technology development:** ICT advancement boosts communication, poorly developed regions lag behind.
- **Government policies:** Policies supporting ICT development encourage better communication infrastructure nationwide.

Application activity 15.2.1

1. Regions with low _____ levels often use fewer communication facilities.
2. Mobile phones and radios are examples of _____ communication.
3. Discuss how technology, government policies, and infrastructure influence communication development in Africa.
4. Explain the major forms and means of communication in Africa and discuss the factors influencing their development.

15.2.2. Importance, problems, and possible solutions for communication in Africa.

Learning activity 15.2.2

Akaliza, a young African student, explored communication across Africa. She learned that telecommunication, the Internet, and mobile phones transformed trade, education, and tourism. Farmers connected to markets, students accessed e-learning, and businesses completed transactions faster. Yet challenges remained: low literacy, limited electricity, poor government policies, high costs, insecurity, and inadequate technology. Solutions included training, infrastructure development, private sector investment, and international cooperation to improve accessibility and efficiency.

1. Communication enables farmers to link directly to _____.
2. One major problem affecting communication in Africa is low _____.
3. Modern communication has promoted _____ between countries.
4. Evaluate the role of communication in promoting economic growth and social development.

a) Importance of communication in Africa include:

- Communication has boosted trade allowing transactions across Africa globally.
- Telecommunication services make business transactions faster, efficient, and convenient.
- Communication facilities enable sending money quickly across different countries worldwide.

- It shortens distances, enabling instant worldwide communication and mobility.
- Communication promotes international relationships and friendships between various countries.
- Employment in communication sector improves living standards for many people.
- It allows faster transmission and reception of ideas and information.
- Communication helps disseminate information to remote African areas effectively.
- It fosters economic cooperation among countries across the continent.
- Communication facilitates education using distance learning and online e-learning platforms.
- It promotes agricultural growth by linking farmers directly to markets.
- Communication enhances tourism by connecting tourists to touristic attractions.

b) Communication Problems in Africa may include the following:

- Small proportion of Africans use modern facilities like TV, Internet.
- Limited capital restricts investment in developing communication infrastructure efficiently.
- Low technological levels hinder modernization of communication systems widely.
- Natural occurrences interfere with normal communication system operations frequently.
- Inadequate skilled ICT labour limits effective communication service delivery.
- High illiteracy levels prevent proper use of communication facilities.
- Low incomes restrict access to communication technologies for many Africans.
- Limited electricity connectivity reduces usage of modern communication technologies.
- Poor government policies limit effective development of communication systems.
- Insecurity disrupts communication functioning in regions with unstable environments.
- Language barriers prevent many Africans from effectively using communication tools.

c) Possible Solutions to Communication Problems

- Train ICT personnel to address shortage of skilled communication workers.
- Popularize ICT in schools and colleges to improve communication literacy.
- Regularly maintain and rehabilitate existing communication equipment and systems.
- Encourage private sector investment to improve communication infrastructure.
- Cooperate with development partners like ADB, WB, EU for support.
- Promote education for all ages to reduce illiteracy and language barriers.

- Improve infrastructure and electricity supply to support communication development.
- Ensure peace and security to maintain stable communication systems.

Application activity 15.2.2

1. The Internet allows students to access _____ learning resources.
2. Mobile phones facilitate faster _____ of goods and services.
3. Discuss the importance of communication in Africa for trade, education, and tourism.
4. Suggest solutions to overcome challenges affecting communication development in African countries.
5. Explain how technology and government policies influence communication development in Africa.

15.3. Trade in Africa

15.3.1. Types and importance of trade in Africa

Learning activity 15.3.1

Kofi, a young trader from Ghana, explored how trade shaped Africa. He observed internal trade in local markets, external trade exporting coffee to Europe, wholesale and retail trade, barter exchanges in rural areas, and e-commerce online. He realized that trade boosts economic growth, creates jobs, encourages specialization, develops infrastructure, improves international relations, and connects farmers, businesses, and consumers across Africa.

1. Internal trade occurs when goods are bought and sold _____ the country.
2. Barter trade exchanges goods directly without using _____.
3. E-commerce trade is conducted primarily over the _____.
4. Trade creates employment in production, selling, and _____ of goods.

a) Types of trade in Africa

Trade is the buying and selling of goods and services to make a profit. **Goods** are tangible items like tea, cattle, and minerals, while services are intangible like education, banking, tourism, and insurance. Historically, communities used **barter trade**, exchanging goods directly, but today trade uses **money** as a medium of exchange.

Trade can be classified into **different types** based on location, participants, and the nature of goods:

- **Internal/Domestic Trade:** Trade occurring within a country, buying and selling goods and services internally. Examples include selling vegetables from Musanze farms to Kigali markets or purchasing school supplies from local shops.

- **External/International Trade:** Trade between countries, involving exports and imports. It can be **bilateral** (between two countries, e.g., Rwanda and Uganda) or **multilateral** (many countries, e.g., world market). Examples: Rwanda exporting coffee and tea to Europe or importing cars and electronics from China. **Exports** sell goods abroad, **imports** bring goods in; in Africa, export values are generally lower than import values, which makes it experiencing trade **imbalance** or **unfavourable** trade.
- **Wholesale trade** involves buying and selling goods in large quantities, usually to retailers rather than direct consumers. Example: A trader in Kigali buying 100 sacks of beans from Musanze farmers to sell to shops in the city.
- **Retail trade** involves selling goods in small quantities directly to consumers. Example: A shop selling sugar, salt, or vegetables to households
- **Barter trade** is the exchange of goods for goods without using money. This form of trade was common in traditional African societies and is still practiced in some rural areas. Example: A farmer exchanging a basket of bananas for a goat
- **E-commerce (Online Trade)** is trade conducted over the internet. Buyers and sellers use websites, social media, or apps to buy and sell goods. Examples: Buying clothes from Jumia or Amazon.

b) Importance of trade in Africa

Trade is very important for the economic, social, and cultural development of Africa. Some key reasons why trade is important include:

- Trade enables a country to acquire goods it does not produce. For example, industrialized nations like Britain, France, Germany, and Japan require raw materials from African countries, while African countries need manufactured goods such as machinery, vehicles, and chemicals not made locally.
- Trade helps countries maintain good relationships with trading partners and encourages business tourism.
- Trade leads to the expansion of towns, e.g., Kigali grows through trading activities.
- Trade promotes economic growth by allowing countries to sell goods and services, earn income, and generate foreign exchange for development projects, e.g., Rwanda exports coffee and tea.
- Trade creates employment opportunities in production, transportation, selling, and marketing; for example, traders, market sellers, and transporters earn livelihoods.
- Trade provides access to goods and services not produced locally, e.g., Rwanda imports cars, electronics, and medicines.
- Trade encourages specialization, letting countries focus on goods they produce best, e.g., Rwanda with tea and flowers, South Africa with minerals and machinery.
- Trade promotes international relations and cooperation, e.g., agreements under the African Continental Free Trade Area (ACFTA).
- Trade encourages infrastructure development such as roads, ports, and

communication systems to transport goods and information efficiently.

- Trade supports agriculture and industry, motivating production for local and international markets, improving productivity and development.

Application activity 15.3.1

1. Trade encourages infrastructure development such as roads, ports, and ____.
2. Discuss the types of trade in Africa and give examples.
3. Explain the importance of trade for Africa's economic and social development.

15.3.2. Problems hindering trade in Africa and possible solutions

Learning activity 15.3.2

Umutesi, a young African entrepreneur, wanted to expand her business across the continent. She realized that political instability, poor transport, tariffs, neocolonialism, limited markets, and low technology slowed trade and increased costs. Determined, she joined regional cooperatives, invested in quality goods, advocated for better roads, and encouraged foreign investment, gradually boosting trade, creating jobs, and connecting communities across Africa.

Read the text above and fill in the blanks using the following words: market access, movement, unequal

- A. Poor transport facilities negatively affect theof goods in Africa.
- B. Neocolonialism createsbargaining power, disadvantaging African countries.
- C. Joining regional cooperatives can improve and trade efficiency.

i. Problems hindering trade in Africa

- Production of similar commodities causes price fluctuations across most African countries.
- Political instability and insecurity hinder production, trade, and delivery systems.
- Trade embargoes prevent countries from trading with other member states.
- Poor transport facilities negatively affect distribution of commodities across Africa.
- Landlocked countries face difficulties accessing international water trade routes.
- Protectionist policies in some countries disturb international and regional trade.

- Tariffs on imports and exports discourage trade and increase prices.
- Neocolonialism creates unequal bargaining, fixing prices to developing countries' disadvantage.
- Low technology usage slows trade flow and international capital movement.
- Limited markets due to poverty reduce purchasing power and trade activity
- Corruption of various forms.

ii. Possible solutions

- Rehabilitation of existing roads and construction of new ones.
- Regional economic grouping and cooperation can promote trade.
- Political negotiations and discussions to reduce political instability and insecurity.
- African governments have to put in place policies and mechanisms of attracting foreign investment. This may overcome the problem of similarity of goods on the market by diversification.
- Improving administrative and bureaucratic arrangements helps to make trading smart. It makes obtaining licenses easier and faster.
- Improving quality of manufactured goods so that they are attractive and competitive on the international market.

Application activity 15.3.2

1. Explain the challenges that hinder trade development in Africa.
2. Suggest solutions that could enhance trade and economic growth

15.3.3. Partnerships and regional integration (EAC, ECOWAS, COMESA, CEEAC)

Learning activity 15.3.3

Partnerships and regional integration in Africa help countries cooperate in trade, reducing tariffs, regulations, and transport costs. Trading blocks like **EAC**, **COMESA**, **ECOWAS**, and **ECCAS** promote peace, security, and economic growth. **COMESA** members export agricultural products, crafts, and minerals while importing machinery and vehicles. Private sector and civil society play key roles in supporting regional integration and development.

Answer by True or False

- a) Partnerships in trade help reduce tariffs and transport costs.
- b) COMESA aims at becoming a free trade area.
- c) The principal exports of COMESA include agricultural products and minerals.

Partnership refers to the special arrangements and organization in which member countries give concessional treatment to trade commodities and resources among the member states. It may be a form of reducing tariffs, trade requirements and regulations, and Transport costs.

Regional integration is a process in which neighboring states enter into an agreement in order to upgrade cooperation in trade through common institutions and rules.

Today a number of countries are increasingly engaging in economic cooperation and have formed trading groups. Trading blocs in Africa are:

- East Africa Community (**EAC**)
- Common Market of Eastern and southern Africa (**COMESA**).
- Economic Community for Central African States (**ECCAS**) or Communauté Economique des Etats de l'Afrique centrale (**CEEAC**)
- Economic Community of West African States (**ECOWAS**)

❖ A case study of comesa

The Common Market for Eastern and Southern Africa (COMESA): This was formed to replace the former preferential Trade Area (PTA) which had existed from the earlier days of day 1981.

❖ The objectives of COMESA

- The Promotion of peace and security in the regional
- Formation of a large economic and trading unit that is capable of overcoming some of the barriers that are faced by individual states.

The member states of COMESA are the following: Burundi, Angola, Comore, DRC, Djibuti, Egypt, Eritrea, Ethiopia, Kenya, Madagascar, Malawi, Mauritius, Namibia, Rwanda, Seychelles, Soudan, Eswatini, Tanzania, Uganda, ZAMBIA, and Zimbabwe.

COMESA aims to become a free trade area and its target is the removal of international trade tariffs and barriers.

The region's principal exports are agricultural products. These include horticulture, tea, coffee, cotton, tobacco, pyrethrum, fish, animal skins etc. Other exports include hand craft and minerals such as gold, diamonds, limestone and so on.

On the other hand, the region's major imports are machinery and other capital equipment, industrial supplies and raw materials, motor vehicles, and motor vehicle parts.

Its trading partners include the European Union, Japan, China, India, United Arab Emirates (UAE)

NOTE: The role of the Private sector and civil society is considered as important to regional integration and development.

Application activity 15.3.3

1. Explain the importance of regional integration and trading blocs in Africa.
2. Discuss the role of the private sector and civil society in COMESA development.

End of unit assessment

1. Identify the main types of transport, communication, and trade practiced in Africa.
2. Outline the importance of transport, communication, and trade in the economic development of African countries.
3. Draw a sketch map of Africa and locate three major transport routes or corridors.
4. Analyse the factors influencing the development of transport and communication systems in Africa.
5. Examine the major problems affecting transport, communication, and trade in Africa and evaluate their effects on development.



Key Unit Competence:

Analyse the impact of tourism on sustainable development in Africa.

Learning objectives: At the end of this unit you should be able to:

Knowledge and understanding

1

- List the different tourist attractions in Africa
- Outline the factors favouring tourism in Africa
- Identify the contributions of tourism to the economy of African states
- Outline the problems affecting tourism in Africa.

Skills

2

- Locate different tourist attraction areas in Africa
- Examine the factors favouring tourism in Africa
- Evaluate the importance of tourism in Africa
- Analyse the problems and prospects for tourism in Africa.

Attitudes and values

3

- Appreciate the beauty of Africa and the need to conserve its environment
- Appreciate the contribution of tourism to the economic development of Africa.

16.0. Introduction

Tourism involves travel for leisure, education, or business, contributing to Africa's economic development through attractions, conservation, employment, foreign exchange, and promotion of sustainable environmental management.

Introductory activity

During a class trip to Rwanda, students from Germany visit Volcanoes National Park and Lake Kivu, observing wildlife, scenic landscapes, and cultural sites. They notice how tourism provides income, jobs, and infrastructure development while highlighting the need for environmental conservation.

1. List four major tourist attractions in Africa.
2. Explain factors favouring tourism in Africa.
3. Analyse how tourism contributes to Africa's economy.
4. Describe problems affecting tourism in Africa.
5. Suggest lessons Rwanda can learn from other African countries to improve sustainable tourism.

16.1. The major tourist areas in Africa

Learning activity 16.1

Impano, a geography student, travels across Africa to learn about tourism. In North Africa, she visits Egypt's pyramids and Morocco's waterfalls. In West Africa, she explores Timbuktu and Djenné's ancient mud houses. Central Africa amazes her with forests and Mountain Gorillas in Rwanda. East Africa offers the richest attractions, including Mount Kilimanjaro, Maasai Mara, Zanzibar, and national parks. In Southern Africa, she enjoys wildlife, Cape Town's coast, and Victoria Falls.

1. Fill the space with the correct word
 - a) The pyramids and the Sphinx are major tourist attractions found in _____
 - b) Victoria Falls is located on River _____ in Southern Africa.
2. Which tourist attraction is found in West Africa?
A. Mount Kilimanjaro B. Timbuktu C. Victoria Falls D. Akagera National Park
3. Which region of Africa is considered the richest in tourist attractions?
A. North Africa B. Central Africa C. East Africa D. Southern Africa
4. Describe the major tourist attractions found in East Africa.

The major tourist attractions in Africa include its natural environments, national parks, historical sites, and socio-cultural activities.



- a) Tourist attractions in **North Africa**: This includes Maghreb countries and Sudan. The major tourist attractions are **historical sites and monuments** such as **pyramids and the Sphinx** of Egypt. The other major tourist attractions in this region include **famous waterfalls** of Morocco.
- b) Tourist attractions in **West Africa**: The major tourist attractions in this region are **historical monuments and sites, and various landforms**. In Mali, the city of Timbuktu is famous for its ancient African architecture. There are also the **mud houses** in the city of Djenné established around 800BC.
- c) Tourist attractions in **Central or Middle Africa**: This region is known for its historical sites, landforms, forests, animals, rivers and associated features. Among the animals we have the Mountain Gorillas in the Birunga Mountains of Rwanda.
- d) Tourist attractions in **East Africa**: This is the richest region in tourist attractions. Some of them include:
 - Mountain Gorillas in Rwandan and DRC (Birunga National Park)
 - The Maasai Mara National Reserve in Kenya which is known for its varieties of birds, mammals such as Zebras, Thomson gazelles.
 - Mount Kilimanjaro which is the largest and the highest in Africa.
 - The Zanzibar's rich history with amazing beautiful beaches and the Stone Town of Zanzibar.
 - The Omo River Region in the south Western side of Ethiopia. This region is known for its rich cultural heritage.
 - Akagera National Park in Rwanda
 - Queen Elizabeth National Park in Uganda
 - The source of River Nile in Rwanda, Burundi, Uganda and Tanzania

e) Tourism in Southern Africa: The major tourist attractions in this region are the National parks with a wide variety of wildlife. There are also scenic beauty, beaches and sea creatures such as dolphins and seals in Cape Town. The other major tourist attraction in this region is **Victoria Falls** on River Zambezi which is the largest waterfall in the world.

Application activity 16.1

1. Match the tourist attraction in **Column A** with its region in **Column B**.

Column A	Column B
i. Mount Kilimanjaro	a. West Africa
ii. Timbuktu	b. East Africa
iii. Victoria Falls	c. Southern Africa
iv. Mountain Gorillas (Birunga)	d. Central Africa

2. Explain the importance of tourism to the economic and social development of African countries.

16.2. Factors and contribution of tourism to the development of Africa.

Learning activity 16.2

Africa welcomes visitors with warm climates, stunning sceneries, and rich wildlife. A traveler admires snow-capped mountains like Kilimanjaro, the Great Rift Valley, waterfalls, lakes, and sandy beaches. Savannahs and forests host elephants, lions, rhinos, and gorillas. Beyond nature, Africa's culture shines through traditional dances, games, and dressing styles. Historical monuments, museums, and pyramids tell ancient stories. Friendly hospitality, political stability, good roads, airports, hotels, and supportive government policies make tourism grow and contribute to Africa's development.

- Fill in the blanks with the appropriate word
 - Snow-capped mountains like Mount Kilimanjaro and Mount Kenya are examples of _____ factors favoring tourism.
 - Friendly and welcoming behavior towards visitors is referred to as _____.
- Which of the following is a physical factor promoting tourism in Africa?

A. Political stability	B. Training in tourism
C. Wildlife diversity	D. Government policy
- Which human factor helps tourists access destinations easily?

A. Warm climate	B. Sceneries
C. Infrastructure development	D. Flora and fauna

❖ **Physical factors influencing tourism in Africa include the following:**

- Good climate with warm and hot climatic conditions
- Presence of sceneries like snow-capped mountains such as Mt. Kenya, Kilimanjaro and Ruwenzori. Other sceneries features are the Great East African Rift Valley, with its associated features like waterfalls, Lakes, Volcanoes, sandy beaches, and so on.
- Presence of wildlife (Flora and Fauna). There is a variety of animals such as elephants, buffaloes, lions, rhinos, leopards, gorillas, etc. There is also a variety of plant species like mangroves, indigenous trees, and so on.

❖ **Human factors influencing tourism in Africa include the following:**

- Traditional culture such as dances, games and dressing styles
- Monuments and historical sites. These include pyramids and museums.
- Hospitality is characterized by a friendly and welcoming culture.
- Political stability in many African countries.
- Well-developed infrastructures in form of roads, airports, and internet connection in touristic areas.
- High-class international hotels and lodges for accommodation.
- Favorable government policy supporting investors in the tourism sector
- Good communication facilities for publicizing tourism
- Training in tourism.

Application activity 16.2

1. Match the factor in **Column A** with its category in **Column B**.

Column A	Column B
(1). Wildlife (lions, elephants)	a. Human factors
(2). Traditional dances	b. Physical factors
(3). Roads and airports	a. Human factors
(4). Warm climate	b. Physical factors

2. Explain the physical factors that favor the development of tourism in Africa.

3. Discuss how tourism contributes to the socio-economic development of African countries.

16.3. Problems affecting tourism in Africa and prospects for sustainable development

Learning activity 16.3

Tourism supports Africa by creating jobs, earning foreign exchange, and improving infrastructure, but it faces serious challenges. A traveler notices poor funding, political instability, insecurity, poaching, environmental destruction, and shortage of accommodation near tourist sites. Social problems like crime and changing values also affect tourism. However, Africa has hope for sustainable development through better security, protecting wildlife, training game rangers, improving infrastructure, allocating funds to tourist sites, and establishing environmental bodies to conserve nature and ensure tourism growth.

1. Fill in the blanks
 - a) Tourism contributes to development by creating _____ opportunities and earning foreign exchange.
 - b) The illegal killing of wild animals for their products is known as _____
2. Which of the following is a problem affecting tourism in Africa?
 - A. Training game rangers
 - B. Allocating funds to tourist sites
 - C. Political instability
 - D. Improving infrastructure
3. Which measure promotes sustainable tourism development in Africa?
 - A. Poaching wild animals
 - B. Environmental destruction
 - C. Establishing environmental protection bodies
 - D. Civil wars.

a) Problems affecting tourism in Africa and prospects for sustainable development

However, tourism contributes to Africa's development by providing a source of foreign exchange and local revenue, creating employment opportunities, improving infrastructure, promoting wildlife conservation through ecotourism, boosting agriculture to meet food demands, and encouraging the establishment of tourism-related training institutions. It is affected by several problems that include:

- Inadequate capital to invest in expanding tourism industry
- Poaching of wild game which lead to extinction of some animal species
- Political instabilities and civil wars
- Insecurity and terrorism
- Destruction of environment due to the need for land for agriculture, mining areas, road construction and so on.
- Shortage of accommodation near the tourist potentials.
- Changes in social values
- Increase in crimes such as prostitution and drug abuse.
- Disturbance of wild animals and environment
- Over-focusing on tourism.

b) Prospects for sustainable development of tourism in Africa

- Allocating funds for the purpose of taking care of tourist sites
- Reinforcing security through cooperation with neighboring countries to combat terrorism
- Training game rangers on measures to curb poaching
- Improving infrastructures in touristic areas for supporting tourism
- Protecting wildlife against encroachment
- Establishing environmental bodies in charge of controlling pollution and degradation of the environment.

Application activity 16.3

1. Match the problem or solution in Column A with the correct category in Column B.

Column A	Column B
i. Poaching of wildlife	a. Sustainable solution
ii. Political instability	b. Problem affecting tourism
iii. Training game rangers	a. Sustainable solution
iv. Environmental degradation	b. Problem affecting tourism

2. Describe the major problems affecting tourism in Africa.
3. Explain the prospects for sustainable development of tourism in Africa.

End of unit assessment

❖ Knowledge and understanding:

1. List the major tourist attractions found in Africa.
2. Distinguish between domestic (internal) tourism and international tourism.

❖ Skills

3. Examine the factors favouring the development of tourism in Africa.
4. Locate and name major tourist attraction areas on a sketch map of Africa.
5. Explain the contributions of tourism to the economic development of African countries

❖ Attitudes and Values

6. Propose strategies for sustainable tourism development.
7. Analyse the problems affecting tourism in Africa.

UNIT 17

AGRICULTURAL CASE STUDIES IN DEVELOPED COUNTRIES: NETHERLANDS AND USA.



Key Unit Competence:

Assess agricultural systems on the Polder Lands of the Netherlands and agricultural systems in the USA in relation to sustainable development.

Learning objectives: At the end of this unit you should be able to:

Knowledge and understanding

1

- Identify the location, size and population of the Netherlands and USA.
- Outline the general agricultural practices on the Polder Lands and forms of agri-business in the USA
- Explain the factors favouring agriculture development on the Polder Lands and in the USA
- Identify the importance of agriculture in the Netherlands.
- Identify the problems hindering agricultural expansion and suggest possible solutions on the Polder Lands and in the USA
- Identify regionalised areas for agriculture in the USA and the major crops grown.

Skills

2

- Examine the agricultural practices on the Polder Lands in the Netherlands and USA
- Evaluate the factors favouring farming on the Polder Lands and USA
- Analyse the general importance of agriculture in the Netherlands and agri-business in the USA.
- Assess the problems or challenges faced by agricultural practices and suggest possible solutions on the Polder Lands and in the USA
- Analyse the regionalisation of agriculture in the USA.

Attitudes and values

3

- Appreciate the level of agricultural development in the Netherlands and USA.
- Appreciate the importance of agricultural systems in Netherlands and USA.
- Show respect for livestock farming (dairy farming) in the Netherlands and USA.

17.0. Introduction

This unit examines agriculture in the Netherlands and the USA, focusing on locations, farming systems, regionalisation, agri-business, challenges, and the factors supporting advanced agricultural development.

Introductory activity

A Rwandan student visits the USA and the Netherlands, observing mechanised farms, greenhouses, dairy systems, and agri-business. He notes advanced technology, irrigation, and high crop yields, contrasting these with Rwanda's small-scale, subsistence farming practices.

1. Identify the location of the USA and the Netherlands.
2. Outline the main agricultural systems practiced in the USA and on the Dutch Polder Lands.
3. Using examples, explain how technology improves agricultural productivity in both countries.
4. Analyse factors favouring agriculture in the USA and the Netherlands.
5. Compare these systems with agriculture in Rwanda and suggest improvements for local farming.

17.1. Agriculture on Polder-Lands in Netherlands

17.1.1. Presentation of Netherlands

Learning activity 17.1

Ishimwe, a geography teacher at GS St. Aloys Rwamagana, explored the Rhine River, which flows through the Rhineland: Netherlands, Germany, Belgium, Luxembourg, and Switzerland. She learned that the Netherlands, also called Holland, lies between the North Sea and Belgium, in low-lying areas of the Great North Europe Plain. The Rhine forms a delta and floodplain, making the land fertile but flood-prone. The Dutch use land reclamation, such as draining swamps and irrigating dry land, to increase farmland and settlements, unlike land rehabilitation which restores land to its original state.

1. What is another name for the Netherlands?
2. Which countries are included in the Rhineland?
3. Where does the Rhine form its delta and floodplain?

Netherlands also called Holland is one of the countries through which the Rhine flows. The Rhineland countries include: Netherlands, Germany, Belgium, Luxembourg, and Switzerland. The Netherlands, also called Holland, is a Western European country through which the Rhine flows, forming its delta and floodplain.

It lies between the North Sea and Belgium, in the low-lying Rhinelands of the Great North Europe Plain, covering 33,717 km² with a 2024 population of about 17,671,125. It is one of the world's most densely populated countries.



Land reclamation is a state where land regarded as wasted land is converted into useful land. It is different from **land rehabilitation** which involves the restoration of land back to its original state. **Reforestation and terracing** to stop soil erosion are examples of land rehabilitation while **drainage of swamps and irrigation of dry land** are typical examples of land reclamation.

Application activity 17.1

1. Why is the Netherlands densely populated?
2. What is land reclamation and how does it differ from land rehabilitation?
3. Give examples of land reclamation and land rehabilitation.
4. Why is land reclamation important in the Netherlands?

17.1.2. Crop cultivation on the Polder Lands in the Netherlands

Learning activity 17.1.2

Land reclamation in the Netherlands, starting in the 17th century, created polders for livestock farming and horticulture. Projects like the Zuyder Zee and Delta Plan addressed land shortages, floods, salinity, and freshwater supply. Polders are reclaimed using dykes, canals, and pumps, then prepared with desalination, ploughing, and crops. The Zuyder Zee Project added 220,000 ha, controlled floods, improved drainage, and supported agriculture, industry, and settlement. Flat relief aids mechanization, canals, wind energy, and urban development.

1. Fill in the blanks

- a) Land reclamation in the _____ started in the _____ century.
- b) Polders are _____ land enclosed by _____ used for farming, settlements, and other purposes.

2. Which of the following is a major land reclamation project in the Netherlands?

- | | |
|----------------------------|------------------------|
| a) Rhine Delta Project | b) Zuyder Zee Project |
| c) North Sea Canal Project | d) Delta Works Project |

3. Explain how flat relief and horticulture contribute to farming, industry, and urban development in the Netherlands.

a) Meaning of Polder Lands

In Netherlands, land reclamation started in the 17th Century. Reclaimed lands, also Polders in Netherlands, are mainly used for: Livestock farming, including dairy farming, ranching and domestication of animals such as pigs, camels, mules and buffaloes. Horticulture, which includes the growing of fruits, vegetables and flowers. This is mainly done in greenhouses.

The following are reasons that led to land reclamation in Netherlands:

- Shortage of land for farming, roads, industries and for urban expansion due to high density of the population
- Need for controlling floods as about half of the country is below sea level.
- Need for improved drainage and quality of existing land
- Reduce salinity in soils along the coast.
- Shorten the coastline by use of dykes and dune.
- Create freshwater lakes such as Lake Yssel so as to provide water for domestic and industrial uses.

The major reclamation projects in the Netherlands are Zuyder Zee project and the Delta plan.

A **"polder"** is a low-lying reclaimed land enclosed or surrounded by dykes that protect the area against higher water levels outside. The polders include **Wieringermeer** (20000 ha), **Markerwaard** (60000 ha), **North eastern polder** (48000ha), **Eastern Fleroland** (54000 ha), **South Fleroland** (44000 ha) and the **fresh water lake Yssel** (125000 ha).

How a polder is reclaimed?

- A ring dyke is constructed to separate the polder from the rest of water body.
- A ring canal is then constructed outside and surrounds the ring dyke.
- Ring canal assists in draining water pumped out of the polder. The ring canal also acts as a reservoir (stores some of the water for irrigation).
- Windmills or diesel engines, or electric power are used for pumping water out of the polder.
- The marshy land enclosed by the dykes is divided into portions by constructing polder ditches.
- Water from the polder, is drained into the ditches and pumped into the ring canal.

b) Methods of land reclamation in the Netherlands

- The sand dunes and dykes protect the land against high water level
- Sand dunes have been strengthened by fencing with stone and planting grass on them to prevent removal of beach material by water.
- Dykes are also built to reinforce the sand dunes.
- Drainage canals (ditches) are constructed over waterlogged or marshy polderland to help drain excess water.
- Pumps raise the excess water from the polder and discharge it into the ring canal. At this stage the polder is left to dry as it is still too wet to be used.

c) Preparation of polder-land for farming

- Desalination of soils is done by flushing the soils with fresh water or using artificial chemicals
- Sometimes fresh soils are spread on polder-lands
- The polder-land is deep ploughed to mix the soil with leached nutrients and fertilizer is added to improve the soils
- To improve the soils, pastures and hardy crops such as barley and oats are first grown. These crops use the excess moisture and help remove salts.
- The land is first farmed by the government for the first few years; then it is demarcated into small and medium size farms ready for settlement by farmers.
- Fresh water is left in canals and reservoirs to help combat the penetration of salt water. This water is used for irrigation during drought.

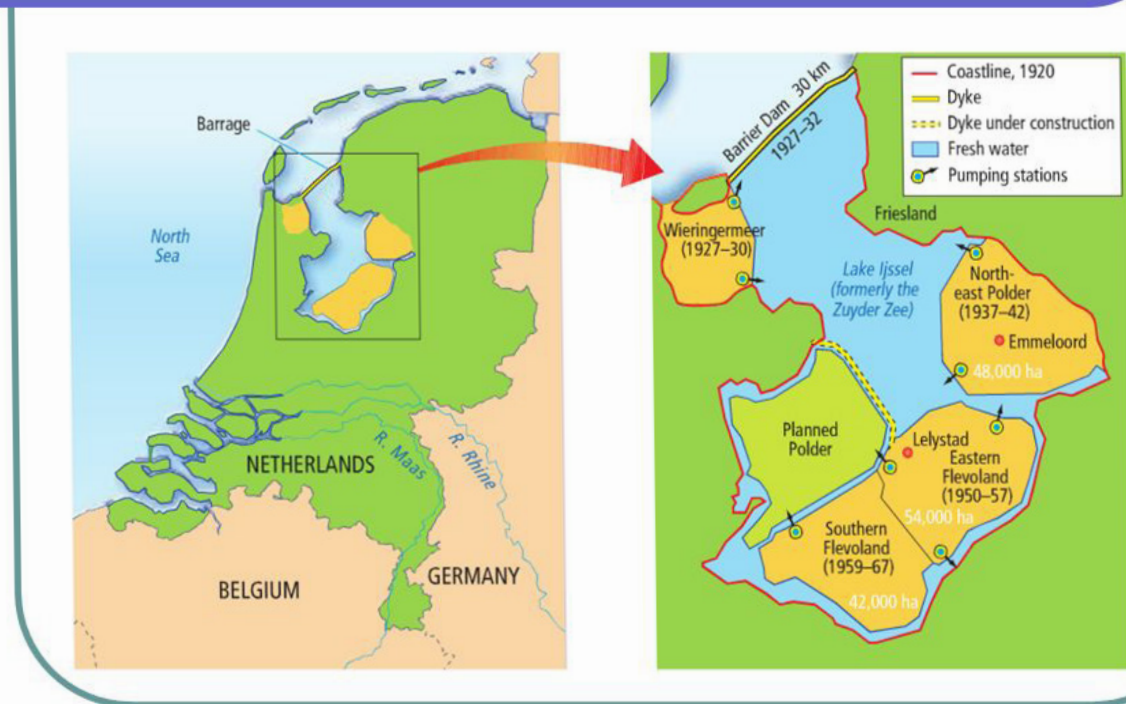
d) Advantages of Netherlands flat relief

- It has enabled the country to use wind power in land reclamation for pumping water from polder-lands
- It has facilitated canal building and construction of transport and communication facilities.

- It has provided suitable site for development of industries and for building of towns
- Farms are easily mechanized.

e) The Zuyder Zee project

The Zuyder Zee project



The Zuyder Zee lands

This project started in 1891. It was aimed at controlling sea invasions floods, providing more land for agriculture and fresh water supply. To reach to this aim; they decided to close off the bay by building a long dyke through the sea. This created Lake Yssel. This project added a total 220,000ha of arable land area to the Netherlands. This is an extra 10% of the original land area of Holland. The Zyder Zee project was designed at the beginning of the 20th century by a Dutchman called Cornelis Lely. It had three main aims:

- To protect Netherlands from the effects of the North Sea
- To increase agricultural land and therefore flood control
- To improve water supply by creating a fresh water lake.

It involved:

- The closing off of the North Sea by use of a massive dam
- The reclamation of four polders and the creation of Lake Yssel, a freshwater lake.

The Barrier dam was completed in 1932. Work on the polders started thereafter, with the Weiringermeer polder completed in 1930, the north east polder of 1942,

the Eastern Flevoland polder by 1957, the Markewaard and the southern Flevoland polders, scheduled to be completed by 1980.

a) Benefits of the Zuyder Zee project

- Creation of freshwater lake (Lake Yssel) which supplies freshwater to the north Holland and Friesland.
- Strong control of flooding from the sea and protection of the old dykes from tidal action that used to increase the cost of maintenance.
- Improving the drainage system around the new polders
- Reduction of salts and chlorine in the soils of the surrounding areas, hence increasing crop yields and expanding area for arable farming and stock breeding.
- The barrier dam was fortified with a road network that connects north Holland to Friesland.
- Freshwater reservoir is used for both agriculture and industry
- It had made easier to construct polders for settlement and for agriculture without fearing of floods.
- It has added 220,000ha for development as industrial, agricultural and even residential areas.

b) Horticulture in Netherlands

Horticulture is the growing of fruits, vegetables and flowers. Netherlands has one of the most highly developed horticultural sectors in the world. Most of crops are grown in green houses.



Application activity 17.1.2

1. What is the main use of reclaimed polders in the Netherlands?
a) Mining and fishing b) Livestock farming and horticulture
c) Industrial waste disposal d) Tourism only
2. Explain why land reclamation was necessary in the Netherlands.
3. Describe the steps taken to prepare polder land for farming.
4. Discuss the Zuyder Zee Project, its objectives, methods, and benefits to the Netherlands.

17.1.3. Livestock farming in the Netherlands: Dairy farming

Learning activity 17.1.3

Irumva, a student interested in agriculture, visited the Netherlands to learn about livestock farming. He discovered that livestock farming involves raising domestic animals such as cattle, pigs, sheep, and horses. In the reclaimed areas, dairy farming is the main type practiced, where farmers keep high-yielding Friesian cows for milk. A single farm may have up to 500 cows. During summer, the cattle graze freely, while in winter, they are fed protein-rich hay, fodder, grass, and silage indoors. Irumva also learned that other livestock includes pigs, sheep, horses, camels, mules, and buffaloes. Dairy products such as cheese, butter, and condensed milk are major exports from the country.

1. Fill in the blanks
 - a) Dairy farming in the Netherlands mainly uses _____ breed of cattle.
 - b) In winter, Friesian cows are fed _____, _____, and silage indoors.
2. Which of the following livestock is mainly reared for dairy products in the Netherlands?
 - a) Pigs
 - b) Friesian cows
 - c) Sheep
 - d) Camels.

Livestock farming refers to the raising of domestic animals such as cattle, pigs, sheep, and horses. Apart from dairy farming which is mainly practiced in the reclaimed areas in the Netherlands; there are other types of livestock reared in the country. Some farmers keep pigs for production of pork and other related products.



Others keep sheep while there are those who keep horses, camels, mules and even buffaloes are other types of livestock reared.

❖ Dairy farming in Netherlands

Dairy farming focuses on milk production. Dairy farmers mainly keep high milk yielding Friesian breed of cattle. A single farmer can keep as many as five hundred cows. The cattle are allowed to graze in free roaming stables in summer. In winter, they are kept indoors where they are fed on protein rich hay, fodder, grass and silage.

The main dairy products exported from the Netherland include cheese, butter and condensed milk.



Application activity 17.3

- Which dairy products are exported from the Netherlands?
a) Cheese, butter, condensed milk b) Yogurt, eggs, meat
c) Milk, wool, honey d) Butter, ice cream, wool
- Explain how dairy cattle are managed differently in summer and winter.
- Name at least four other types of livestock raised in the Netherlands besides dairy cows.

17.1.4. Factors influencing agriculture on the Polder lands in the Netherlands

Learning activity 17.1.4

Umumararungu studied Dutch agriculture and discovered that polder farming thrives due to fertile soils like Zuyder Zee clay, a temperate climate, flat landscapes, and abundant fresh water from rivers and lakes. Skilled labor, advanced technology such as greenhouses, good communication networks, high-quality products, government support, and strong international and local organization make agriculture intensive, profitable, and well-integrated with markets.

- Complete the statements below with the appropriate concept
a) Fertile soils such as _____ of Zuyder Zee support polder agriculture.
b) The flat landscape and _____ allow mechanization of farms.
- Which factor has enabled Dutch farmers to adopt new farming methods?
a) Abundant rainfall b) Skilled labor force
c) Import of livestock d) Dry climate
- What role does government support play in Dutch polder agriculture?
a) It provides security, infrastructure, and economic stability
b) It supplies fertilizers only
c) It imports crops from other countries
d) It controls livestock breeds.

❖ Factors influencing agriculture on the polder Lands in the Netherlands

Fertile soils: The presence of fertile soils, such as the young marine clay of the Zuider Zee, provides essential nutrients for crops, enhancing agricultural productivity.

Temperate climate: The Netherlands has a temperate climate with cold summers and mild winters, which is suitable for a wide range of crops and livestock farming.

Availability of fresh water: Abundant fresh water from rivers, lakes (e.g., IJssel), and canals supports irrigation in drier areas, ensuring consistent crop growth.

Flat landscape: The flat terrain of the polder lands facilitates mechanization, allowing for the use of modern farming equipment and efficient land management.

Highly skilled labour force: Dutch farmers are well-trained and skilled, enabling the adoption of innovative farming methods and technologies.

Good communication and transport networks: Well-developed roads and waterways allow easy transportation of inputs and agricultural produce to domestic and international markets.

Large ready market: A strong demand for products such as butter, cheese, vegetables, and fruits encourages farmers to expand production and diversify crops.

Advanced technology in production: Use of modern technologies, including greenhouses, processing equipment, and land reclamation methods, enables intensive and profitable farming.

Large capital base: Significant investment in land reclamation and mechanization supports large-scale, efficient agricultural operations.

High quality of agro-products: The excellent quality of Dutch agricultural products enhances market competitiveness and encourages continuous improvement in farming practices.

Supportive government policies: Government support in areas like economic stability, infrastructure development, security, transport, and power supply fosters a conducive environment for agricultural growth.

Application activity 17.1.4

1. Name three natural factors that influence agriculture on the polder lands.
2. How does advanced technology improve agriculture on polder lands?
3. Discuss the key factors that have contributed to the success of agriculture on Dutch polder lands.
4. Explain how both natural and human factors interact to make polder agriculture intensive and profitable.

17.1.5. Importance, challenges and possible solutions of agriculture in Polder lands

Learning activity 17.1.5

Mukamana visited Holland and learned that agriculture provides food, employment, foreign exchange, and supports agro-industries. Dutch farms face unique challenges on polder lands, including infertile soils, salinity, waterlogging, and dry spells. Farmers address these by using fertilizers, hardy crops, maintaining dykes, pumping water, adopting greenhouses, and increasing organic manure. Agriculture remains labour-intensive but vital for Holland's economy and development.

1. Fill in the blanks
 - a) Agriculture in Holland provides _____, employment, and foreign exchange.
 - b) Farmers use _____, hardy crops, and maintain _____ to overcome polder challenges.
3. Which of the following is a unique problem of Dutch polder farms?
 - a) Low rainfall
 - b) Infertile soils and salinity
 - c) Pests and diseases
 - d) Lack of machinery
4. How do Dutch farmers prevent waterlogging in polders?
 - a) By using fertilizers
 - b) By pumping water and maintaining dykes
 - c) By planting flowers
 - d) By importing soil

❖ Importance of agriculture in Holland

- It is a regular source of foreign exchange to Holland, as butter, cheese, pork, cereals are widely exported to the European Union, Belgium and Britain;
- The sector meets over 50 percent of the food requirements of Holland;
- Agriculture in Holland is labour intensive and carried out on large scale;
- The sector provides a large market to industrial products such as fertilizers, herbicides and pesticides;
- Many people are employed in agriculture
- Provision of income to the people by ensuring a high standard of living;
- Agriculture has encouraged the development of agro-based industries such as creameries;
- It is a good source of capital for development to the people and, to the government through tax revenue.

Unique problems/challenges hindering the agricultural expansion on the Polder Lands and possible solutions.

Although the Dutch use highly advanced technology in agriculture, they do experience some problems that affect the sector. Some of the problems are unique because most of the farms are found on land that has been reclaimed from the sea. This is not so in many other countries in the world.

Problems	Possible Solutions
– Areas with soils made of peat and sands are infertile – Seepage of sea water causes salinity in the soil – Collapse of the dykes could lead to flooding of the farms. – Some areas experience water-logging because the land is low-lying – Prolonged dry spells lead to shortage of water needed for irrigation – Increased use of chemicals is an environmental hazards	– Use of fertilizer and manure to increase the fertility. – To improve the soils, pastures and hardy crops such as barley and oats are first grown. – Regular maintenance of dykes – Improving technology by regular pumping of water from the polders – Glass houses are adapted to use in crop cultivation. – Increase the use of organic manure and reduce overdependence on chemicals used in agriculture.

Application activity 17.1.5

1. Explain why agriculture is vital for Holland's economy.
2. Name two methods used to improve polder land for farming.
3. Discuss the importance of agriculture in Holland, including its contributions and economic benefits.
4. Explain the unique challenges of farming on polder lands and the solutions adopted by Dutch farmers.

17.2. Agriculture in the USA

17.2.1. General presentation of USA and regionalisation of agriculture in USA

Learning activity 17.2.1

Sophia and Mazimpaka studied the United States' geography and agriculture. The USA stretches 4,500 km from the Atlantic to the Pacific, covering 9.83 million km² across 50 states. Alaska is the largest state, twice the size of Texas, and Hawaii lies 320 km from the mainland. The contiguous forty-eight states are bordered by the Atlantic and Pacific Oceans, with Washington D.C. on land donated by Maryland. Agriculture is regionalized according to climate and landscape: the northern plains grow wheat, oats, and barley; the South produces cotton, corn, and tobacco; the Northeast grows vegetables and fruits; the Great Lakes region raises livestock, and California uses irrigation for fruits and vegetables, supporting domestic and export markets.

Questions:

1. Complete the following:
 - (a) The USA covers approximately __ million km² across ____ states.
 - (b) Alaska is ____ times larger than Texas, and Texas is twice as big as ____.
2. Which oceans bound the contiguous forty-eight states of the USA?
 - a) Atlantic and Indian
 - b) Atlantic and Pacific
 - c) Pacific and Arctic Sea
 - d) Gulf of Mexico and Indian
3. Name three major crops grown in the northern plains region.
4. Explain how agriculture in the USA is regionalized, giving examples of crops and livestock.

a) General presentation of USA: *Location, Shape and Size*

The USA is one of the countries that make up North America. USA stretches for 4,500km from the Atlantic to the Pacific Ocean. It covers an area of 9.3 million square Kilometres in 50 states. Hawaii State is 320km from the main land in Pacific Ocean. The largest state is Alaska followed by Texas. Alaska is twice as big as Texas yet Texas is twice as big as France. Other islands which are part of the USA are Marshall and Mariana in the Pacific Ocean near Japan.

The USA is a country in the Western hemisphere. It consists of forty-eight contiguous states on the North American continent. By land area (exclusive of waters), the USA is the World's third largest country following Russia and China. In total area (+ inland water), Russia and Canada are larger than the USA. The total surface area of USA is nearly 9,826,630km².



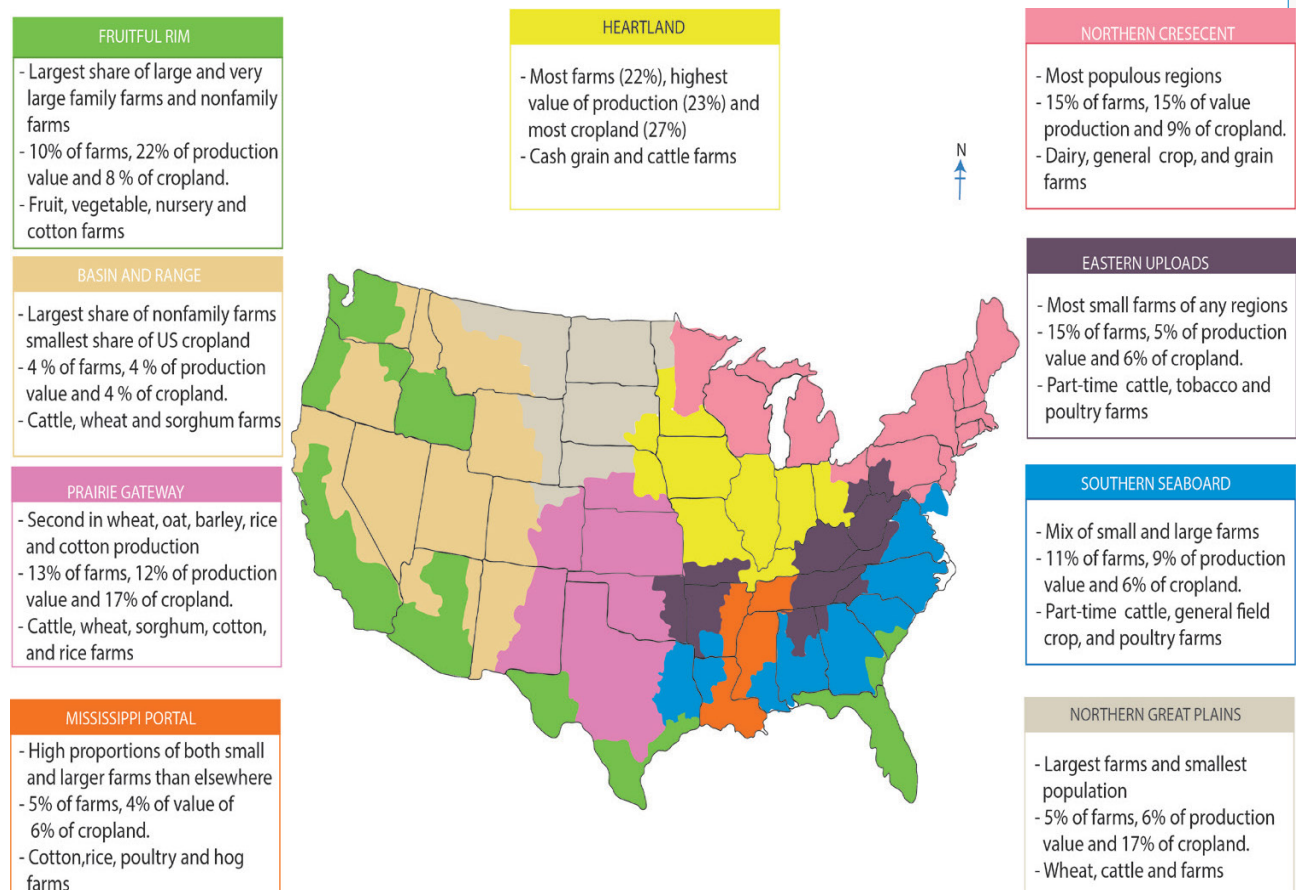
The United States share land borders with Canada to the North and Mexico to the South, and a territorial water border with Russia in the Northwest. The contiguous forty-eight states are otherwise bounded by the Pacific Ocean on the West, the Atlantic Ocean on the east, and the Gulf of Mexico to the Southeast. Alaska borders the Pacific Ocean to the South, the Bering Strait to the West, and the Arctic Ocean to the North, while Hawaii lies far to the Southwest of the mainland in the Pacific Ocean.

The capital city, **Washington**, District of Columbia, is a federal district located on land donated by the state of Maryland. (Virginia has also donated land, but it was returned in 1847).

b) Regionalisation of agriculture in USA

Agriculture in the USA has been regionalized based on the climate of the regions that make up the country.

Regionalization of agriculture refers to the zoning of agricultural types and systems which are carried out mainly due to varied climate and nature of landscape in different areas in the country. Each zone is unique in the characteristics of agriculture.



The main agricultural zones in the USA are summarized in the following table:

Region	Agricultural system	Crops grown and livestock kept
The prairie region	Extensive/Plantation farming	Wheat
The south	Mixed farming, extensive farming	Cotton, corn, tobacco, Hogs
The Northeast and Atlantic Coastlands	Intensive farming: Truck farming	Vegetables and fruits
The St Lawrence Valley and Atlantic seaboard	Intensive farming: Market gardening	Vegetables and flowers
The Great Lakes Lowland region	Livestock farming; Dairy and Poultry farming	Cattle, Hogs, Poultry
The Central Valley	Irrigation, Market gardening and Truck farming	Citrus fruits, vegetables, wheat, corn, rice and some cotton.

- **Fruitful region:** Fruitful region specializes in fruits, vegetables, nursery crops and cotton, with production value from large farms.
- **Basin and range:** Basin and range region has nonfamily farms and limited cropland, focusing on cattle, wheat and sorghum.
- **Prairie gateway:** Prairie gateway region produces wheat, sorghum, cotton and rice, with extensive cropland and significant cattle farming.
- **Mississippi portal:** Mississippi portal region combines small and large farms, producing cotton, rice, poultry and hogs on land.
- **Heartland:** Heartland region has most farms, highest production value, cropland, and specializes in cash grains and cattle.
- **Northern crescent:** Northern crescent region is densely populated, with dairy, general crops and grain farming, contributing significant production.
- **Eastern uplands:** Eastern Uplands region has many small farms, with part-time cattle, tobacco and poultry production on uplands.
- **Southern Seaboard:** Southern seaboard region mixes small and large farms, with part-time cattle, poultry and field crop production.
- **Northern great plains:** Northern great plains region has largest farms, sparse population, and focuses on wheat and cattle farming

Northern great plains region has largest farms, sparse population, and focuses on wheat and cattle farming.

USA is a major exporter of food because of her highly developed agriculture. In addition, each region produces for domestic and export market. The major crops grown in northern plains where the population is fairly low include wheat, oats, barley and corn. Here, farms are very large and highly mechanized.

It is important to note that in the very dry areas such as parts of California and the neighboring states, crop production is through irrigation. The main crops are fruits and vegetables.

Application activity 17.2.1

1. Complete the statement below with an appropriate concept
 - a) Agriculture in the USA is regionalized based on _____ and _____ of the regions
 - b) The Central Valley mainly relies on _____ to grow fruits and vegetables.
2. Which region in the USA specializes in wheat and cattle with large mechanized farms?
 - a) Northern Great Plains
 - b) Southern Seaboard
 - c) Eastern Uplands
 - d) Mississippi Portal
3. What type of farming is common in the South of the USA?
 - a) Dairy farming
 - b) Mixed and plantation farming
 - c) Truck farming only
 - d) Intensive horticulture
4. Discuss why the USA is a major food exporter, referencing regional production systems and mechanization.

17.2.2. Agricultural systems and agribusiness in USA: Agrichemicals, farm machinery, crop production, processing, and seed supply

Learning activity 17.2.2

Maya studied agriculture in the USA and learned that farming adapts to different environments. Systems include commercial farming, small-scale intensive farming, large-scale arable farming, ranching, and mixed farming. Agribusiness supports farming through production of inputs, machinery, seed supply, crop processing, and distribution. It creates jobs, improves infrastructure, and facilitates marketing. Farms vary from family-owned small plots to large, mechanized TNC-owned farms.

1. Fill in the blanks
 - a) Large-scale arable farms in the USA are highly _____ and usually produce a _____ crop for export.
 - b) Agribusiness includes production of inputs, machinery, seed supply, and _____ of agricultural products.
3. Which farming system is mainly practiced in open grasslands like Montana and North Dakota?
 - a) Mixed farming
 - b) Ranching
 - c) Intensive small-scale farming
 - d) Commercial crop farming

4. What is the main goal of small-scale intensive farming in the USA?

- a) Maximize yields using high labor or capital input
- b) Export a single crop only
- c) Raise livestock on open ranges
- d) Reduce farm inputs.

❖ Agricultural systems in the USA

In the USA agricultural systems have been adapted to the different environments in the different regions.

- a) **The commercial farming system:** here, the production is mainly aimed at the market. Farms range from small family farms to large Trans National Corporation (TNC) owned farms. A small family farm in the USA is maybe at least 20ha while large TNC owned farms maybe at least 1000ha.
- b) **Small scale intensive farming system:** this is one where a huge amount of inputs is used in the farming process to ensure a high output. In areas with large population, intensive farming uses a high labour input while in areas with a low population intensive farming uses a high capital input and high technology. It aims at maximizing the yields through means such as application of manure and fertilizer. The land is always under cultivation and is characterized by high yields. Some farms have both livestock and crops.
- c) **Large scale arable farming:** characterized by very large farms that are highly mechanized. Each farm produces a single crop for export. It is common in areas with gentle slopes and flat lands to facilitate mechanization in activities such as land preparation, planting, application of pesticides and harvesting.
- d) **Ranching:** Ranching is rearing livestock in large numbers on a rangeland. It is mainly practiced in the open grasslands of states like Montana and North Dakota. Animals are looked after by cowboys riding on horses. Some farmers keep cattle while others keep sheep in ranches.
- e) **Mixed farming:** This involves the practice of carrying out both crop and livestock farming concurrently or at the same time.
- f) **Agribusiness** in the USA Agribusiness is a business operation that may include farming that is done as a business and any other businesses that are directly or indirectly dependent on agriculture.

Agribusinesses may produce agricultural commodities, process agricultural products; distribute agricultural produce or supply goods and services to farmers. Agribusinesses in USA include:

- Making of agrichemicals
- Manufacturing of farm machinery such as tractors, milking machines, automatic egg collectors, and so many others.
- Commercial crop production
- Production of animal feeds.
- Food processing of fruits, vegetables, meat canning, milk processing, and so many others.

- Seed supply. This provides various seeds in various regions depending on seasons.

❖ Importance of agribusiness in USA

Agribusiness is an important economic sector without which agricultural activities would not succeed. Agribusiness facilitates the supply of agricultural input to the farms. It also enables inputs to the farms. It also enables farmers to market their produce. Other than benefiting those directly involved in farming, there are many other people who earn their living through businesses that relate indirectly to agriculture. This content has been summarized.

Apart from the above; agribusiness in the USA has led to:

- Provision of farm inputs
- Creation of employment opportunities
- Encouraging the development of infrastructures
- Development of transportation.

Application activity 17.2.2

1. Name three services provided by agribusinesses in the USA.
2. Discuss the different agricultural systems in the USA, giving examples of crops or livestock for each system.
3. Explain the importance of agribusiness to farming and the wider economy in the USA
4. Explain the difference between commercial farming and small-scale intensive farming.

17.2.3. Factors favouring agriculture, challenges facing agriculture and possible solutions in USA

Learning activity 17.2.3

Manzi explored agriculture in the USA and learned it thrives due to favorable climate, fertile soils, flat topography, skilled labor, high technology, government support, industrialization, and good transport and banking systems. However, challenges include water depletion, soil deterioration, urban expansion, fossil fuel overuse, and global competition. Farmers use irrigation, mechanization, fertilizers, cloud seeding, and migrant labor to overcome these challenges.

1. Fill in the blanks
 - a) Agriculture in the USA is favored by climate, _____, and _____ as natural factors.
 - b) Cloud seeding and irrigation are used to overcome problems caused by _____ and low rainfall.

2. Which human factor encourages farmers to produce more because they are assured of selling produce at good prices?
 - a) Industrialization
 - b) Accessibility to large and rich markets
 - c) Government subsidies
 - d) Skilled manpower.
3. Which problem is unique to USA agriculture compared to other countries?
 - a) Fluctuation of world commodity prices
 - b) Soil deterioration due to monoculture
 - c) Pests and diseases
 - d) Drought.

a) Factors favoring agricultural development in the USA

Generally, factors that favour agriculture in the USA can be categorized into **natural factors** and **human factors**.

❖ Natural factors

From the perspective of natural factors, agriculture in the USA has been favoured by factors such as climate, soils and topography.

Climate: USA has varied climates which favours a wide range of agricultural systems. In some places, agricultural activities are determined by prevailing climate conditions. For instance, in areas that have ranches and the areas with large scale arable farms, farmers depend on natural rainfall for growth of pastures or crops.

Soils: Originally, most of USA's agricultural areas depended on natural soil fertility. However, with continued use of the land, soil fertility is maintained by adding chemical fertilizers so that high yields can be sustained.

Topography: The topography of USA is ideal for agriculture because land is gently sloping or generally flat. A part from the Rocky and Appalachian mountain ranges, the rest of the country has a favorable landscape.

❖ Human factors:

- i. The application of fertilizers increases soil fertility and crop yields, enabling farmers to produce more food and raw materials for both local consumption and markets.
- ii. An improved transport network makes it easier to move farm inputs to rural areas and transport agricultural products to markets, reducing losses and increasing farmers' incomes.
- iii. Accessibility to large and rich markets encourages farmers to produce more because they are assured of selling their produce at better prices.
- iv. Government policies that provide subsidies to farmers lower production costs by making inputs such as seeds, fertilizers, and machinery more affordable.

- v. High levels of industrialization support agriculture by supplying farm machinery, processing facilities, and providing ready markets for agricultural raw materials.
- vi. An improving banking system enables farmers to access loans and credit, which helps them invest in modern farming methods and expand production.
- vii. High levels of technology increase efficiency in farming through mechanization, irrigation, and improved crop varieties, leading to higher productivity.
- viii. The availability of skilled manpower ensures proper management of farms and the effective use of modern agricultural techniques, improving overall agricultural performance.

❖ **Problems hindering agricultural expansion and possible solutions in USA**

Farmers in various parts of the world face common problems such as weather changes and fluctuation of prices of commodities in the world market. However, in USA, there are some unique problems that hinder expansion of agriculture.

❖ **Problems/challenges facing agriculture in USA**

- High usage of water which depletes the water resources.
- Soil deterioration due to Monoculture
- Excessive use of fossil fuels.
- Restrictive migration policy.
- Expansion of urban centres
- Competition in the global market.

❖ **Possible solutions to problems facing agriculture in USA**

- Irrigation of the land with water from the San Joaquin and Sacramento rivers is done to reduce the impact of low rainfall
- Building of dams for storage of water and supplying of water to different farms at different times to avoid excessive use at particular time.
- Land consolidation to reduce land fragmentation
- Employment of migrant labour in the busy periods and seasons of the year
- Use of fertilizers and manures to restore fertility and structure
- Mechanization is done to consolidate plots to solve the problem of land shortage
- Spraying with chemicals is done in order to combat pests and diseases
- Cloud seeding to supplement natural rainfall. This is done by planting rain inducing chemical tablets like silver iodide and dry ice into the clouds using high flying air crafts
- Eradication of pests and diseases is accomplished by spraying with chemicals and pesticides

Scarcity of manual labour during the busy periods is being rectified by hiring unskilled immigrants from Mexico and Latin America.

Application activity 17.2.3

1. Name three human factors that favor agricultural development in the USA.
2. List three solutions used by US farmers to address water shortage and soil fertility problems.
3. Discuss the natural and human factors that have favored agricultural development in the USA.
4. Explain the major challenges facing agriculture in the USA and the solutions employed to overcome them.

End of unit assessment

❖ Knowledge and understanding:

1. Which of the following is a unique problem of Dutch polder farms?
a) Low rainfall b) Infertile soils and salinity
c) Pests and diseases d) Lack of machinery
2. Which dairy products are exported from the Netherlands?
a) Cheese, butter, condensed milk b) Yogurt, eggs, meat
c) Milk, wool, honey d) Butter, ice cream, wool
3. Which state lies 320 km from the mainland in the Pacific Ocean?
a) Texas b) Hawaii c) Alaska d) Maryland
4. Which region in the USA specializes in wheat and cattle with large mechanized farms?
a) Northern Great Plains b) Southern Seaboard
c) Eastern Uplands d) Mississippi Portal

❖ Skills

5. Discuss the Zuyder Zee Project, its objectives, methods, and benefits to the Netherlands.
6. Explain the unique challenges of farming on polder lands and the solutions adopted by Dutch farmers.
7. Explain the different agricultural systems in the USA, giving examples of crops or livestock for each system.

❖ Attitudes & Values

8. Evaluate why the USA is a major food exporter, referencing production systems and mechanization.
9. Analyze the major challenges facing agriculture in the USA and the suggested solutions applied.

UNIT 18

INDUSTRIAL DEVELOPMENT CASE STUDIES OF JAPAN AND CHINA



Key Unit Competence:

Examine the importance of industrial development to the economy of Japan and the reasons for rapid growth of industrial development of China.

Learning objectives: At the end of this unit you should be able to:

Knowledge and understanding

1

- Give the location, size and population of Japan and China.
- Identify the major industries found in Japan and China.
- Outline the factors favouring the development of industries in Japan and China.
- Identify the importance of industrialisation in Japan and China.
- State the problems affecting industrialisation in Japan and China.

Skills

2

- Locate the major industries found in Japan and China.
- Describe the factors that favour industrial development in Japan and China.
- Analyse the importance of various industries in Japan and China.
- Examine the problems affecting industrialisation and suggest possible solutions for Japan and China.

Attitudes and values

3

- Appreciate the factors for industrial development in Japan and China
- Show respect for the economic contribution of industrialisation in the development of Japan and China

18.0. Introduction

This unit examines industrial development in Japan and China, focusing on location, major industries, development factors, economic importance, challenges, and solutions shaping their industrial growth.

Introductory activity

Students learned that Japan, an island nation in East Asia, and China in East Asia's mainland host automobile and electronics industries, grew through skilled labour, technology, capital, and markets; industries boost jobs and exports, face pollution and congestion, and teach Africa planning, innovation, sustainability lessons

1. Outline factors that favoured industrial development in Japan and China.
2. Give examples showing how industries contribute to economic growth in Japan and China.
3. Examine the major problems affecting industrialisation in Japan and China.

18.1. Industrial development in Japan

18.1.1. General presentation, industrial regions and major industries of Japan

Learning activity 18.1.1

Japan is an island nation in East Asia in the Northwest Pacific Ocean, made up of over 6,000 islands and covering 377,835 km². By 2025, it had a population of about 123.1 million, mainly concentrated in Tokyo, and is administratively divided into 47 prefectures. Industrial development in Japan began in the early 1800s and expanded rapidly between 1919 and 1945, leading to major industrial districts such as Tokyo–Yokohama, Osaka–Kobe, Nagoya, and Northern Kyushu. Key industrial regions include Keihin, Hanshin, Ise Bay, and Kitakyushu. Japan's economy is supported by diversified industries including shipbuilding, motor vehicles, electronics, aerospace, petrochemicals, food processing, fisheries, telecommunications, and microelectronics, making the country a leading global industrial power.

1. Which of the following is a major industrial district in Japan?
a) Hokkaido b) Tokyo–Yokohama c) Okinawa d) Shikoku
2. Which industrial region is associated with heavy manufacturing and port activities?
a) Keihin Region b) Okinawa Region c) Tohoku Region d) Shikoku Region
3. Which company is part of Japan's motor vehicle industry?
a) Canon b) Sony c) Toyota d) Panasonic
4. Japan is administratively divided into _____ prefectures.
5. One major industrial region of Japan is the _____ Region.
6. Describe the main industrial regions of Japan and explain why they developed along coastal areas.
7. Explain how Japan's major industries have contributed to the country's economic development and global influence.

a) General presentation of Japan

Japan is an island nation in East Asia. It is located in the Northwest Pacific Ocean off the coast of the Asian continent. It lies to the east of the East sea/ Sea of Japan and to the north of East China Sea.

Japan is made up of over six thousand Islands in the Pacific Ocean. Only four hundred and thirty of these islands are inhabited.

By 2020, the population of Japan was estimated at 126.3 million people. However, Japan has been experiencing population decline and by 2025, the population had gone down to about 123 million. The country is the world's eleventh in population size. The largest concentration of population is in Tokyo which is the capital city. The many islands of Japan have a total land area of 377,835 square kilometres.



Map showing the location of Japan

Tokyo, the capital city of Japan

Japan is divided into 47 administrative divisions called prefectures. Each prefecture has its own administrative capital city and the capital city of Japan is Tokyo.

b) Industrial regions and major industries of Japan

Industrial areas/zones.

Industrial development in Japan began in the 1809s. Between 1919 and 1945 there are four major industrial districts: Tokyo-Yokohama, Osaka-Kobe, Nagoya, and Northern Kyushu.

Major industrial zones in Japan include the following: The Keihin Region, Hashin Region, Isle Bay Region and Kitakyushu Region

Major industries of Japan include: Ship-building, motor vehicle and machinery industry (Companies Toyota, Nissan and Subaru), aerospace, petrochemicals, electronics (companies Canon, Casio, Panasonic and Sony), food processing, fish industry, telecommunication equipment and computers and microelectronics, etc.

Application Activity 18.1.1

1. Fill in the Blanks
 - a) Between 1919 and 1945, Japan developed four major industrial districts, namely Tokyo–Yokohama, Osaka–Kobe, Nagoya, and _____.
 - b) Japan is administratively divided into _____ prefectures.
2. Between 1919 and 1945, Japan developed four major industrial districts. Which one of the following is NOT among them?
 - a) Tokyo–Yokohama
 - b) Osaka–Kobe
 - c) Nagoya
 - d) Hokkaido
3. Which of the following is a major industrial zone in Japan?
 - a) Keihin Region
 - b) Okinawa Region
 - c) Shikoku Region
 - d) Hokkaido Region
4. Which city has the largest population concentration in Japan?
 - a) Osaka
 - b) Nagoya
 - c) Tokyo
 - d) Kobe
5. Describe the location, size, and population distribution of Japan.
6. Explain Japan's administrative divisions and discuss the role of prefectures in governance.

18.1.2. Factors for industrial development in Japan

Learning activity 18.1.2

Japan's industrial development began in the 1800s and accelerated after World War II. By 1945, major industrial districts such as Tokyo–Yokohama, Osaka–Kobe, Nagoya, and Northern Kyushu had emerged. Japan's rapid industrial growth has been driven by several factors, including advanced technology, skilled labour, availability of capital, supportive government policies, efficient transport and communication systems, availability of energy, and a large domestic and international market. These factors have enabled Japan to develop strong industries such as shipbuilding, motor vehicles, electronics, machinery, and petrochemicals, with global companies like Toyota, Sony, and Canon.

Questions

1. Fill in the blanks
 - a) One factor that contributed to Japan's industrial development is advanced _____.
 - b) The availability of skilled _____ has supported industrial growth in Japan.

2. Multiple-choice questions

- i. Which of the following is a major factor that promoted industrial development in Japan?
 - a) Abundant mineral resources
 - b) Advanced technology
 - c) Large agricultural land
 - d) Low population
- ii. Which government-related factor supported the growth of industries in Japan?
 - a) Supportive industrial policies
 - b) Political isolation
 - c) Traditional farming systems
 - d) Dependence on imports.

❖ Factors favouring industrial development in Japan.

- **Supportive government policy:** Government policies favor industrial growth, investment, and regulation.
- **Supportive government policy:** The Japanese government implemented strong industrial policies through MITI (Ministry of International Trade and Industry) to support industrial growth, protect local industries, and guide investment into key sectors like automobiles and electronics.
- **Presence of capital:** Japan has strong financial institutions and large corporations (zaibatsu/keiretsu such as Mitsubishi and Sumitomo) that provided capital for establishing and expanding industries.
- **Presence of energy:** Despite limited natural energy resources, Japan developed reliable energy supply through hydroelectric power, thermal power, nuclear energy, and imported oil to sustain continuous industrial production.
- **Availability of labour:** Japan has a disciplined, educated, and skilled labour force that supports industrial efficiency, especially in manufacturing and technology-based industries.
- **Technological development:** Japan is a global leader in advanced technology, robotics, automation, and precision manufacturing, improving efficiency, quality, and industrial productivity.
- **Presence of market:** Japan has a large domestic market due to its high population and strong purchasing power, as well as access to international markets through exports of vehicles, electronics, and machinery.
- **Mergers and take-over:** The Japanese government enacted anti-monopoly laws after World War II, which encouraged mergers and takeovers, leading to the formation of strong and competitive industrial firms capable of managing risks in production and distribution.
- **Easy transport:** Japan has a highly developed transport network including modern ports, highways, railways, and high-speed Shinkansen trains, facilitating the movement of raw materials and finished goods.

- **Research and development:** Heavy investment in research and development by the government and private companies has promoted innovation in industries such as automobiles, electronics, and biotechnology.
- **High number of population:** Japan's large population provides a ready labour force and a steady internal market for industrial goods.
- **Accessibility to the sea:** Being an island nation, Japan has excellent access to the sea, which facilitates international trade, importation of raw materials, and export of manufactured products.
- **Innovation and creativity:** Japanese industries emphasize innovation, quality control, and creativity (e.g., Kaizen and Just-In-Time production), enhancing competitiveness in global markets.
- **Import and export government policy:** Japan's favorable trade policies encourage the import of raw materials and export of finished industrial goods, supporting industrial growth and global trade integration.

Application activity 18.1.2

1. Which factor has most contributed to Japan's industrial development despite its limited natural resources?
 - a) Large mineral deposits
 - b) Advanced technology
 - c) Extensive agricultural land
 - d) Low population density
2. Which of the following government actions supported industrial growth in Japan?
 - a) Restricting foreign trade
 - b) Supportive industrial policies
 - c) Reducing skilled labour training
 - d) Discouraging technological innovation
3. State two factors that have contributed to industrial development in Japan.
4. Why is a skilled labour force important for Japan's industrial growth?
5. Explain how technology and capital have influenced Japan's industrial development.
6. Assess the importance of transport, communication, and markets in supporting Japan's industries and international trade.

18.1.3. Importance, problems and solutions of industrialisation in Japan

Learning activity 18.1.3

Manzi studied why Japan has become a leading industrial nation. He found that supportive government policies, abundant capital, energy resources, skilled labor, advanced technology, research and development, and a large population all favor industrial growth. Easy transport, access to the sea, innovation, and government import-export policies further enhance industrial development. Mergers and takeovers led to large firms managing manufacturing, distribution, and sales efficiently.

1. Fill in the blanks

- a) Industrial development in Japan is favored by _____, presence of capital, energy, and availability of _____.
- b) Mergers and takeovers helped Japanese firms to manage _____, distribution, and sales efficiently.
- c) Industries in Japan help earn foreign exchange, create employment, and promote _____ and technology.
- d) One way Japan combats pollution is by introducing strict _____ laws and advanced pollution-control technologies.

2. Which of the following factors supports industrial development in Japan?

- a) Limited population
- b) Lack of transport
- c) Supportive government policy
- d) Isolation from the sea

3. What role does technological development play in Japanese industry?

- a) Reduces population growth
- b) Enhances research, innovation, and production efficiency
- c) Limits trade opportunities
- d) Prevents merger

❖ **The importance of industry in the development of Japan.**

- Earning foreign exchange
- Creation of employment opportunities
- Creation of internal co-operation earning revenue to the government
- Promotion of research and technology.
- Development of transport and communication network
- Greater political and economic strength
- Raising of living standards to the people
- It stimulates the growth of other sectors of the economy
- Improvement of the social amenities, schools and hospitals
- Exploitation of natural resources and high production of energy
- Development of urbanization.

❖ **Problems of industrialization in Japan**

- Pollution: Industrial activities cause air, water, and soil pollution
- Scarcity of raw materials: Limited availability of raw materials hinders industrial production
- Shortage of unskilled labour: Insufficient unskilled labour affects industrial operations.

- Market competition: Strong competition with other highly industrialized countries
- Overproduction: Automation and robots lead to overproduction of goods
- Deforestation: Industrial expansion leads to widespread deforestation of forests
- Global warming: Industrial emissions contribute significantly to global warming

❖ Possible solutions taken by Japan to address the industrial problems

Below are possible solutions taken by Japan to address each of the listed industrial problems, with one clear solution for each problem, suitable for geography notes or examinations.

- Pollution of air, water, and soils: Japan introduced strict environmental laws and advanced pollution-control technologies to reduce industrial emissions and treat industrial waste.
- Scarcity of raw materials: Japan solves raw material scarcity by importing raw materials from other countries and promoting recycling and efficient use of resources.
- Shortage of unskilled labour: Japan addressed labour shortages by investing in automation and robotics and providing technical training to improve workers' skills.
- Competition for markets with other industrialized countries: Japan improves competitiveness by producing high-quality goods, promoting innovation, and focusing on export-oriented industries.
- Overproduction due to use of automated robots: Japan manages overproduction by diversifying products, expanding export markets, and adjusting production to market demand.
- Deforestation: Japan reduced deforestation by importing timber, promoting afforestation, and enforcing forest conservation policies.
- Global warming: Japan combats global warming by promoting energy efficiency, using renewable energy, and reducing greenhouse gas emissions through clean technologies.

Application activity 18.1.3

1. Name three natural or human factors that favor industrial development in Japan.
2. Explain how mergers and takeovers contributed to industrial growth in Japan.
3. Discuss the key factors that have contributed to industrial development in Japan.
4. Explain how government policies, innovation, and population support industrial growth in Japan.

18.2. Industrial development in China

18.2.1. General presentation, industrial regions and major industries of China

Learning activity 18.2.1

Rukundo studied China, a leading industrial nation in East Asia under a socialist system led by the Communist Party. About 15–18% of the workforce is in manufacturing, mainly in coastal and north-eastern regions, guided by government policies, modernization, exports, and technology. Kaliza found Manchuria focuses on coal, iron, steel, and machinery; Shanghai on automobiles, shipbuilding, electronics, and petrochemicals; Tianjin, Qingdao, and Wuhan on steel, chemicals, textiles, and machinery, while southern coastal zones produce export-oriented consumer goods.

1. Fill in the blanks
 - a) China is governed under a ____ political system led by the Communist Party.
 - b) Major industrial regions in China are mainly concentrated in the ____ and ____ regions.
 - c) The Manchuria industrial region in north-eastern China specializes in ____, coal, steel, and heavy machinery.
 - d) Coastal and export-processing zones in southern and eastern China focus on __, electronics, and consumer goods.
5. What percentage of China's workforce is employed in manufacturing?
a) 5–10% b) 10–12% c) 15–18% d) 20–25%
6. Which of the following cities is a major municipality contributing to industrial administration in China?
a) Guangzhou b) Chengdu c) Tianjin d) Kunming
7. Which industrial region is located in the Yangtze River Delta?
a) Manchuria b) Shanghai c) Wuhan d) Tianjin
- h) The Wuhan Industrial Region mainly specializes in:
a) Textiles and apparel b) Automobiles, iron, steel, and machinery
c) Electronics and consumer goods d) Petrochemicals and shipbuilding.

a) General presentation



China is one of the world's leading industrial nations, located in East Asia and governed under a socialist political system led by the Communist Party of China. The country is divided into provinces, autonomous regions, and major municipalities such as Beijing, Shanghai, and Tianjin, which play a key role in industrial administration.

Manufacturing employs about 15–18% of China's workforce, showing the importance of industry in the national economy. Guided by strong government policies and national development plans, China has developed its industrial sector in a planned and organized manner, with major industries concentrated in coastal and north-eastern regions as shown on the political and industrial map of China, focusing on modernization, export production, and technological advancement.

b) Industrial regions and major industries of China

- **Manchuria (North-Eastern China – Liaoning, Jilin, Heilongjiang):** This region specializes in iron and steel production, coal mining, aluminum processing, heavy machinery, chemicals, and armaments, supported by rich mineral resources.
- **Shanghai Industrial Region (Yangtze River Delta):** Shanghai is a leading center for automobiles, shipbuilding, electronics, petrochemicals, machinery, textiles, and high-tech industries, supported by a major port and advanced infrastructure.

- **Tianjin Industrial Region (Bohai Gulf):** Tianjin specializes in steel manufacturing, chemicals, machinery, petroleum refining, electronics, and aerospace industries, benefiting from its port and proximity to Beijing.
- **Qingdao Industrial Region (Shandong Peninsula):** This region is known for shipbuilding, marine engineering, food processing, electronics, textiles, and chemical industries, supported by fishing and port activities.
- **Wuhan Industrial Region (Central China – Yangtze River Basin):** Wuhan specializes in iron and steel, automobiles, machinery, chemicals, cement, and educational-based technology industries, using river transport and skilled labour.
- **Coastal and Export-Processing Zones (Southern and Eastern China):** These areas focus on textiles and apparel, electronics, consumer goods, household appliances, and assembly industries, mainly for export markets.

Application activity 18.2.1

1. Name three major municipalities in China that play a key role in industrial administration.
2. Explain why China is considered a leading industrial nation and discuss the role of modernization and export production in its industrial growth
3. Describe the major industrial regions of China and the types of industries they specialize in.

18.2.2. Factors for industrial development in China

Learning activity 18.2.2

Mei explored why China has become a major industrial nation. She discovered that abundant mineral resources like coal and iron, agricultural raw materials, and a large skilled labor force support industrial growth. Well-developed transport and port facilities, government planning and policies, domestic and foreign capital, and rapid technological advancement also play key roles in modernizing China's industries and boosting production.

1. Fill in the blanks
 - a) Industrial development in China is supported by mineral resources such as _____, iron ore, and ferro-alloys.
 - b) Strong government planning, policies, and _____ have contributed to the growth of Chinese industries.
2. Which of the following is a human factor favoring industrial development in China?
 - a) Coal and iron ore
 - b) Large and skilled labor force
 - c) Fertile soils
 - d) Rivers for fishing
3. Rapid technological advancement in China has led to:
 - a) Decline in industrial output
 - b) Industrial modernization and higher production efficiency
 - c) Reduction in transport networks
 - d) Less foreign investment.

❖ Factors for industrial development in China

- The availability of rich mineral resources such as coal, iron ore, and ferro-alloys provides essential raw materials and energy for heavy industries, for example coal mining in Shanxi and iron ore use in Anshan steel industries.
- Abundant agricultural raw materials support the growth of agro-processing industries and promote value addition, as seen in rice milling, tea processing, and food industries in eastern China.
- A large and relatively skilled labour force ensures high productivity, efficiency, and industrial competitiveness, especially in manufacturing hubs like Guangdong and Shanghai.
- A well-developed transport network enables the efficient movement of raw materials and finished products, through extensive railways, expressways, and inland waterways such as the Yangtze River.
- Modern port facilities facilitate international trade, exports, and the importation of industrial inputs, notably at ports like Shanghai, Shenzhen, and Ningbo-Zhoushan.

- Strong government planning and industrial policies provide clear direction, regulation, and support for industrial growth, as shown in China's Five-Year Development Plans.
- Industrial reforms and economic liberalization encourage private investment, improve efficiency, and promote industrial expansion, for example the economic reforms introduced after 1978.
- The availability of capital from domestic savings finances the establishment, expansion, and modernization of industries, through state-owned banks and national development funds.
- Foreign Direct Investment (FDI) provides capital, advanced technology, and managerial expertise, as seen in foreign companies operating in China's coastal industrial zones.
- Rapid technological advancement and industrial modernization increase production efficiency, quality, and overall industrial output, especially in electronics and automobile industries.
- A large domestic market ensures constant demand for locally manufactured goods, due to China's population exceeding 1.4 billion people.
- An export-oriented industrial strategy expands foreign markets and increases industrial competitiveness, with China exporting electronics, textiles, and machinery worldwide.
- The availability of energy resources ensures a reliable power supply for continuous industrial production, using coal, hydroelectric power from the Three Gorges Dam, and renewable energy.
- Special Economic Zones (SEZs) attract investors through incentives and promote industrial concentration, such as Shenzhen, Zhuhai, and Xiamen.
- Improved education and technical training produce skilled manpower required for modern industrial activities, through universities and technical institutes across China.
- Urbanization and the development of industrial cities concentrate labour, infrastructure, markets, and industrial services, as seen in cities like Beijing, Guangzhou, and Chongqing.
- The adoption of digital technologies and automation improves efficiency, accuracy, innovation, and industrial productivity, for example in smart factories and robotics manufacturing.
- Investment in research and development promotes innovation, new products, and industrial competitiveness, evident in companies like Huawei and BYD.
- Political stability and effective governance create a secure environment that attracts long-term industrial investment, making China attractive to both local and foreign investors.
- The shift towards green and sustainable industries reduces pollution while supporting sustainable industrial development, through solar energy, electric vehicles, and green manufacturing policies.

Application activity 18.2.2

1. Name two natural factors that favor industrial development in China.
2. Explain how government policies and capital have contributed to industrial growth in China.
3. Discuss the natural and human factors that have contributed to industrial development in China.
4. Explain how technological advancement and government planning have influenced China's industrial modernization.

18.2.3 Problems and prospects of industrialisation in China

Learning activity 18.2.3

1. Fill in the blanks
 - a) Rapid industrialization in China has caused ____, ____, and ____.
 - b) China's industries are adopting ____ and ____ to reduce environmental impact.
2. Which of the following is a major problem affecting industries in China?
 - A. Large domestic market
 - B. Technological innovation
 - C. Environmental pollution
 - D. Government support
3. What is a key prospect for China's industries?
 - A. Rising labor costs
 - B. Dependence on coal
 - C. Regional imbalance
 - D. Investment in renewable energy and high-tech manufacturing
4. True or False
 - a) China's industrial development is concentrated only in inland regions.
 - b) Over-reliance on exports can make industries vulnerable to global economic changes.
 - c) Research and innovation provide prospects for high-value industrial products.
 - d) Environmental pollution has no effect on China's industrial growth.

a) Problems faced by industries of China include the following:

- High demand raw materials
- Regional imbalance in industrial development
- Strong competition in international markets
- Environmental pollution: Rapid industrialization has caused air, water, and soil pollution, affecting public health and sustainability.
- High energy consumption: Many industries rely heavily on coal and fossil fuels, leading to high energy costs and environmental degradation.

- Labor issues: Rising labor costs in urban areas make manufacturing less competitive internationally.
- Skilled labor shortages in certain high-tech sectors.
- Over-reliance on export markets: Many industries depend heavily on foreign demand; global economic slowdowns can affect them.
- Technological gaps: Although China is advancing in high-tech, some industries still rely on imported technology for critical processes.
- Resource constraints: Scarcity of certain raw materials and pressure on water resources in industrial regions.
- Regional imbalances: Industrial development is concentrated in coastal areas, leaving inland regions underdeveloped.

b) Prospects of industries of China include the following:

- Increased investment in clean and renewable energy
- Growth of high-technology and knowledge-based industries
- Expansion of automation and green manufacturing
- Continued access to global and export markets
- Technological advancement: Investment in AI, robotics, electronics, and green technologies enhances productivity and global competitiveness.
- Government support: Policies like "Made in China 2025" aim to upgrade industries, promote high-tech manufacturing, and reduce pollution.
- Infrastructure development: Advanced transport networks, ports, and logistics support industrial growth.
- Large domestic market: Growing population and middle class provide a ready market for industrial products.
- Foreign investment: Policies encouraging foreign direct investment (FDI) boost industrial expansion and knowledge transfer.
- Renewable energy Integration: Industries are shifting towards solar, wind, and hydro power, reducing environmental impact.
- Innovation and research: Strong focus on research and development to produce high-value goods and compete in global markets.

Application activity 18.2.3

1. Match Column A with Column B:

Column A	Column B
i. "Made in China 2025"	a. Environmental challenge
ii. Heavy coal use	b. Policy to upgrade industries
iii. Rising labor costs	c. Problem affecting competitiveness
iv. Renewable energy integration	d. Sustainable solution for industries

2. Explain the problems facing industrial development in China.
3. Discuss the prospects for China's industries and how government policies and technology can address challenges.
4. Suggest ways to make China's industrial growth more sustainable.

End of unit assessment

❖ Knowledge and understanding:

1. Japan is located in:
A. East Asia B. South Asia C. Central Europe D. West Asia
2. China's population is approximately:
A. 1.4 million B. 1.4 billion C. 125 million D. 900 million
3. Japan's land area is approximately:
A. 377,975 km² B. 9.6 million km² C. 41,293 km² D. 500,000 km²
4. The Manchuria industrial region in China specializes in:
A. Textiles and electronics B. Iron, steel, coal, and heavy machinery
C. Food processing D. Automobiles and consumer goods
5. A major industry in Shanghai, China, includes:
A. Shipbuilding and electronics B. Coal mining
C. Forestry D. Tourism
6. Industrialisation in Japan and China helps to:
A. Promote research, technology, and innovation B. Reduce foreign trade
C. Increase subsistence farming D. Discourage urbanization
7. Industrialisation encourages:
A. Dependence on imports B. Growth of other economic sectors
C. Decline in manufacturing D. Limited transport development

❖ Skills

8. Locate and name the main industrial regions of Japan and China on a map
9. Summarise and analyse the factors favouring industrial development in Japan and China

❖ Attitudes & Values

10. Compare the problems affecting industrialisation in Japan and China and propose possible solutions applicable to developing countries like Rwanda

UNIT 19

TOURISM CASE STUDY OF SWITZERLAND



Key Unit Competence:

Analyze the role played by the tourism industry in the economic transformation of Switzerland.

Learning objectives: At the end of this unit you should be able to:

Knowledge and understanding

1

- Give the location, size and population of Switzerland.
- Name the major tourist attractions in Switzerland.
- State the factors favouring the development of tourism in Switzerland
- Identify the importance of tourism in Switzerland
- State the problems affecting the tourism industry in Switzerland

Skills

2

- Examine the tourist attractions sites of Switzerland.
- Describe the factors that favour the development of the tourism industry
- Evaluate the importance of the tourism industry
- Assess the problems facing the tourism industry

Attitudes and values

3

- Appreciate the importance and level of tourism development in Switzerland
- Show respect for the economic contribution of tourism to the economy of Switzerland.

19.0. Introduction

This unit studies tourism in Switzerland, focusing on location, attractions, development factors, economic importance, challenges, and comparisons with developing countries to understand advanced tourism systems.

Introductory activity

Manishimwe observes that Switzerland, located between Germany, France, Italy, and Austria, attracts tourists to the Alps, Lake Geneva, ski resorts, and cities; tourism creates jobs, earns foreign exchange, thrives on good transport, security, good scenery, and service quality.

1. State the location of Switzerland in relation to its neighboring countries.
2. Name four major tourist attractions found in Switzerland.
3. Using Jean's visit, explain how tourism contributes to Switzerland's economy.
4. Analyse the factors that favour the development of tourism in Switzerland.

19.1. General presentation of Switzerland and tourist attractions in Switzerland

Learning activity 19.1

1. Fill in the blanks
 - a) Switzerland is a _____ country located in Central Europe.
 - b) The capital city of Switzerland is _____.
2. Which country does **NOT** border Switzerland?
A. Germany B. France C. Spain D. Italy
3. Which city is the largest economic hub of Switzerland?
A. Bern B. Zurich C. Lausanne D. Neuchâtel
4. Name two major lakes found in Switzerland
5. Match the feature in **Column A** with the correct description in **Column B**.

Column A	Column B
i. Bern	a. Largest city and financial centre
ii. Zurich	b. Capital city
iii. Lake Geneva	c. International city
iv. Geneva	d. Major Lake in western Switzerland

a) General presentation of Switzerland

- Switzerland is a small, landlocked country in the southern region of the Rhineland. It has an area of 41,293Km². Its capital city is called Berne.
- It is located in central Europe, with Germany to her north, Italy to her south, France to West and Northwest and Austria to the East.

- Switzerland is a landlocked country in Central Europe, well known for its political neutrality, high standard of living, and strong economy. It is bordered by Germany to the north, France to the west, Italy to the south, and Austria and Liechtenstein to the east.
- The country lies roughly between 46°–48° North and 6°–10° East, as shown on the map.
- The capital city is Bern, while major cities include Zurich, the largest city and economic hub, Geneva, an important international city, and Lausanne.
- Switzerland has many large lakes, such as Lake Geneva, Lake Neuchâtel, and Lake Constance, which play important roles in transport, tourism, and water supply.
- Physically, Switzerland is dominated by the Alps, the Swiss Plateau, and the Jura Mountains, giving it spectacular scenery and making tourism a key economic activity.
- The country experiences a temperate climate, varying with altitude.
- Switzerland is famous for banking, precision industries (watches, machinery), dairy farming, tourism, and hosting international organizations. It is a federal republic with strong democracy, multilingual society (German, French, Italian, Romansh), and excellent infrastructure, making it one of the most developed countries in the world.



b) Tourist attractions in Switzerland

The tourism sector of Switzerland is one of the most developed in the world with following touristic attractions:

- Alpine Mountains Ranges have beautiful sceneries mostly snowcapped peaks and u-shaped valleys
- A variety of Climates with cold winters enabling the formation of snow on the Mountain sides which attract tourists for skiing. Climates with warm summers are good for sight-seeing
- Presence of Rivers and lakes. Switzerland has many lakes such as Geneva, Lucerne, Constance and Zurich. Also, the beautiful waterfalls along River Rhine attract tourists.

- Presence of Fauna and flora. There is a variety of vegetation types like a mixture of deciduous and coniferous forests. There are also animals such as deer, stag, fox, wolves, and so on. Also, there are a variety of birds like the black redstars, blackbirds, robins, pigeons, etc.

Application activity 19.1

1. Mention two official languages spoken in Switzerland.
2. Explain how Switzerland's location and physical features have contributed to the development of tourism.
3. Assess the importance of political neutrality to Switzerland's economic and international role.
4. Evaluate how Switzerland's infrastructure and governance system support its high standard of living.
5. Using the map, discuss how Switzerland's landlocked position influences its transport and trade.

19.2. Factors for the development of tourism industry in Switzerland

Learning activity 19.2

A group of students visiting Switzerland were amazed by its natural beauty and efficient transport. They learned how rivers generate hydro-electric power, how tunnels and trains reduce traffic, and how political neutrality attracts international organizations and tourists. The trip showed how Switzerland balances infrastructure, safety, and nature to remain a top tourist destination.

1. Fill in the blanks
 - a) Switzerland generates abundant hydro-electric power from rivers such as the _____ and _____.
 - b) Switzerland's policy of _____ has made it a home for people from many countries.
3. Which factor mainly helps reduce traffic congestion in Switzerland?

A. High-grade hotels	B. Alpine Road tunnels and electrified railways
C. Banking services	D. Tourist concessions
4. Which international organizations are based in Switzerland

A. AU and ECOWAS	B. WHO and ICRC
C. IMF and World Bank	D. UNEP and FAO

❖ Factors in the development tourism industry in Switzerland include:

- The efficient transport and communication network by road, railway, water and air

- Abundant hydro-electric power generated on numerous rivers such as Rhone, Rhine
- The availability of accommodation facilities characterized by high grade hotels and lodges
- A wide range of tourist attractions ranging from the beautiful landscape, glacial features cascading rivers, lakes and a natural park
- The Swiss policy of neutrality made Switzerland a home to nearly all people of the world
- The WHO (World Health Organization), ICRC (International Committee of Red Cross) are based in Switzerland
- The diversity of languages such as German, English, French, Italian, Latin and so many others.
- Switzerland's variety of climates
- The Swiss hospitality where Swiss are very friendly people always willing to serve tourists.
- The high technology used in the easy management of the industry attracts many tourists
- The large, rich banks of universal type have made Switzerland an international finance centre
- Tourist guides are used in order to minimize the risk of accidents
- National road network, tunnels such as alpine Road Tunnel, electrified double-track routes are being emphasized to reduce on traffic congestion and joining Switzerland to Rhine fast motorways (Autobahn).
- Labour shortage is being minimized by importing labour from Italy, German and Poland.
- Granting concessions to tourists, especially to those who go there in large, organized groups/clubs.

Application activity 19.2

1. Match Column A with Column B:

Column A	Column B
i. Political neutrality	a. Safe management of tourists
ii. Tourist guides	b. International attraction
iii. Hydro-electric power	c. Reliable energy supply
iv. High-grade hotels	d. Quality accommodation

2. Explain the physical and human factors that have contributed to the development of tourism in Switzerland.

3. Assess how transport infrastructure and political neutrality have promoted Switzerland as a leading tourist destination.

19.3. The importance of tourism, problems affecting tourism and possible solutions in Switzerland

Learning activity 19.3

1. Fill in the blanks

- Tourism in Switzerland promotes the growth of _____ as locals make antiques and crafts for tourists.
- Tourism provides employment in _____, _____, and _____.
- One challenge to winter tourism in Switzerland is _____.

2. What is the main foreign exchange earner in Switzerland?

- A. Agriculture B. Tourism C. Banking D. Manufacturing

3. Which of the following is a solution to tourism problems in Switzerland?

- A. Reducing hotel standards B. Diversifying tourism activities
C. Closing ski resorts permanently D. Discouraging foreign tourists

a) Importance of tourism in Switzerland include:

- It has promoted the growth of craftsmanship as people develop antiques and crafts pieces that are on a high demand among the tourists
- Source of foreign exchange earning (the main foreign exchange earner in Switzerland).
- Provides employment in hotels, transport, and tourism service
- Promotes development of infrastructure like roads, railways, and cable cars
- Encourages conservation of natural landscapes and wildlife
- Promotes Swiss culture and international relations

b) Problems affecting tourism in Switzerland

- Due to global warming, the snow has been reducing such that in some areas, tourists cannot engage in skiing any more
- Competition from other countries
- Occurrences of accident in the mountains
- The strong Swiss franc
- High cost of tourism services
- Seasonal nature of tourism activities
- Environmental degradation in tourist areas
- Effects of climate change on winter tourism

c) Possible solutions to tourism problems in Switzerland

- Promotion of sustainable and eco-tourism
- Diversification of tourism activities beyond winter sports
- Investment in modern transport and tourism infrastructure
- Enforcement of strict environmental protection laws

Application activity 19.3

1. Say whether the following sentence is True or False
 - a) Tourism in Switzerland encourages conservation of wildlife and natural landscapes.
 - b) Global warming has no impact on winter tourism in Switzerland.
 - c) High-quality infrastructure like cable cars helps support tourism.
2. Explain the importance of tourism to Switzerland's economy and culture
3. Discuss the problems affecting tourism in Switzerland and suggest possible solutions.

End of unit assessment

❖ Knowledge and understanding:

1. Switzerland is located in:
A. Central Europe B. East Asia C. South America D. West Africa
2. Switzerland is bordered by:
A. Germany, France, Italy, Austria, Liechtenstein,
B. Spain, Germany, Austria, Poland
C. France, Netherlands, Belgium, Italy,
D. Norway, Sweden, Denmark, Finland

3. The capital city of Switzerland is: A. Zurich B. Geneva C. Bern D. Lausanne
4. One of the most famous mountain ranges attracting tourists is:
A. Alps B. Rockies C. Himalayas D. Andes
5. Switzerland's tourism is supported by:
A. Desert landscapes B. Snow-capped mountains, lakes, and rivers
C. Tropical beaches D. Savannah grasslands
6. Switzerland is famous for:
A. Banking, tourism, dairy farming, precision industries B. Oil production
C. Rubber plantations D. Coffee and tea exports
7. Tourism in Switzerland promotes:
A. Foreign exchange earnings B. Decrease in employment
C. Closure of hotels D. Urban-rural migration
8. Tourism provides employment in:
A. Hospitals and schools B. Hotels, transport, and tourism services
C. Factories D. Mining
9. Tourism encourages:
A. Conservation of natural landscapes and wildlife B. Deforestation
C. Desertification D. Industrial pollution

❖ Skills

10. Draw a sketch map of Switzerland showing its location in relation to Germany, France, Italy, and Austria
11. Describe the factors that favour the development of tourism in Switzerland.:

❖ Attitudes & Values

12. Explain the importance of tourism to Switzerland's economy and assess problems affecting the industry.
13. Compare the level of tourism development in Switzerland with Rwanda and suggest lessons Rwanda can learn

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